Aerospace Technology

Study notes for the whole course

Notes by **Eden R.S.**

Master's Degree in Unmanned Aircraft Systems
and Associated Technologies · UPV

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Summary of the course slides, written for the exam, with extra explanations for readers new to the field. It covers the first-midterm syllabus: the slides, up to propulsive systems. Red boxes mark what you must know and blue boxes are worked examples, many taken from past exams.

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UNIT 1

Introduction to aerospace technology

1 Course overview

Aerospace Technology (code 11888, UPV, academic year 2026–2027) is a **general overview** course: it does not go deep into a single topic, but gives a view of all the systems that make up an aircraft or a spacecraft, with particular emphasis on propulsion. It bridges the basic subjects (physics, mathematics, thermodynamics) and the applied ones (aircraft design, propulsion, structures). It is worth 4.5 ECTS credits (2.5 theory and 2.0 practical), and a maximum of 20 % absences is allowed, which weighs especially on the practical sessions.

1.1 Content and assessment

The slides divide the syllabus into five blocks, and these notes follow that order. The available slides cover only the first two blocks, and these notes cover only those:

Course block	Content	Unit of these notes
Introduction	industry, vehicles, infrastructure	1
Fixed-wing aircraft	environment, design, structures, flight, propulsion	2 to 6
Space vehicles	launchers, space environment, satellites	not covered (no slides)
Missiles	classification, guidance	not covered (no slides)
Miscellaneous aircraft	rotorcraft, special aircraft	not covered (no slides)

Assessment	Weight and date	Details
Multiple-choice test 1	20 %, 05/11/2026, 12:30	up to section 2.5
Multiple-choice test 2	20 %, 15/01/2027, 8:00	from section 2.6 onwards
Open-answer test	35 %, 15/01/2027, 9:00	all sections
Oral defense	20 % (group)	30/11 to 02/12/2026; plus a debate on 18/12/2026
Observation	5 %	throughout the course
Resit	27/01/2027, 11:30	any combination of the three written tests; the resit mark replaces the original one

WHAT THE MULTIPLE-CHOICE TESTS ARE LIKE
According to past exams: multiple-choice questions with **a single correct answer**, and **each wrong answer subtracts a third** of what a correct one adds. Blank answers score nothing. It is best to answer only when you can rule out at least one option.

Note: the slides do not state how sections 2.5 and 2.6 map onto these units, and no mapping has been assumed (source: course presentation, slides 6 to 11).

2 The aerospace sector

Aerospace covers two worlds: **aeronautics** (flight within the atmosphere) and **astronautics** (flight outside it). The slides divide the activity into six groups: industry, airlines, aeronautical organizations, space agencies, research centers and the military.

2.1 The aerospace industry

It is a strategic activity with very particular characteristics:

Characteristics (learn them: they appear in tests)	
Cutting-edge technology	**Long** delivery times
Highly qualified staff	High investment required
Small scales	Government intervention
Difficult automation	International cooperation

The logic is a chain: few units are built (small scales) and each one is complex (difficult automation), so each aircraft takes a long time to deliver and requires a lot of money. That is why governments intervene and projects are shared among countries.

> **COMMON TEST MISTAKE**
> "*Short* delivery times" is **not** a characteristic of the aerospace industry (it appeared in the 2018/19 and 2019/20 exams). They are long.

A fact to remember the cost. Pound for pound, an aircraft engine is worth the same as silver, and a car the same as a hamburger (Northwest Aerospace Alliance, cited on slide 5). In the 2022/23 exam, the cost per kg of an A320 turbofan was asked; the official key as collected was *"about 100 times that of a car piston engine"* (the option "like gold" was not marked), so it is worth remembering that the cost per kilogram is an order of magnitude higher than that of a car engine, not the exact figure.

Breakdown of the industry (slide 6; Figure 1.1):

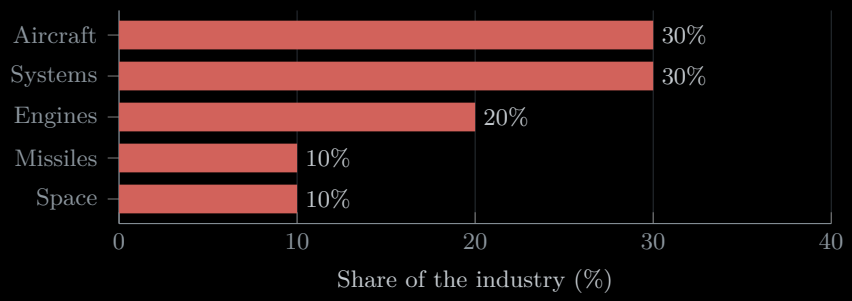


Figure 1.1. Approximate breakdown of aerospace industry activity. Source: T1, slide 6.

The teams are **highly multidisciplinary** (flight physics, propulsion, structures, avionics…). Each program requires: design and development; advanced testing and calculations (wind tunnel, flight tests, CFD, *Computational Fluid Dynamics*); production (manufacturing, assembly and certification); and maintenance.

2.2 Airlines

Airlines are the industry's **customers**. They are distinguished by their target audience: traditional (Iberia, Lufthansa), low-cost (easyJet, Ryanair, Vueling) and regional (Air Nostrum). It is a market

with a large volume of money but small profits and fierce competition, so airlines form alliances, merge, and many will disappear. For the designer this means that the customer demands **low operating cost**.

2.3 Aeronautical organizations

They were created mainly to **maintain air traffic safety**. It is worth knowing what each one does and its scope (Figure 1.2):

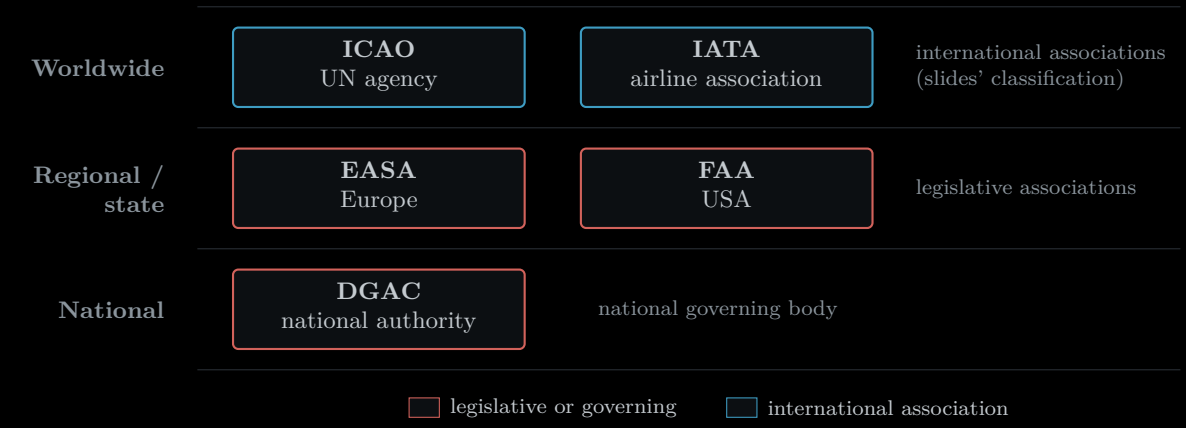


Figure 1.2. Who is who in the aeronautical organization, following the slides' classification. Source: T1, slide 9.

- **ICAO** (International Civil Aviation Organization): specialized UN agency. It is **not** a research center, nor the agency that legislates for Spain.
- **IATA** (*International Air Transport Association*): partnership of airlines for cooperation.
- **EASA** (*European Union Aviation Safety Agency*): main legislative organization in **Europe**. **FAA** (*Federal Aviation Administration*): the one in the **USA**.
- **DGAC** (General Directorate of Civil Aviation): national governing body.

> **BEWARE OF EXAM MIX-UPS**
>
> - **INTA** is a research and testing center and does **not** legislate (2023/24 question).
> - **ESA** is a **space** agency; it does not define European aviation legislation: that is EASA (2023/24 and 2024/25).
> - ICAO is an international organization **created by the UN** (2020/21). IATA does **not** legislate in Europe. DGAC is not a US body.

2.4 Space agencies and research centers

The main space agencies are **ESA** (*European Space Agency*) and **NASA** (*National Aeronautics and Space Administration*); some countries have their own. Research centers include the space agencies themselves, the French **ONERA** (*Office National d'Études et de Recherches Aérospatiales*) and **INTA** (*National Institute of Aerospace Technology*, the Spanish public institute for R&D in aeronautics, space and naval operations).

INTA tests materials, components, equipment and systems, and provides services to industry. Milestones cited in the slides: *Intasat* (the first Spanish satellite) and collaboration with NASA's Mercury and Apollo projects. Its facilities (slide 11) include a turbojet testing center, the El Arenosillo Testing Centre and space tracking stations.

2.5 Military activity

Air forces are both **customers and users** of aerospace equipment, and have maintenance and research centers. The Spanish Air Force is organized into a Force (material and human resources to prepare for and carry out operations) and a Force Support (management, administration and control of resources), according to slide 12.

3 Classification of aerospace vehicles

Three major families are used: **aircraft**, **missiles** and **space vehicles**, organized as in Figure 1.3.

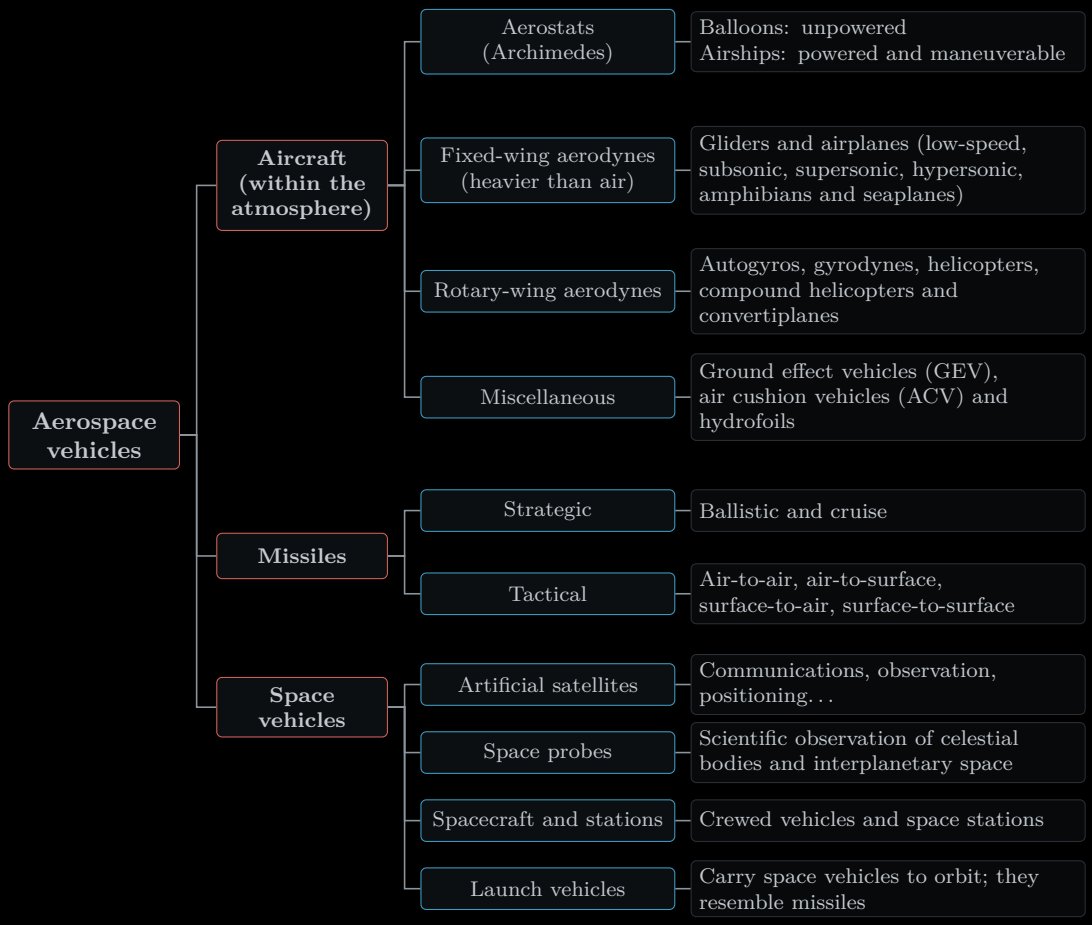


Figure 1.3. Classification tree of aerospace vehicles. Source: T1, slides 14 to 23 (slide 21 repeats "air-to-surface"; it has been completed with surface-to-air).

3.1 Aircraft: how they overcome weight

The criterion on slide 14 is *how they generate the force that overcomes gravity*. There are two fundamentally different ways. Figure 1.4 shows what each family looks like and Figure 1.5 where its supporting force comes from:

- **Aerostats**: they use **Archimedes' principle** ("any body submerged in a fluid experiences an upward vertical force equal to the weight of the fluid displaced"). They are large containers filled with a light gas, so that their average density is **lower than that of the surrounding air**. Two types: **balloons** (unpowered) and **airships** (powered and maneuverable).
- **Aerodynes** (the slide simply calls them *aircraft*): they are **heavier than air** and are supported by aerodynamic forces generated by fixed or rotating wings.

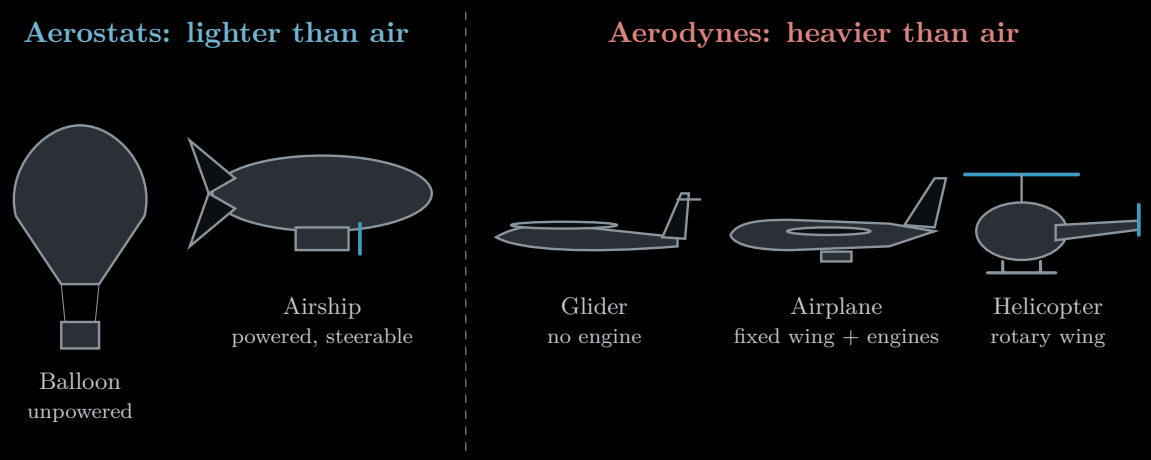


Figure 1.4. Aircraft by how they overcome weight: aerostats (lighter than air) and aerodynes (heavier than air). Side views, schematic silhouettes, not to scale.

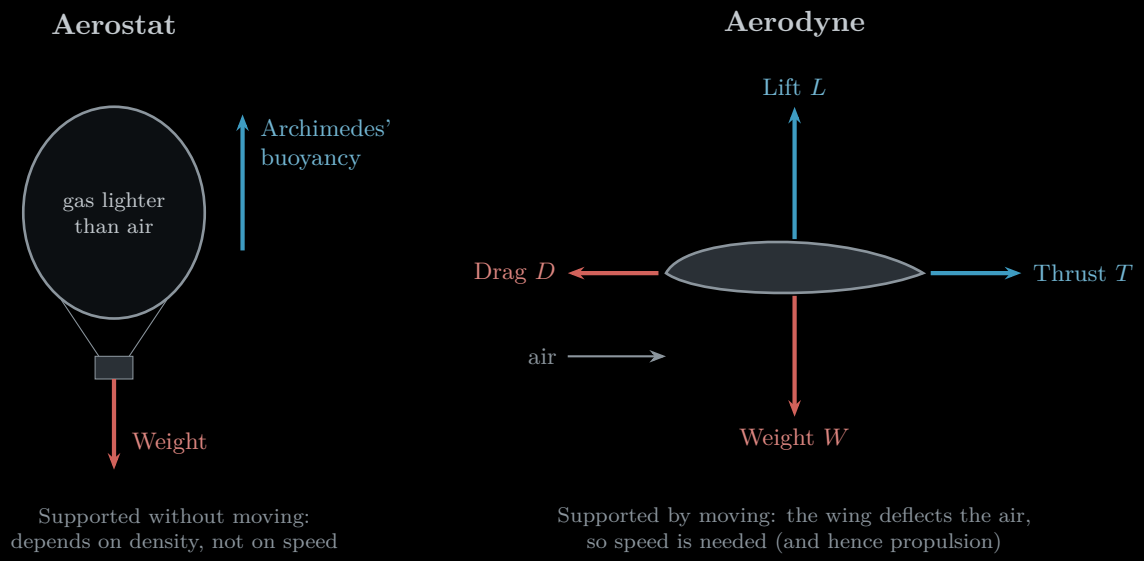


Figure 1.5. Aerostat versus aerodyne: where the force that supports the aircraft comes from.

- **Miscellaneous**: Ground Effect Vehicles (GEV), which fly very close to the surface on a layer of air; Air Cushion Vehicles (ACV, or *hovercraft*), which create a volume of pressurized air under the vehicle; and hydrofoils, wing sections that work submerged in water instead of air (Figure 1.6).

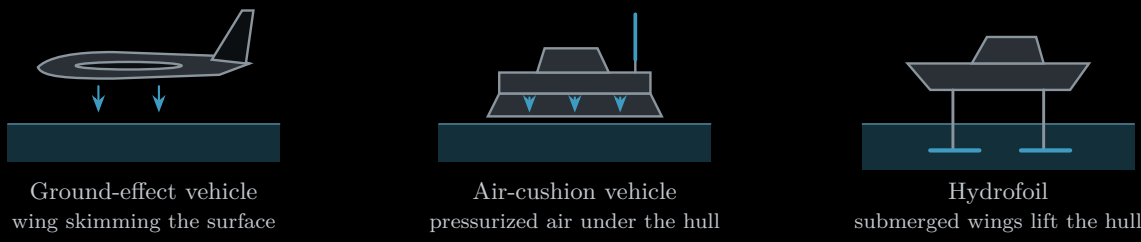


Figure 1.6. Miscellaneous vehicles: ground-effect vehicle (cyan arrows = air trapped between wing and surface), air-cushion vehicle and hydrofoil. Schematic.

Analogy. An aerostat floats like a cork in water and does not need to move. An aerodyne is supported like a planing boat: it only does so while moving forward, because the wing needs air in relative motion.

Aerodynes are subdivided into:

- **Fixed wing**: gliders (unpowered) and airplanes. They are also classified by speed regime (low speed, subsonic, supersonic, hypersonic), and there are amphibians and seaplanes.
- **Rotary wing**: autogyros, gyrodynes, helicopters, compound helicopters and convertiplanes.

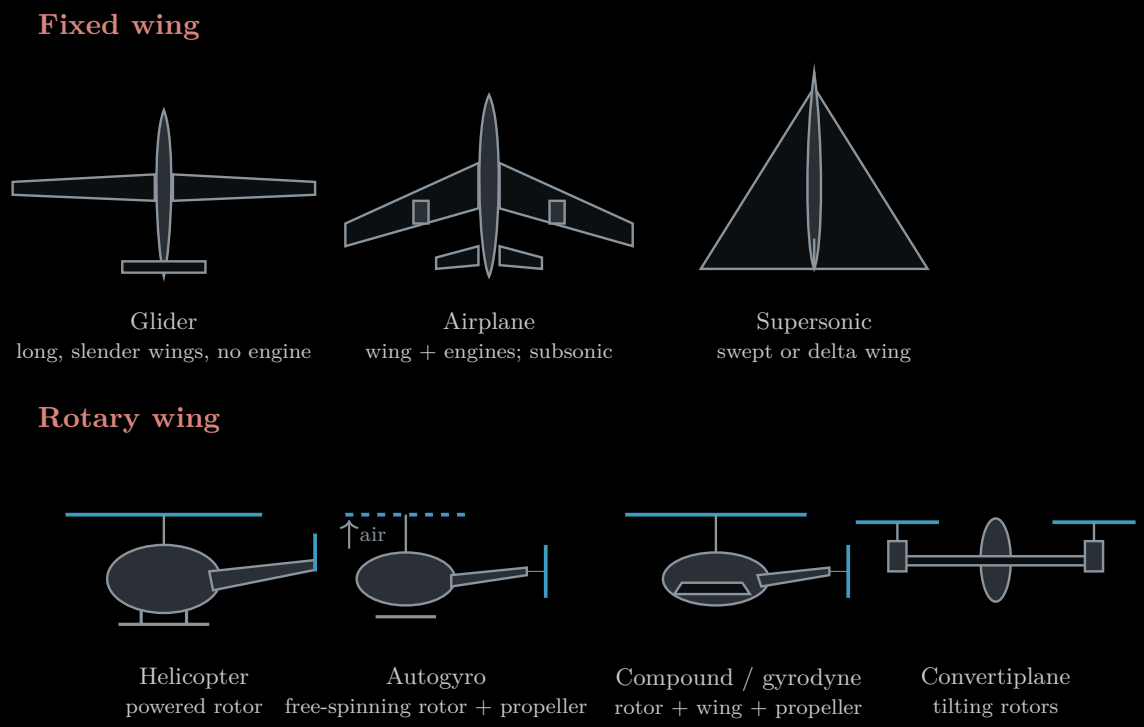


Figure 1.7. Aerodyne families: fixed wing (top views) and rotary wing (side views, blue = rotor). Schematic silhouettes, not to scale.

RECURRING EXAM QUESTION: AEROSTATS VERSUS AERODYNES
Aerostats do **not** need airflow over airfoils (they are not supported by a wing), do **not** deflect flow toward the ground, and the A320 or the B737 are **aerodynes**. It is not true that aerodynes must be lighter than air (it is the opposite: they are heavier) nor that aerostats with a less dense gas need shorter runways (they need no runway to take off). The popular explanation "the air travels a longer path over the upper surface" is not valid either (rejected in 2022/23). In 2024/25 it was marked that aerostats *cannot fly at high speeds because of their aerodynamic drag* (large volumes).

3.2 Missiles

A missile is an **unmanned, powered vehicle that guides a military payload to a target**. It is different from a guided bomb (which has no propulsion of its own) and from a rocket (which is not guided). The most common classification is by military function:

- **Strategic**: ballistic (long range, leave the troposphere) and cruise.
- **Tactical**: air-to-air, air-to-surface, surface-to-air and surface-to-surface.

In a 2020/21 test the false option was *"cruise missiles commonly reach supersonic speeds in cruise"*: they usually fly subsonic, like a small unmanned airplane.

3.3 Space vehicles

They are designed for **extra-atmospheric** flight: artificial satellites (which orbit the Earth: communications, observation, positioning…), space probes (scientific observation of celestial bodies and interplanetary space), crewed spacecraft and space stations, and **launch vehicles** (which carry the others to orbit; they resemble missiles). Figure 1.8 shows the five types.

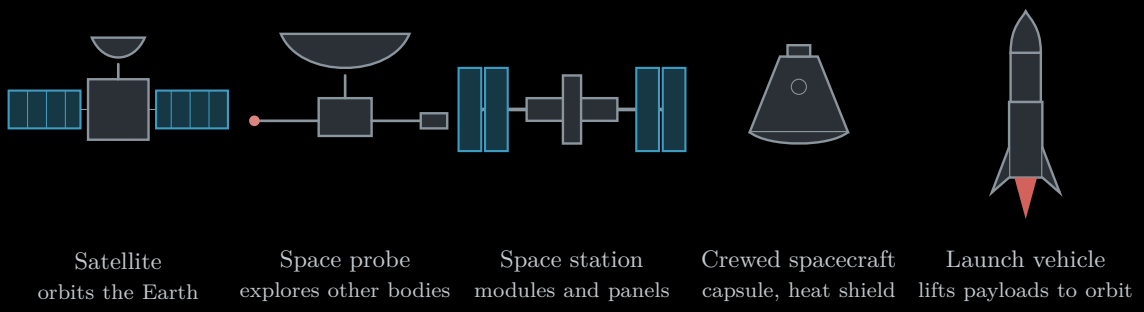


Figure 1.8. Types of space vehicles. Schematic silhouettes, not to scale.

4 Aerospace infrastructure

4.1 For aircraft

Aerodromes and airports. *Aerodrome*: a defined area used wholly or partly for the departure, arrival and surface movement of aircraft. *Airport*: an aerodrome with permanent facilities and services open to the public. It is divided into the **airside** (runways, aircraft parking), the **landside** (passenger terminal), the industrial zone and the flight control and navigation system (Figure 1.9).



Figure 1.9. Schematic airport layout in plan view: the airside (where aircraft move) and the landside (where passengers arrive).

Air Traffic Control (ATC). Three levels of control that hand the aircraft over from one to the next (Figure 1.10):

- **TWR** (control tower): takeoffs, landings and movements at the airport.
- **Terminal Control Centers**: approach/departure area around the airport.
- **En-route Air Traffic Control Centers**: the aircraft in cruise.

Navigation aids. They are the "lighthouses and roads" of the sky. The slides list them; in Unit 2 (Section 5, low-visibility flight) it is explained how each one works.

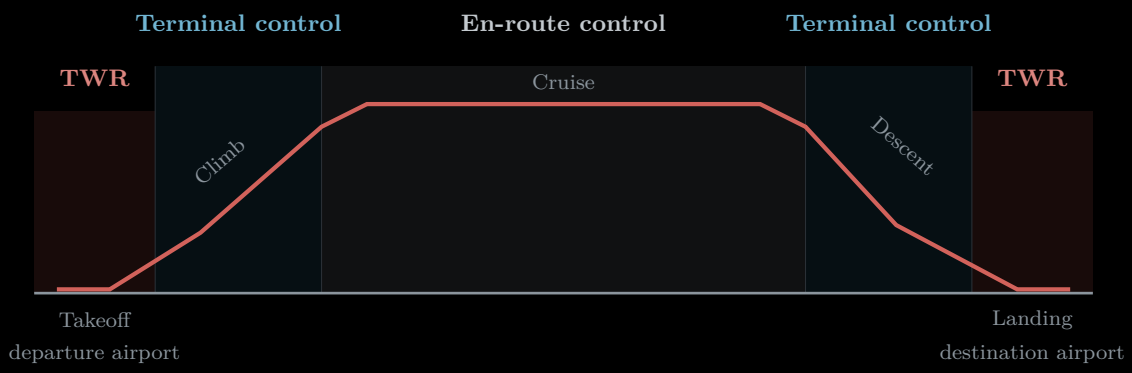


Figure 1.10. Who controls the aircraft in each phase of flight. Source: T1, slide 27 (the TWR tower is the airport's own control; the phase diagram is explanatory).

- Ground stations: **VOR** (*VHF Omni-directional Range*), **DME** (*Distance Measuring Equipment*), **NDB** (*Non-Directional Beacon*), **ILS** (*Instrument Landing System*), **PAPI** (*Precision Approach Path Indicator*).
- Satellites: the GPS constellation (and GNSS in general).

Figure 1.11 sketches the idea behind each ground aid.

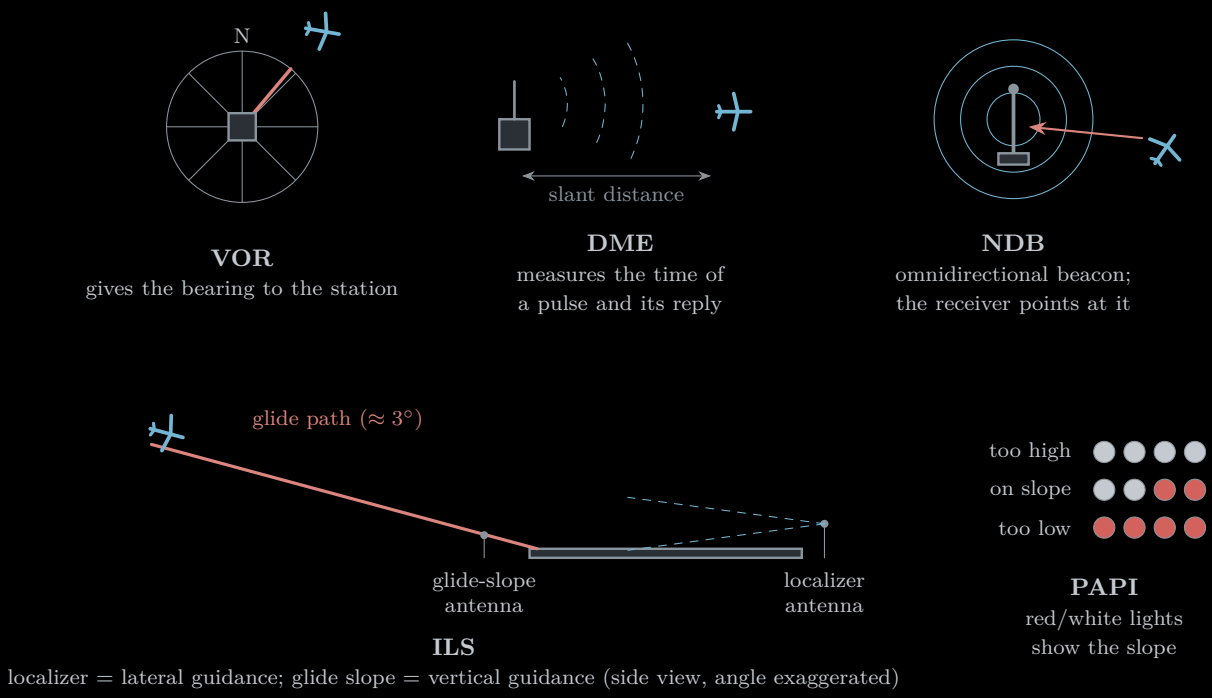


Figure 1.11. Ground navigation aids, schematic (not to scale). Detail of each in Unit 2.

4.2 For space vehicles

- **Launch sites**: they must be close to the **equator** (the Earth's rotation contributes speed: about 0.46 km/s at the equator, supplementary data) and to the **coast**, in an **uninhabited** region (if the rocket fails, it should fall into the sea). Nearby they need an industrial zone, final assembly facilities, power supply, a port and airport, and the launch pad.
- **Control centers**: they monitor the trajectory, check the operation of instruments and systems, and send commands (cameras, orbit corrections). Since a satellite moves relative to the center, **tracking stations** spread around the world are needed (example on the slide: those of the ISS).

FROM THE 2020/21 TEST
"Launch sites: (a) close to inhabited areas; (b) close to the equator to take advantage of the Earth's motion; (d) require industrial zones in the surroundings": the collected key marked "all the others are correct", but (a) contradicts slide 29 (uninhabited region). Remember: equator, coast, uninhabited region and industrial zone.

5 Future challenges

Once durability and safety are "guaranteed", the slides cite these challenges: in **propulsion**, more efficient concepts (outer rotor, boundary layer ingestion, distributed electric propulsion), alternative fuels and more electric engines; in **materials and structures**, the adoption of composite materials, new materials (metallic foams, nanomaterials) and non-destructive testing; in **logistics and manufacturing**, optimizing the supply chain and automating more; in **operations**, more efficient air traffic management and navigation; and **data management** (*big data*).

6 Summary and what usually appears on the exam

WHAT USUALLY APPEARS ON THE EXAM
This part contributes **one or two stand-alone test questions** per exam, almost always concept review (2018/19 to 2024/25 midterms):

- Characteristics of the industry ("short delivery times" is the false one) and organizations (EASA, ICAO, IATA, INTA, ESA), always as matching an organization to its function.
- Aerostats (Archimedes) versus aerodynes (aerodynamic lift); classification of missiles and launch sites.
- Navigation aids: Unit 2. There are no numerical problems in this unit.

UNIT 2

The atmospheric flight environment

1 Why the atmosphere matters

An aircraft is an object that "swims" in the air. Everything it can do (be supported, move forward, climb) depends on **what the air is like** around it: how much mass there is in each cubic meter, what temperature it is at, whether there is wind, clouds or ice. This unit describes that environment and builds a simple **model** so that calculations can be made.

1.1 Forces in cruise

In straight and level flight at constant speed (*cruise*) there are four forces: **lift** L (upward, the force that supports the aircraft), **weight** W (downward), **drag** D (which slows it down) and **thrust** T (pushes it forward, provided by the engine). In cruise they balance: $L = W$ and $T = D$ (Figure 2.1).

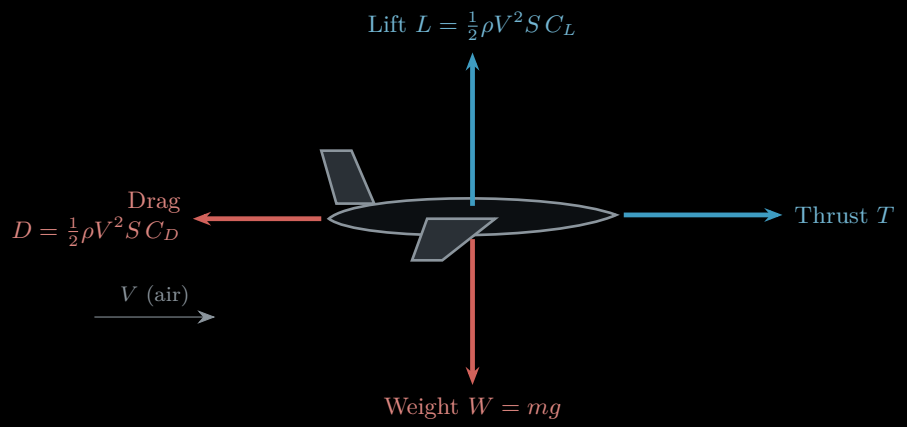


Figure 2.1. Forces in cruise flight (simplified version). Source: T2, slide 4.

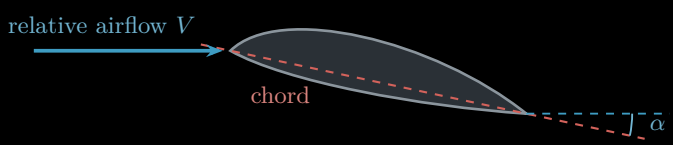


Figure 2.2. Angle of attack α: the angle between the wing chord and the relative airflow V. Schematic.

$$L = \tfrac{1}{2}\,\rho\,V^2\,S\,C_L \qquad D = \tfrac{1}{2}\,\rho\,V^2\,S\,C_D \qquad W = m\,g$$

Glossary of symbols: ρ is the air density (kg/m^3); V is the aircraft's speed relative to the air (m/s); S is the wing area (m^2); C_L and C_D are the lift and drag coefficients (dimensionless numbers that depend on the shape of the wing and the angle of attack α, defined in Figure 2.2); m is the mass (kg); g is gravity (9.81 m/s^2).

The key point: L and D are proportional to the density ρ. Engine thrust and power have a more complex dependence, but, broadly speaking, thrust also increases with density (the higher you fly, the less air enters the engine). In addition to the above, density influences:

- the **buoyancy** of aerostats: it is a direct function of the air density;
- **cabin pressurization**: a breathable pressure must be maintained inside while outside there is almost none;
- the **weather**: headwind and tailwind, icing and visibility;
- other effects specific to speed or altitude: **kinetic heating** and **radiation**.

The effects of altitude are important, day-to-day variations are significant and geographic position matters quite a lot. This is why an **atmosphere model** is needed that gives, at least, the properties of the air as a function of flight altitude.

> **WHAT HAPPENS TO THE ANGLE OF ATTACK WHEN CLIMBING (2023/24 EXAM)**
> An aircraft flies at 10 km at 850 km/h and climbs to 12 km **without changing its speed**. What does its angle of attack α (the angle between the wing and the incoming air) do?
> **Setup.** In level flight, $L = W$, that is, $\frac{1}{2}\rho V^2 S C_L = W$. With V, S and W constant, the required coefficient is $C_L = \dfrac{2W}{\rho V^2 S} \propto \dfrac{1}{\rho}$.
> **Step 1.** The higher you climb, the less dense the air is: $\rho(10\text{ km}) = 0.413$ and $\rho(12\text{ km}) = 0.311$ kg/m^3. The ratio is $0.413/0.311 \approx 1.33$: a C_L 33 % higher is needed.
> **Step 2.** In the linear region of the wing's curve, C_L grows with α, so α must **increase during the 2 km**. Answer: **it increases over the two km** (option b).
> **Common mistake.** The temperature is constant above 11 km, but that does not mean the density stops decreasing: the pressure keeps falling and therefore so does $\rho = p/(RT)$.

2 The International Standard Atmosphere (ISA)

2.1 Layers of the atmosphere

The atmosphere is divided into layers according to how the temperature varies with altitude (Figure 2.3):

Layer	Altitude	T gradient	What happens
Troposphere	0 to 11 km	−6.5 K/km	weather occurs here; small aircraft fly here, and airliners in its upper part
Stratosphere (1)	11 to 20 km	0 (isothermal, 216.65 K)	airliner cruise
Stratosphere (2)	20 to 32 km	+1 K/km	
Stratosphere (3)	32 to 47 km	+2.8 K/km	the temperature rises to ≈ 270.65 K at the stratopause
Mesosphere, thermosphere	above	—	outside the scope of the course model (slide 8)

The boundary between the troposphere and the stratosphere is the **tropopause** (11 km); the one that closes the stratosphere is the **stratopause** (47 km in ISA).

Why the temperature drops in the troposphere. The ground, heated by the Sun, warms the air from below; the farther from the ground, the colder. In the stratosphere the ozone layer absorbs ultraviolet radiation, which is why the temperature stops dropping and then rises (supplementary explanation, not in the slides).

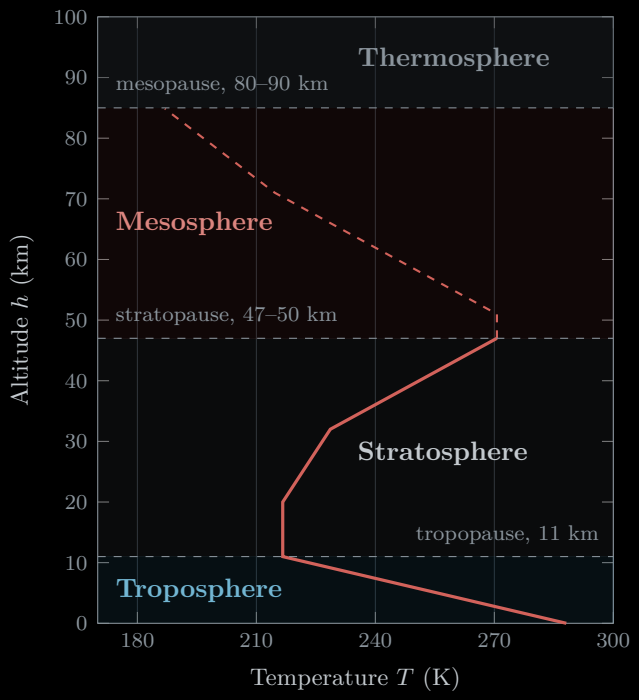


Figure 2.3. Layers of the atmosphere and ISA temperature profile (solid line, up to 47 km; the dashed lines above are the standard continuation beyond what the slides require, 47 to 85 km). Source: T2, slides 8 and 9; the upper limits (stratopause, mesopause) are approximate, as in the slide figure.

2.2 The ISA model (ISO 2533:1975)

The *International Standard Atmosphere* is a **fictitious, average** atmosphere that lets all engineers use the same numbers (design, testing, altimeters). Its assumptions:

- It is defined as a function of **altitude**, measured from sea level. **Careful**: this is not the same as **height**, which is the distance to the ground (an aircraft 3 000 m above an airport located at an altitude of 600 m is flying at an altitude of 3 600 m; Figure 2.4).
- Air behaves as an **ideal gas**: $p = \rho R T$ (with $R \approx 287$ J/(kg K), the gas constant of air; supplementary data).
- There is **hydrostatic equilibrium** (fluid at rest): the pressure at a point supports the weight of the air above it, $\dfrac{\mathrm{d}p}{\mathrm{d}h} = -\rho g$.
- The temperature follows the piecewise law in the table above.
- Sea-level values: $T_0 = 288.15$ K (15 °C), $p_0 = 101\,325$ Pa, $\rho_0 = 1.225$ kg/m^3.
- The altitude is **geopotential**, not geometric: $h = \dfrac{1}{g_0}\displaystyle\int_0^h g\,\mathrm{d}z$, with $g_0 = 9.80665$ m/s^2. This avoids having to account for g changing slightly with height (at aircraft flight altitudes the difference is negligible; for problems, $g = g_0$ is taken as constant).

Where the formulas come from. Combining the two equations: from $\dfrac{\mathrm{d}p}{\mathrm{d}h} = -\rho g$ and $\rho = \dfrac{p}{RT}$ we get $\dfrac{\mathrm{d}p}{p} = -\dfrac{g}{RT}\,\mathrm{d}h$. There are two cases:

- **Troposphere** ($T = T_0 - \lambda h$, with $\lambda = 6.5 \cdot 10^{-3}$ K/m): integrating gives a power law.
- **Isothermal layer** (T constant): integrating gives an exponential.

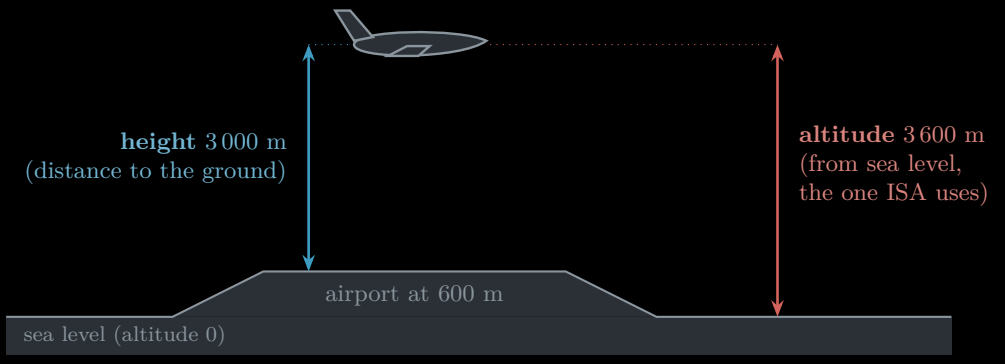


Figure 2.4. Altitude (from sea level) versus height (above the ground) for an aircraft over an airport at 600 m.

Troposphere ($0 \le h \le 11$ km), with $A = \dfrac{g_0}{6.5 \cdot 10^{-3}\, R} \approx 5.256$:

$$T(h) = 288.15 - 6.5 \cdot 10^{-3}\, h$$

$$p(h) = 101\,325 \left(1 - \frac{6.5 \cdot 10^{-3}\, h}{288.15}\right)^{A}$$

$$\rho(h) = 1.225 \left(1 - \frac{6.5 \cdot 10^{-3}\, h}{288.15}\right)^{A-1}$$

(with h in meters; p in Pa; ρ in kg/m^3). Note that the exponent of ρ is $A - 1 \approx 4.256$, one less than that of p, because $\rho = p/(RT)$ and T also decreases. **These are the formulas the slides ask for**; for the isothermal stratosphere, $p = p_{11} \exp\left[-\dfrac{g_0(h - 11000)}{R\,T}\right]$ with $T = 216.65$ K and $p_{11} = 22\,632$ Pa (supplement), and $\rho = p/(RT)$ in any case. Figure 2.5 plots these profiles.

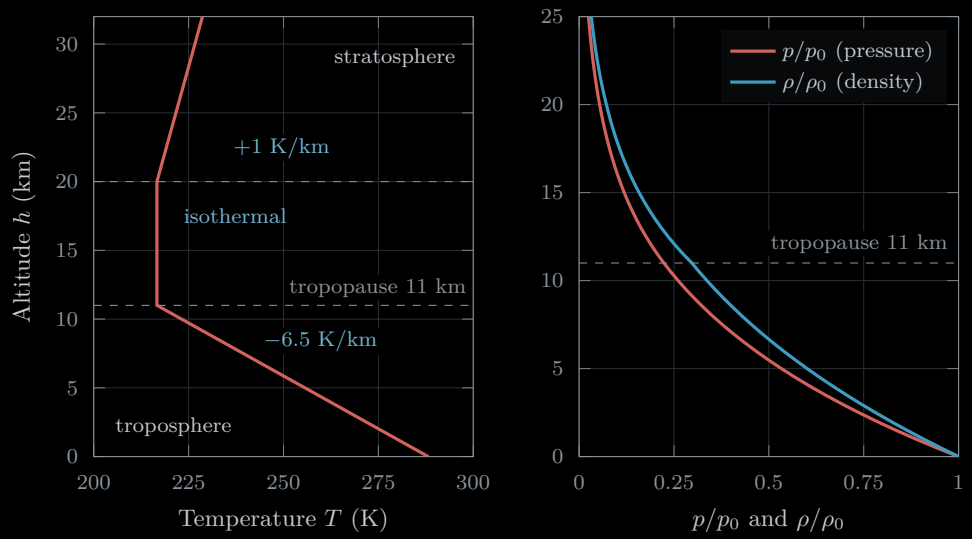


Figure 2.5. ISA profiles. Left: temperature (red). Right: relative pressure and density, which fall *nonlinearly*. Source: T2 model, slides 9 and 10.

h (km)	T (K)	p (kPa)	ρ (kg/m^3)	ρ/ρ_0
0	288.15	101.33	1.225	1.000
1	281.65	89.87	1.112	0.907
2	275.15	79.50	1.007	0.822
3	268.65	70.11	0.909	0.742
4	262.15	61.64	0.819	0.669
5	255.65	54.02	0.736	0.601
6	249.15	47.18	0.660	0.539
8	236.15	35.60	0.525	0.429
10	223.15	26.44	0.413	0.337
11	216.65	22.63	0.364	0.297
12	216.65	19.33	0.311	0.254
15	216.65	12.04	0.194	0.158
20	216.65	5.47	0.088	0.072

Table computed with the model formulas ($R = 287.053$ J/(kg K)); it is not in the slides and serves to check calculations. Rule of thumb: at 5.5 km (about 18 000 ft) half the pressure remains, and at 11 km a little more than one fifth.

COMMON MISTAKES WITH ISA

- The **temperature** drops **6.5 K per kilometer** (not per meter) in the troposphere. The pressure does **not** drop linearly ("10 kPa/km" is false; it was tested in the 2024/25 exam). It falls almost exponentially.
- Above the tropopause the **pressure keeps dropping**; only the temperature is constant up to 20 km. The pressure at 15 km is **lower** than at the tropopause (2023/24 exam).
- The temperature does **not** always decrease with altitude: in the stratosphere it rises (2024/25 exam).
- In the stratosphere "the temperature decreases 6.5 K/km" is false: it is isothermal in its first part (2022/23).

ISA PROPERTIES AT 8 KM

Compute T, p and ρ at $h = 8000$ m.

Step 1 (temperature): $T = 288.15 - 6.5 \cdot 10^{-3} \cdot 8000 = 236.15$ K ($-37\,^\circ$C).

Step 2 (ratio): $\dfrac{T}{T_0} = 1 - \dfrac{6.5 \cdot 10^{-3} \cdot 8000}{288.15} = 0.8195$.

Step 3 (pressure): $p = 101\,325 \cdot 0.8195^{5.256} = 101\,325 \cdot 0.3514 \approx 35\,600$ Pa $= 35.6$ kPa.

Step 4 (density): $\rho = 1.225 \cdot 0.8195^{4.256} = 0.525$ kg/m^3. Check: $\rho = \dfrac{p}{RT} = \dfrac{35\,600}{287 \cdot 236.15} = 0.525$. It is 43 % of the sea-level value.

COMPARING TEMPERATURES WITH ISA±Δ (2023/24 EXAM)

"The temperature at 3 km on a cold day (ISA−15) is lower than at 5 km on a standard day". True?

$T(3\text{ km}, \text{ISA} - 15) = 288.15 - 19.5 - 15 = 253.65$ K; $T(5\text{ km}, \text{ISA}) = 288.15 - 32.5 = 255.65$ K. Since $253.65 < 255.65$: **true**.

"The temperature at 10 km in ISA−10 is higher than at 13 km in ISA": $223.15 - 10 = 213.15$ K versus 216.65 K: **false**. Rule of thumb: ISA±Δ adds or subtracts Δ to the whole temperature curve.

3 Takeoff and landing at high altitude and in hot weather

3.1 High altitude

The density at a high-altitude airport can be estimated with ISA. Example from slide 12: Daocheng Yading airport (China), at 4411 m above sea level, has $\rho_{ISA} = 0.64\,\rho_{SL}$ (SL: sea level).

Consequence: with lower density, **more speed** is needed to support the same weight, so **long runways and higher takeoff and landing speeds** are required. Quantitatively: the stall speed (the minimum speed at which the wing can still support the weight) is

$$V_s = \sqrt{\frac{2W}{\rho\, S\, C_{L,max}}} \;\propto\; \frac{1}{\sqrt{\rho}},$$

so that $\dfrac{V_s}{V_{s,0}} = \sqrt{\dfrac{\rho_0}{\rho}}$. This result has been asked in exams (2019/20: "the stall speed **increases** with altitude"). Figure 2.6 shows how much it grows with altitude.

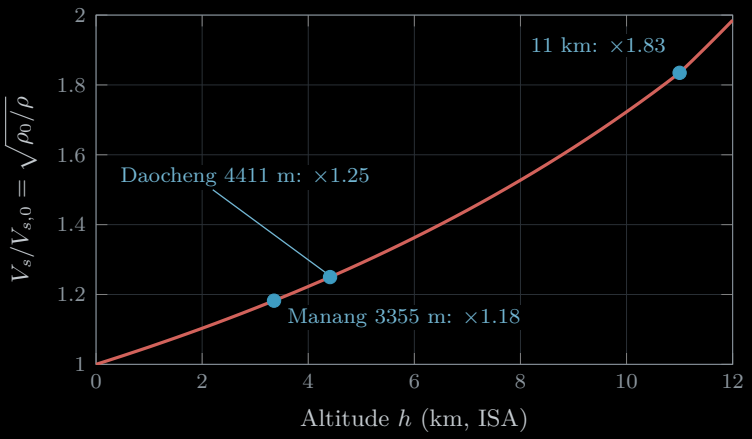


Figure 2.6. How much the stall speed rises with altitude (ISA). Daocheng: $\sqrt{1/0.64} = 1.25$. Own calculation from the ISA model.

MANANG (2020/21 EXAM)
An aircraft has a stall speed of 160 km/h at sea level. What is it at Manang airport (3355 m)?

Step 1: density at 3355 m. $T/T_0 = 1 - \dfrac{6.5 \cdot 10^{-3} \cdot 3355}{288.15} = 0.9243$; $\rho/\rho_0 = 0.9243^{4.256} = 0.715$.
Step 2: $V_s = 160\sqrt{1/0.715} = 160 \cdot 1.182 = 189$ km/h. **Answer: 189 km/h** (option b).

3.2 Hot and cold days

A nonstandard day is described as ISA+Δ or ISA−Δ (examples: ISA+15, ISA−10). This is done as follows:

- The **sea-level temperature is modified** (303.15 K for ISA+15; 278.15 K for ISA−10) and the whole T curve is shifted by the same Δ.
- The **pressure is computed as usual** (the ISA value for that altitude).
- The **density is computed with the ideal gas law**: $\rho = \dfrac{p_{ISA}(h)}{R\,(T_{ISA}(h) + \Delta)}$.

Hot day ⇒ less dense air ⇒ harder to take off and land. A cold day is the opposite (denser air), as Figure 2.7 shows.

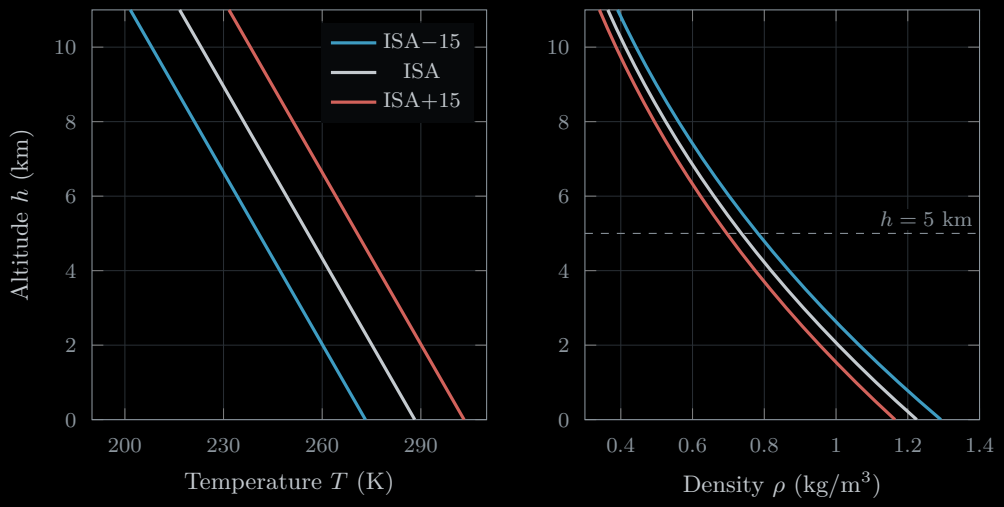


Figure 2.7. Effect of ISA±15 on temperature (left) and density (right) in the troposphere: ISA−15 is the cold day (blue) and ISA+15 the hot day (red). At the same altitude the hot day is less dense.

DENSITY ON A HOT DAY

Sea-level density in ISA+15: $\rho = \dfrac{101\,325}{287.05 \cdot 303.15} = 1.164$ kg/m^3, 5 % less than the standard value (1.225). In ISA−10: $\dfrac{101\,325}{287.05 \cdot 278.15} = 1.269$, 3.6 % more.

At 5 km with ISA+20: $\rho = \dfrac{54\,020}{287.05 \cdot (255.65 + 20)} = 0.683$ kg/m^3, versus 0.736 in ISA.

4 Clouds, winds and icing

4.1 Clouds

They form with **high humidity and low temperature**. Effects: poor visibility (instrument flight rules are needed, Section 5); possible storms, with the risk of lightning strikes (aircraft are protected against lightning, but it causes damage) and gusts; and possible icing. The risk is **lower above the tropopause**: almost all weather phenomena occur in the lower part of the troposphere (2024/25 exam) and in the stratosphere they are not significant (2022/23). That is why airliners fly high, above about 9 km: humidity and turbulence are avoided (2022/23 exam).

4.2 Winds

- They are **critical in takeoff and landing** (headwind, tailwind or crosswind). It is usual to take off and land with a **headwind**: the incoming air already gives relative speed to the wing without the aircraft moving relative to the ground, so less runway is needed.
- **Jet stream**: strong, narrow winds at high altitude. It is exploited with a **tailwind**, because it increases the ground speed; it is not "always avoided" nor does it "increase lift" (2018/19 exam).
- **Gusts**: turbulence, abrupt changes in lift and drag and possible loss of control. They are also lower above the tropopause.

Figure 2.8 illustrates the headwind on takeoff and the jet stream in cruise.

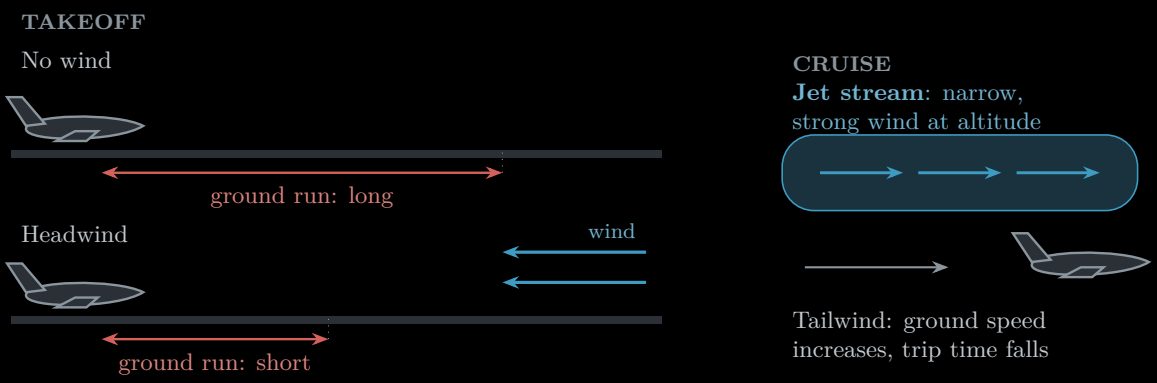


Figure 2.8. Left: with a headwind the wing reaches flying speed after a shorter ground run. Right: a jet stream used as a tailwind raises the ground speed. Schematic.

4.3 Icing

It occurs at **very low temperatures and high humidity**. When ice forms on the surface, it **changes the shape of the aircraft: lift decreases, drag and weight increase**. It can damage engines and instruments (example of the ATR-72 flight in 1994: the iced horizontal stabilizer stalled).

Prevention and protection: ice detectors and deicing systems on the ground and in flight (pneumatic boots, fluids, engine bleed air, electro-thermal, electromechanical, passive systems...). They are used on the wing, the engines and several sensors (2019/20 exam), not only on the ground. Figure 2.9 shows the effect of ice on the profile and how a pneumatic boot (an inflatable rubber strip on the leading edge) breaks it off.

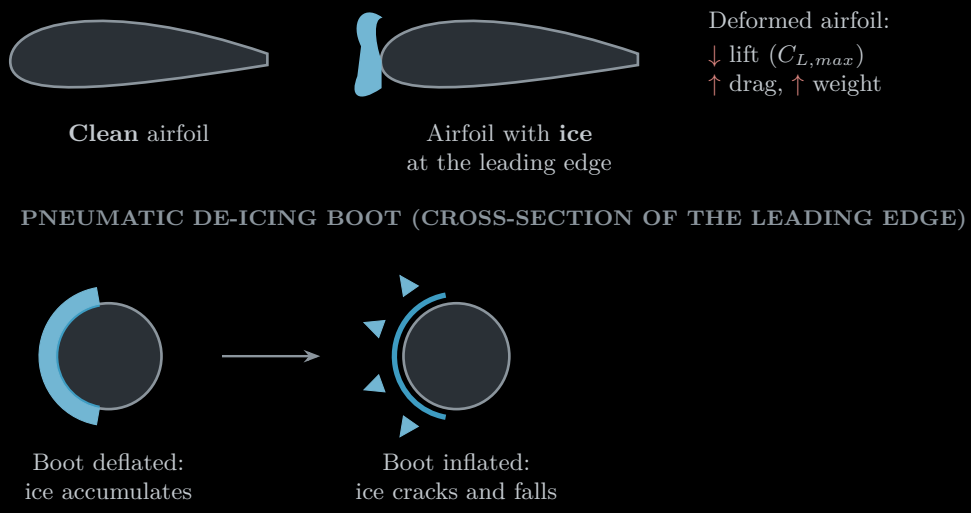


Figure 2.9. Ice on the leading edge deforms the airfoil; the pneumatic boot inflates and cracks the ice, which the airflow then carries away. Schematic.

ABOUT ICE IN THE TESTS

- Ice formation is **not** more typical at 10 km than at 6 km "because of humidity": almost all the water vapor is in the lower part of the troposphere. This was the false option in 2024/25.
- Ice on the leading edge is a possible cause of **stall**: it deforms the airfoil and lowers $C_{L,max}$ (2023/24: the option "all the others are valid").

5 Low-visibility flight

When the pilot cannot see, flight is conducted under **Instrument Flight Rules** (IFR) instead of **Visual Flight Rules** (VFR). For each aid, the slides (17 to 19) ask: "what is its operating principle and how do they differ?". Each one is summarized here (the detailed explanation is supplementary); Figure 2.10 shows where each aid operates.

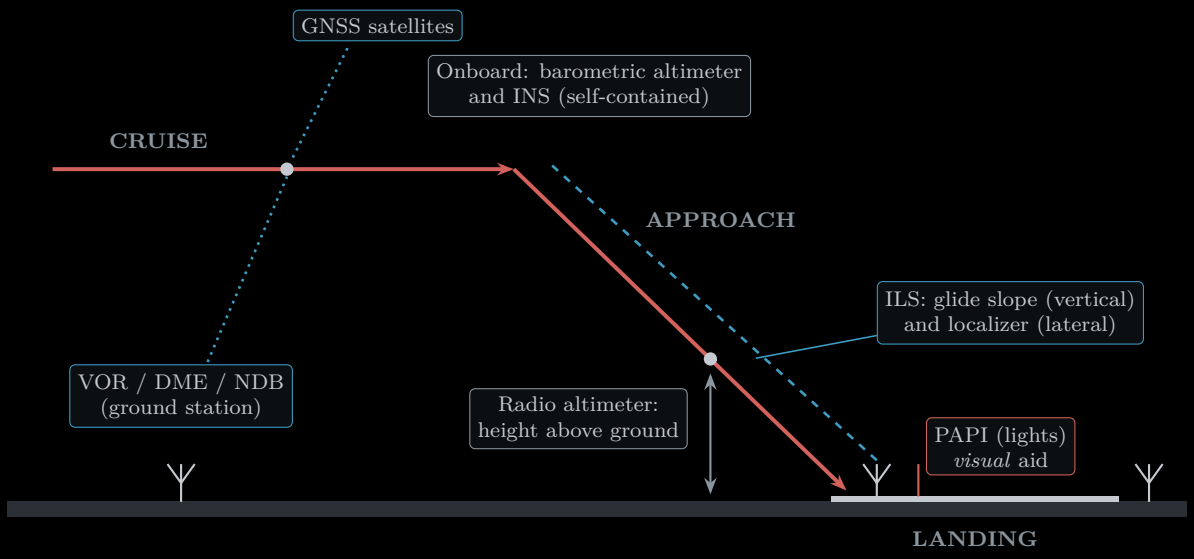
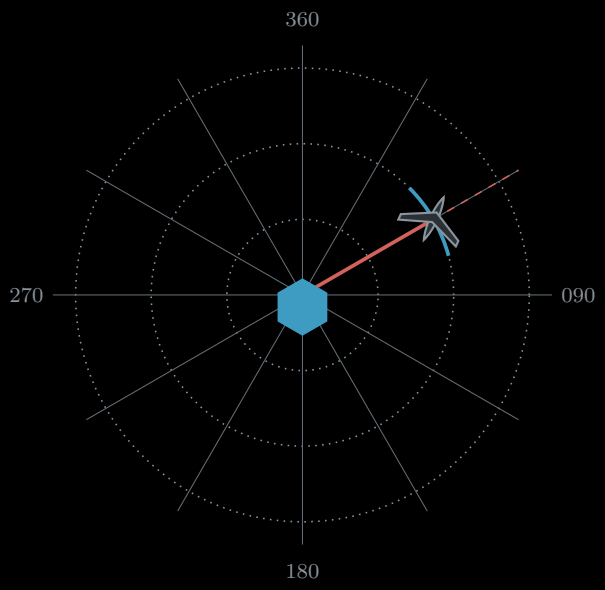


Figure 2.10. Where each aid operates. In blue, aids that work in low visibility; in red, the one that needs visibility (PAPI). Source: T2, slides 17 to 19 (own diagram).

In cruise

Aid	Principle	Remarks
Barometric altimeter	measures static pressure and converts it into altitude using ISA	cruise; it is miscalibrated if the day is not standard
NDB (*Non-Directional Beacon*)	ground beacon that transmits in all directions; the onboard receiver indicates the direction toward it	cruise; low accuracy
VOR (*VHF Omni-directional Range*)	ground station in the VHF band that lets the aircraft know on which radial (bearing from the station) it is	cruise
DME (*Distance Measuring Equipment*)	the aircraft interrogates and the station replies; the round-trip time gives the distance	cruise; with VOR it gives the position
INS (*Inertial Navigation System*)	accelerometers and gyroscopes integrate the motion from a known position	self-contained; accumulates error over time
GNSS (*Global Navigation Satellite System*)	receives signals from several satellites and computes the position from time differences (GPS is a GNSS)	cruise; depends on satellites

Figure 2.11 shows how a VOR and a DME combine to fix the position.



VOR: which **radial** (bearing seen from the station) the aircraft is on: here 060°.

DME: **distance** to the station, from the round-trip time of a radio pulse (circle of equal distance).

VOR + DME = intersection of a radial and a circle = **position**.

Figure 2.11. VOR and DME. Top view, north up; the aircraft is on radial 060 at the distance given by the DME circle. Schematic.

In approach and landing

Aid	Principle	Remarks
Radio altimeter	downward-pointing radar that measures the height **above the ground**; not to be confused with the barometric one	landing; short range
ILS (*Instrument Landing System*)	two radio beams: one marks the runway centerline (lateral) and the other the glide slope (vertical)	landing; works with almost zero visibility
Augmented GNSS (LAAS/GBAS)	GNSS corrected by a ground station for precision approaches	landing
PAPI (*Precision Approach Path Indicator*)	red and white lights beside the runway that indicate whether the aircraft is on the glide path	**visual aid**: landing in good visibility

Figure 2.12 shows the two ILS beams and Figure 2.13 the PAPI light signals.

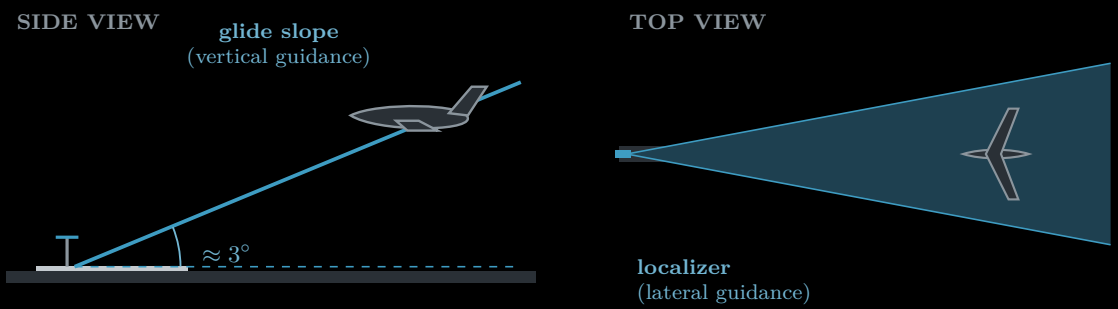


Figure 2.12. ILS: the localizer beam guides laterally along the runway centerline and the glide slope beam guides vertically (about 3°; angle exaggerated). Schematic.



Figure 2.13. PAPI as seen by the pilot: the number of white and red lights shows whether the aircraft is above, on or below the glide path.

RECURRING QUESTION: NAVIGATION AIDS

- The **PAPI** is only a **visual** aid (VFR); the **ILS** is an instrument aid (IFR). Therefore, with visibility < 30 m the PAPI cannot be used, but the ILS, the radio altimeter or GNSS can (2023/24 exam); and in low visibility "requires good visibility" $\Rightarrow$ PAPI (2024/25).
- The **VOR** is not suitable for following the glide slope (that is done by the ILS). The barometric altimeter does not provide guidance in landing (2019/20). The ILS does not guide the takeoff.
- False in 2020/21: "the radio altimeter cannot be used in cruise because of the high speed" (its limitation is the short range above the ground, not the speed), "the barometric one cannot be used at high altitude" (it can) and "the PAPI is used in takeoff" (it is for landing).

6 Kinetic heating

When the air slows down on reaching the aircraft's surface (nose, leading edge of the wings…), its **kinetic energy is converted into internal energy** (heat). The temperature reached by air brought completely to rest is called the **total or stagnation temperature** T_0. If the deceleration is **adiabatic and isentropic** (no heat exchange and no losses), energy is conserved:

$$T_0 = T + \frac{V_\infty^2}{2\,c_p} \qquad \Longleftrightarrow \qquad T_0 = T\left(1 + \frac{\gamma - 1}{2}\,M^2\right), \qquad \gamma = \frac{c_p}{c_v} \approx 1.3 \text{ to } 1.4$$

where T is the static air temperature (the ISA value at that altitude), V_∞ the flight speed, c_p the specific heat at constant pressure (air: ≈ 1005 J/(kg K)) and $M = V/a$ the **Mach number** (the ratio of the aircraft's speed to the speed of sound $a = \sqrt{\gamma R T}$). With $\gamma = 1.4$: $T_0 = T\,(1 + 0.2\,M^2)$. It is a **major problem at high speeds**: it grows with M^2 (Figure 2.14).

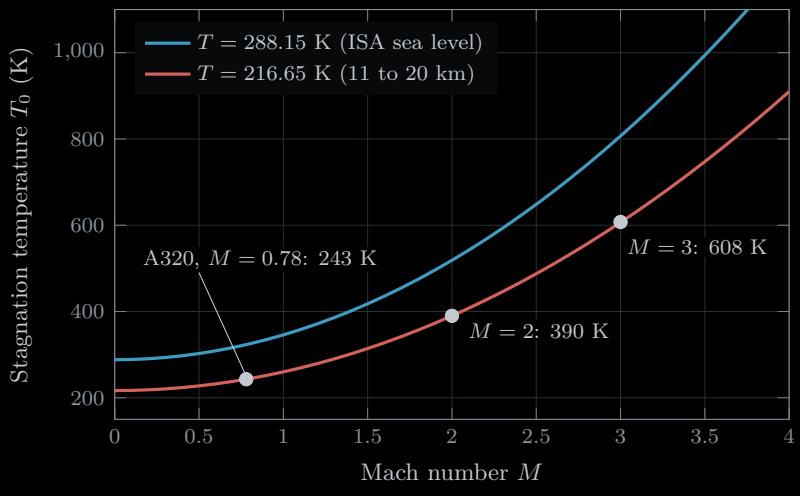


Figure 2.14. Stagnation temperature as a function of Mach number, for $\gamma = 1.4$. The two curves correspond to the ISA static temperature at sea level and at 11–20 km. The SR-71 example below (Mach 3 at 24 km, $T = 220.65$ K) gives 618 K, slightly above the lower curve.

SR-71 BLACKBIRD AT MACH 3 AND 24 KM (2020/21 EXAM)
Step 1: $T(24\text{ km}) = 216.65 + 1 \cdot (24 - 20) = 220.65$ K (the +1 K/km segment between 20 and 32 km).

Step 2: $T_0 = 220.65\,(1 + 0.2 \cdot 3^2) = 220.65 \cdot 2.8 = 618$ K. **Answer: 618 K** (option c), about 345 °C. Common mistake: forgetting the +1 K/km above 20 km and using 216.65 K, which would give 607 K.

AIRBUS A320NEO AT 11 000 M (2018/19 EXAM)
Implicit data: cruise speed $M \approx 0.78$.
$T = 216.65$ K; $T_0 = 216.65\,(1 + 0.2 \cdot 0.78^2) = 216.65 \cdot 1.122 = 243$ K (−30 °C). The collected key was "around 215 K" (the closest of the options 115, 215, 315 and 415 K). Note that 215 K is the *static* temperature; with kinetic heating about 27 K more results. At subsonic speeds the effect is small; it only matters at high speeds.

Note: in the 2023/24 exam it was stated that at Mach 2 kinetic heating "can give surface temperatures of about 600 K". With the formula: at 11–20 km, $216.65 \cdot 1.8 = 390$ K; at sea level, 519 K. Therefore 600 K does not fit Mach 2 but Mach 3 (608 K). The collected key marked it as correct; it is left here as a doubt.

7 Radiation at high altitude

Here "radiation" refers to **cosmic rays**, high-energy particles that come from space: protons (≈ 90 %), alpha particles (≈ 10 %) and heavy particles (1 to 2 %) (the figures are those on the slide and do not add up to exactly 100 %; they are orders of magnitude). Figure 2.15 sketches how the atmosphere attenuates them.

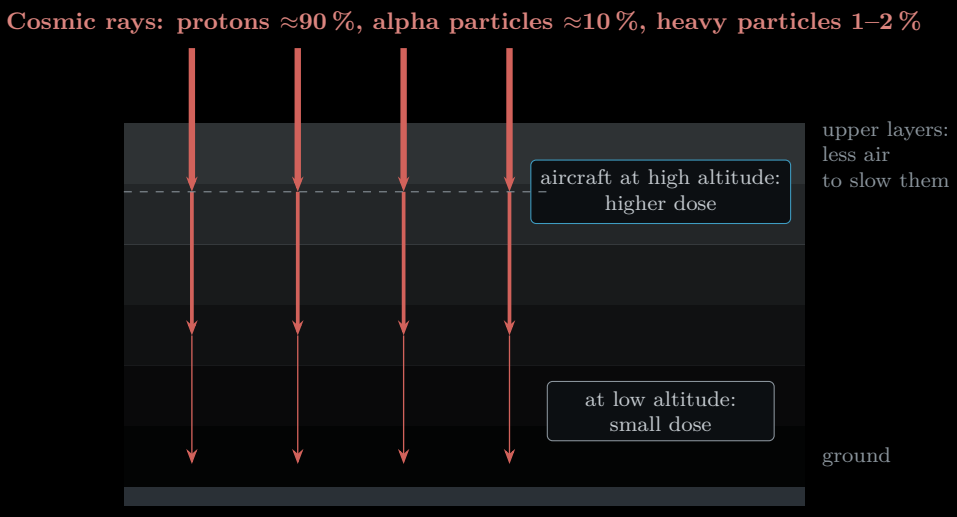


Figure 2.15. The atmosphere slows down cosmic rays; at higher altitude there is less air to stop them. Qualitative diagram.

- **Why it matters**: the atmosphere acts as a shield, but the upper layers are less dense, and there is less protection near the **poles**.
- **Effects**: dangerous for electronics and for health; higher cancer risk.
- **Trend and a poor solution**: aircraft fly ever higher, which increases the dose. Shielding would be the solution, but the weight penalty is prohibitive.
- **Mitigation**: dosimeters and regular medical checkups; rerouting (for example, polar routes during a solar storm).
- **Crews** are the group of workers most exposed to radiation (excluding nuclear accidents and the like).

FROM THE TEST
The absorbed radiation **increases with altitude** and is **higher near the poles** than at the equator (2019/20 exam). Excessive doses can be avoided by avoiding critical routes such as polar ones (2023/24). It is not reduced by cabin pressurization.

8 What usually appears on the exam

WHAT USUALLY APPEARS ON THE EXAM
This is **the unit with the most questions among Units 1 to 4** (only Unit 5 contributes more): several per midterm. Types observed (2018/19 to 2024/25):

- **Layers and ISA (conceptual)**: comparing temperatures or pressures at different altitudes (ISA±Δ); gradients (6.5 K/km; isothermal from 11 to 20 km; nonlinear pressure).
- **Recurring numerical problems**: total temperature (SR-71: 618 K; A320: $T_0 = T(1 + 0.2M^2)$); stall speed at another altitude ($V \propto 1/\sqrt{\rho}$; Manang: 189 km/h); angle of attack when climbing at constant speed.
- **Hot/high takeoff**: lower density, longer runways, higher speeds.
- **Weather**: headwind in takeoff and landing; jet stream with tailwind; ice and deicing systems; effects of lightning.
- **Navigation aids**: PAPI (visual) versus ILS; radio altimeter versus barometric.
- **Radiation**: grows with altitude and toward the poles.

The essentials to memorize:

- $T_0 = 288.15$ K, $p_0 = 101\,325$ Pa, $\rho_0 = 1.225$ kg/m^3; -6.5 K/km up to 11 km; 216.65 K from 11 to 20 km; $+1$ K/km from 20 to 32 km; $+2.8$ K/km up to 47 km.
- $p = p_0(T/T_0)^{5.256}$ and $\rho = \rho_0(T/T_0)^{4.256}$ in the troposphere.
- ISA+Δ: same pressure, shifted T, $\rho = p/RT$; heat ⇒ less dense.
- $T_0 = T(1 + 0.2M^2)$.

UNIT 3

Aircraft design concepts

1 What aircraft design is

For Daniel P. Raymer (a reference author in the slides), *design* is **"creating the geometric description of a thing to be built"**. Aircraft design is carried out by highly multidisciplinary teams (aerodynamics, propulsion, structures, avionics...), but it is a discipline of its own, with its own history, methods and rules of thumb.

The classical problem is **iterative**: the weight depends on the engine and the aerodynamics, and these depend on the weight. It is a circle, so the question is *where do we start?* The answer, which appears on exams: **with the design specifications** (what the aircraft must do), not with the engine, the weights or the aerodynamics.

2 Phases of aircraft design

Starting from the mission specifications there are three phases, followed by manufacturing (Figure 3.1):

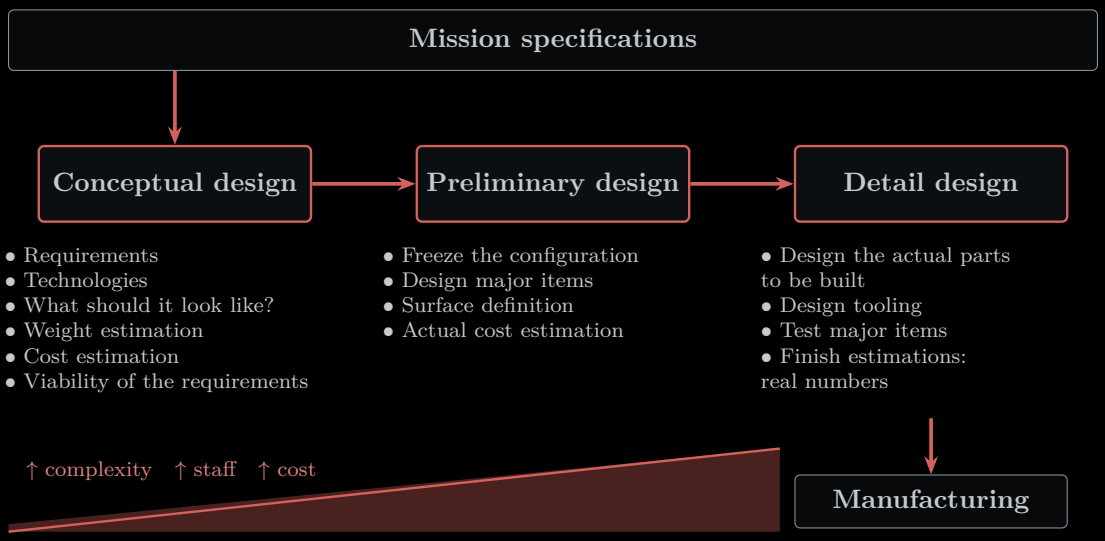


Figure 3.1. Phases of aircraft design: each phase adds complexity, staff and cost. Source: T3, slide 6.

Phase	What is done
Conceptual design	requirements, available technologies, "what it should look like", first estimate of weights and costs, and viability of the requirements. It is cheap and fast, and it is where almost everything is decided.
Preliminary design	the **configuration is frozen**; the major items are designed, the surfaces are defined and the actual cost is estimated.
Detail design	the parts to be built and the tooling are designed, the major items are tested, and the estimates are closed with real numbers.

Analogy. An analogy is building a house: first a sketch with an approximate budget (conceptual), then the plans with the layout already fixed (preliminary) and, finally, the detail of each installation (detail). Changing the idea is cheap at the sketch stage and very costly once the plans are closed.

COMMON TEST MISTAKES ABOUT THE PHASES

- The **initial weight estimate** belongs to **conceptual** design, not preliminary.
- The design is **frozen** in the **preliminary** phase (not in the conceptual one) (2022/23 exam).
- The preliminary phase **requires more staff and time** than the conceptual one (2018/19 and 2019/20): complexity, staff and cost grow with each phase.
- The main requirements are set **earlier** (specifications/conceptual). Each part is designed in the **detail** phase.
- Saying that the conceptual phase is "the most critical because it is the most expensive" is false: it is the cheapest, although its decisions condition everything.
- Available technology is critical already in the conceptual phase (it is identified in that phase), not only in the detail phase.

3 Conceptual design overview

Conceptual design is organized into four blocks that are repeated until the aircraft is viable (Figure 3.2):

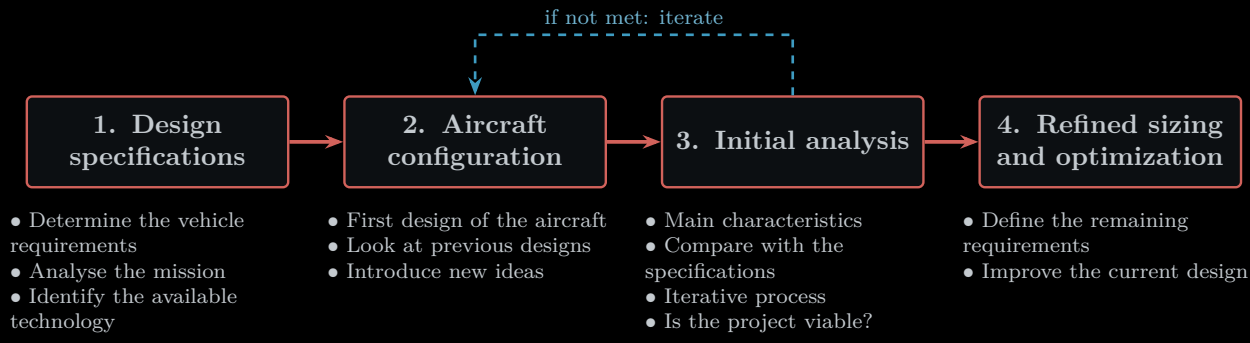


Figure 3.2. The four blocks of conceptual design. Source: T3, slide 9.

4 Design specifications

These are the **mission requirements**. Those listed in the slides (10):

- **Range** and **payload**: how much and how far.
- **Cruise speed**.
- **Runway**: length available for takeoff and landing.
- **Manoeuvrability** and stability.
- **Cost**.
- Others: number of desired operations, fashion, development time...

The mission rules: an agricultural aircraft and an airliner have such different specifications that they end up with very different shapes.

5 Aircraft configuration

This is the **first design**: where the wing, the engines, the control surfaces and the landing gear go. Previous similar aircraft are examined and new ideas are introduced. The decisions are tied to cruise speed, runway, stability/manoeuvrability, range, aerodynamic efficiency and payload.

5.1 The wing

Wing location: low, medium or high (Figure 3.3, which also shows the dihedral angle).

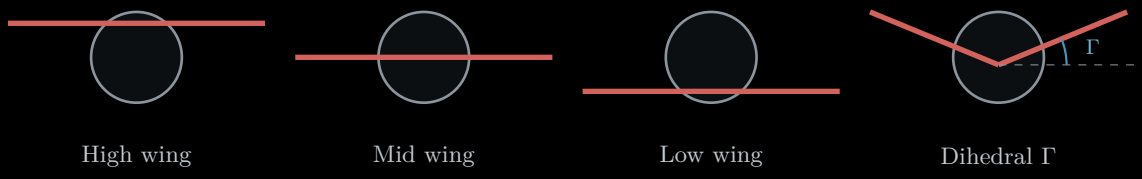


Figure 3.3. Wing position relative to the fuselage (front view) and dihedral angle. Own diagram.

Rule of thumb (supplement): cargo aircraft usually have a high wing, to keep the fuselage close to the ground, load easily and leave room for the engines; airliners usually have it low. In the 2022/23 exam the statement "a low wing is chosen for cargo aircraft" was rejected.

Wing planform: the shape of the wing seen from above (Figure 3.4).

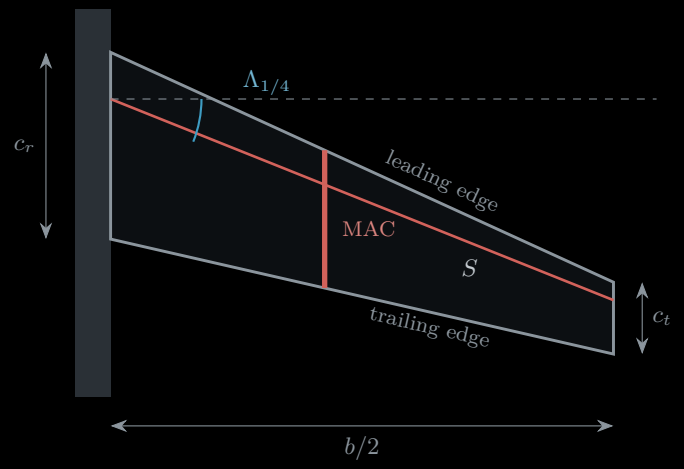


Figure 3.4. Half-wing planform with its parameters. Source: T3, slide 12 (own diagram, with MAC computed with the formula below).

Symbol	Name	Meaning
S	wing area	area of the wing seen from above
b	wingspan	from tip to tip
c_r, c_t	root chord and tip chord	
$\lambda = c_t/c_r$	taper ratio	how much the wing tapers
$AR = b^2/S$	aspect ratio	how long and narrow it is
$\Lambda_{1/4}$	sweep angle	angle of the 25 % chord line
Γ	dihedral angle	upward inclination of the wings seen from the front
$\bar{c} = S/b$	mean geometric chord	chord of a rectangular wing with the same S and b
MAC	mean aerodynamic chord	"equivalent" chord in aerodynamic effects

$$AR = \frac{b^2}{S} \quad \lambda = \frac{c_t}{c_r} \quad \bar{c} = \frac{S}{b} \quad S_{\text{trapezoidal}} = \frac{b}{2}(c_r + c_t)$$

The **MAC** (*Mean Aerodynamic Chord*), supplement: for a trapezoidal wing it is

$$\text{MAC} = \tfrac{2}{3}\, c_r\, \frac{1+\lambda+\lambda^2}{1+\lambda}.$$

PARAMETERS OF A WING PLANFORM

Trapezoidal wing with $b = 30$ m, $c_r = 6$ m and $c_t = 1.5$ m (example data).

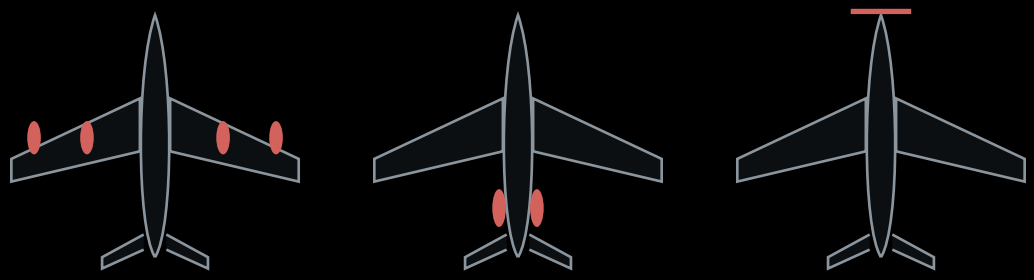
$S = \frac{b}{2}(c_r + c_t) = 15 \cdot 7.5 = 112.5$ m^2; $\lambda = 1.5/6 = 0.25$; $AR = \frac{30^2}{112.5} = 8$; $\bar{c} = \frac{112.5}{30} = 3.75$ m; $\text{MAC} = \frac{2}{3} \cdot 6 \cdot \frac{1 + 0.25 + 0.0625}{1.25} = 4.2$ m.

Check with approximate real data (A320, from memory): $b \approx 34.1$ m and $S \approx 122.6$ m^2 give $AR \approx 9.5$.

What these parameters do is studied in Unit 5, on aerodynamics (sweep, aspect ratio, taper ratio). What has been asked in tests: the typical taper ratio of an airliner is **"around 0.6"** among the options $\{0,\ 0.6,\ 1,\ >1\}$ (2023/24 exam; real values are usually somewhat lower, but 0 and 1 must be ruled out, as they correspond to the delta wing and the rectangular wing). A low taper ratio improves structural stresses and makes it easier to stow the landing gear and fuel in the wing (2022/23 exam).

5.2 The engine

The **type** (propeller, turbofan, turbojet), the **location** (under the wing, on the fuselage) and the **number** must be decided; Figure 3.5 shows three typical arrangements.



Under the wing (like the A380, 4 turbofans) | On the rear fuselage (like the Citation VII, 2 turbofans) | Propeller on the nose (like the Piper PA-18, 1 engine)

Figure 3.5. Three engine arrangements, with the examples from the slides. Source: T3, slide 13 (own silhouettes).

5.3 Control surfaces and landing gear

- **Control surfaces**: the tail ones (elevator and rudder) and the ailerons. The surface that controls roll is the **aileron** (2024/25 exam). Figure 3.6 shows where they are.
- **Landing gear**: position and type. Examples from slide 15: BAe Jetstream 31, PZL-106 Kruk (agricultural) and An-225 (very heavy cargo). **Fixed gear** is usually the best solution in **low-speed** aircraft: it weighs less and is simpler, at the cost of more drag (2022/23 exam). Figure 3.7 compares the three solutions.

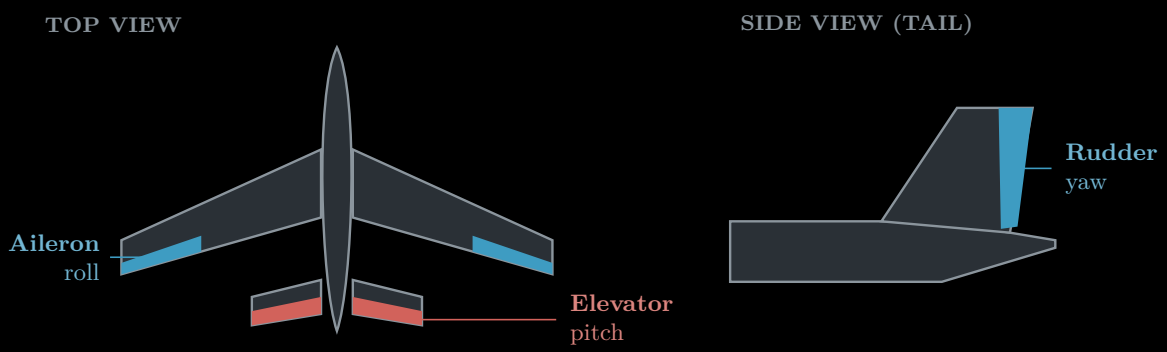
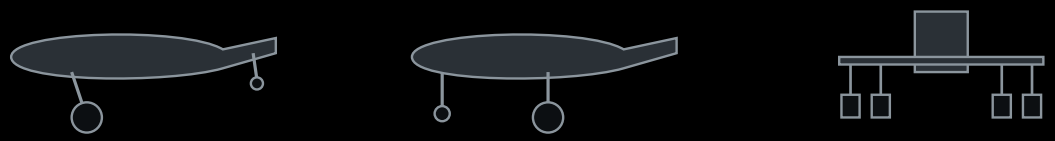


Figure 3.6. Primary control surfaces: ailerons (roll, on the wing trailing edge), elevator (pitch, on the horizontal tail) and rudder (yaw, on the fin). Schematic.



Fixed gear
light, simple, more drag
(PZL-106 Kruk)

Retractable gear
folds into the body; less drag, heavier
(BAe Jetstream 31)

Many wheels
spread a very heavy load
(An-225, 32 wheels)

Figure 3.7. Landing gear types, with the examples from the slides. Schematic.

5.4 Roskam classification

There is no single configuration: it depends on the mission. One possible classification (Jan Roskam, slide 16; "not by any means the only one"):

1. Homebuilt propeller aircraft
2. Single-engine propeller
3. Twin-engine propeller
4. Agricultural aircraft
5. Business jet
6. Regional turboprop
7. Jet transport
8. Military: trainer, transport, patrol, tanker, search and rescue
9. Military: fighters and bombers
10. Seaplanes
11. Supersonic cruise aircraft
12. Rotary-wing aircraft

6 Initial analysis

With the sketch, the main characteristics are determined and compared with the specifications. For cruise only, the loop is the one in Figure 3.8:

6.1 Weights

$$W_{TO} = W_{OE} + W_F + W_{PL}, \qquad W_{OE} = W_E + W_{crew} \quad \Rightarrow \quad W_{TO} = W_E + W_{crew} + W_F + W_{PL}$$

Figure 3.9 shows how these terms add up.

- W_{TO}: takeoff weight (its maximum is the MTOW, *Maximum Take-Off Weight*).
- W_{OE}: operating empty weight $= W_E + W_{crew}$.
- W_E: empty weight: structure, engines, systems, *seats*...
- W_{crew}: flight crew (pilots, flight attendants...).

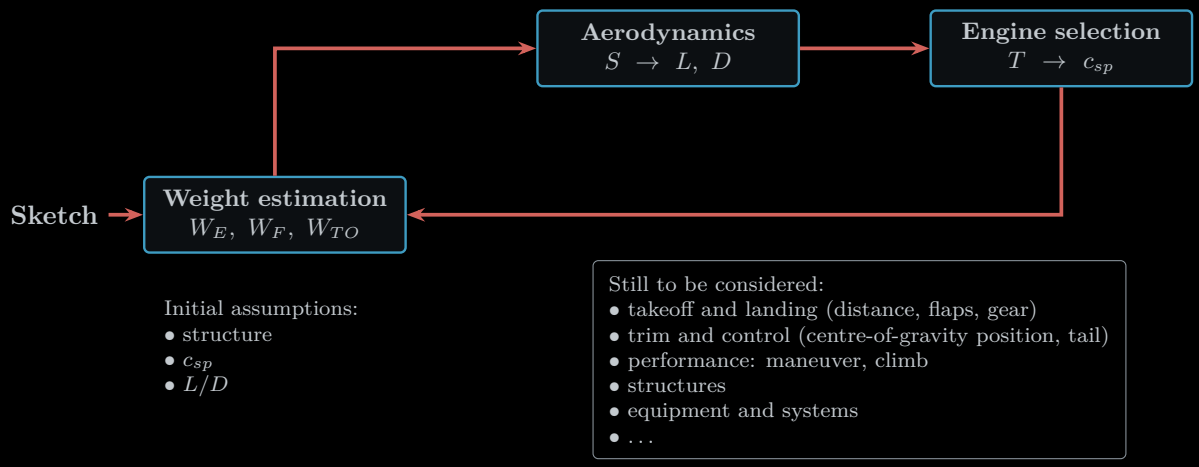


Figure 3.8. Iterative design (cruise only). Source: T3, slide 17. "The Dash-one is never built": the first design always changes.

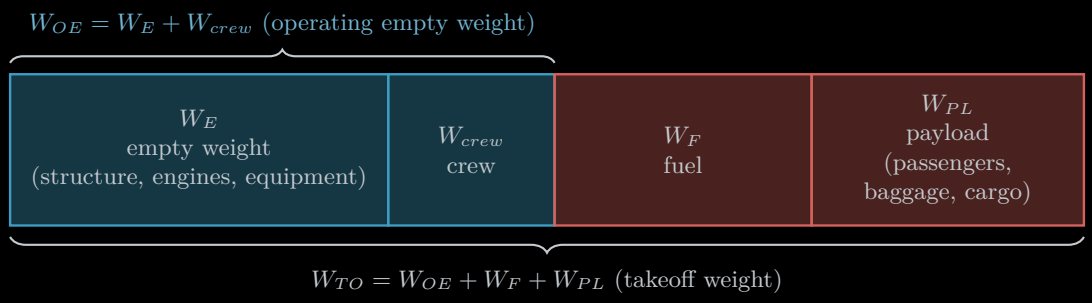


Figure 3.9. Breakdown of the takeoff weight (not to scale). Source: T3, slide 18.

- W_F: fuel. W_{PL}: payload (passengers, baggage, cargo).

Each one depends on the mission. (The slides write "weight" and the exam "mass": for problem purposes they are the same except for the factor g.)

RECURRING QUESTION: WHAT GOES INTO EACH WEIGHT?

- The **crew** is in the operating empty weight (2023/24) and **not** in the payload (2018/19). Fuel, passengers and suitcases are **not** in W_{OE}.
- **Seats** do not go in the payload: they are empty weight (2019/20).
- Empty weight fraction: the fraction W_E/W_{TO} changes with the mission and **tends opposite to the total weight** (the greater W_{TO}, the smaller the fraction; 2020/21 exam). New materials reduce it and leave more room for fuel and payload.

6.2 Fuel: mission profile and the Breguet equation

The fuel depends on the **mission profile** (Figure 3.10) and is minimized with a good specific fuel consumption and good aerodynamic efficiency.

The **Breguet equation** (under the assumptions that V, L/D and c_{sp} are constant) gives the range of a jet aircraft:

$$R \approx \frac{V}{c_{sp}} \frac{L}{D} \ln \frac{W_i}{W_f}$$

where V is the cruise speed, c_{sp} the **specific fuel consumption** (fuel burned per unit of thrust

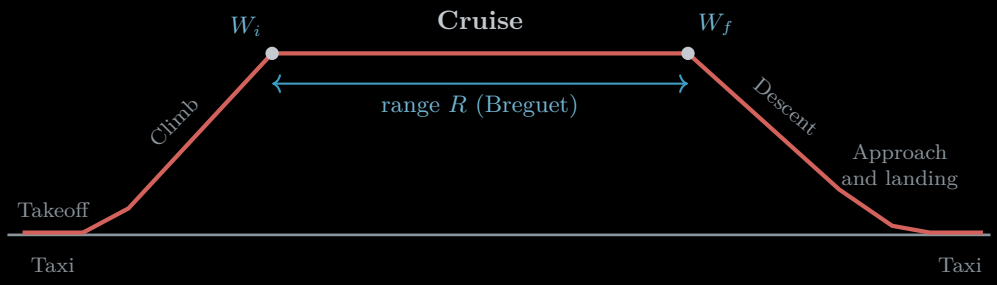


Figure 3.10. Typical mission profile. Source: T3, slide 20.

and time; here in units of 1/s, referred to weight), L/D the **aerodynamic efficiency** (lift divided by drag) and W_i, W_f the weights at the start and end of cruise (the difference is the fuel burned).

How to read it: more range with more speed, more efficiency L/D, less consumption c_{sp} and more fuel ($\ln W_i/W_f$ grows slowly, because it is a logarithm). Solving, the fuel fraction is $1 - \dfrac{W_f}{W_i} = 1 - \exp\Big(-\dfrac{R\,c_{sp}}{V\,L/D}\Big)$.

NUANCES OF THE BREGUET EQUATION (2020/21 TO 2024/25 TESTS)

- The range is proportional to the **logarithm** of the weight ratio, not to W_f/W_{TO} nor directly to the fuel.
- In a jet it is proportional to the **flight speed**; in a propeller engine it is **not** (for a propeller, the propeller efficiency and the power-specific consumption appear, and the speed cancels; supplement).
- In the absence of wind, the range of a jet is directly proportional to the average speed (2024/25). It is *inversely* proportional to the specific fuel consumption.
- The operating empty weight and the payload **do** influence the range (via W_i): the opposite statement was false (2020/21).

RANGE AND FUEL OF AN AIRLINER (ILLUSTRATIVE DATA)

$V = 230$ m/s, $L/D = 17$, $W_i/W_f = 1.25$ and $c_{sp} = 1.67 \cdot 10^{-4}$ s^{-1} (equivalent to about 0.6 lb/(lbf h), typical of a turbofan).

$R = \dfrac{230}{1.67 \cdot 10^{-4}} \cdot 17 \cdot \ln 1.25 = 1.377 \cdot 10^6 \cdot 17 \cdot 0.2231 = 5.2 \cdot 10^6$ m ≈ 5200 km. The fuel fraction is $1 - 1/1.25 = 0.20$, that is, 20 % of the weight at the start of cruise.

2023/24 EXAM: LOWER CONSUMPTION, LOWER AERODYNAMIC EFFICIENCY

Jet airliner: the specific fuel consumption drops by 3 % and the aerodynamic efficiency drops by 5 %, with the same speed and the same design mission. What happens?

Step 1. With V and R constant, $\dfrac{L/D}{c_{sp}} \ln \dfrac{W_i}{W_f}$ must remain the same. The ratio $\dfrac{L/D}{c_{sp}}$ becomes $\dfrac{0.95}{0.97} = 0.979$ of the previous one: it has worsened by 2 %.

Step 2. To compensate, $\ln \dfrac{W_i}{W_f}$ must be $1/0.979 = 1.021$ times greater: **more fuel** is needed.

Step 3 (example numbers). If it was previously 20 % ($\ln = 0.2231$), it is now $\ln = 0.2278$, that is $1 - e^{-0.2278} = 0.204$, that is 20.4 %. With the payload fixed and, for example, $W_E/W_{TO} = 0.5$, $W_{TO} = \dfrac{W_{PL} + W_{crew}}{1 - 0.5 - W_F/W_{TO}}$ goes from $\dfrac{x}{0.300}$ to $\dfrac{x}{0.296}$: it rises 1.4 %.

Answer: the **takeoff mass will have increased** (option c). The payload does not drop

because the aircraft is designed for the load the mission requires.

INITIAL WEIGHT SIZING (MADE-UP NUMERICAL EXAMPLE)
An aircraft must carry $W_{PL} = 18\,000$ kg and a crew of $W_{crew} = 500$ kg. From similar aircraft, $W_E/W_{TO} = 0.50$ is estimated and, with Breguet, $W_F/W_{TO} = 0.20$. What is W_{TO}?
$W_{TO} = W_E + W_{crew} + W_F + W_{PL} \;\Rightarrow\; W_{TO}\,(1 - 0.50 - 0.20) = 18\,500$, hence $W_{TO} = \dfrac{18\,500}{0.30} = 61\,667$ kg, with $W_E = 30\,833$ kg and $W_F = 12\,333$ kg. The calculation is circular: if W_E/W_{TO} turns out greater than assumed, W_{TO} increases and it must be repeated with the new estimate.

2019/20 EXAM: VALENCIA–PARIS AND BACK
An aircraft takes off at its MTOW on VLC–CDG, with 15 000 kg of cargo, and burns **all** its fuel. After refueling it flies the return leg with 10 000 kg.
The MTOW is a **maximum**. For the return it will take on the same fuel (same route), with $W_{TO} = W_{OE} + W_F + 10\,000 < W_{OE} + W_F + 15\,000 =$ MTOW: it **takes off at a weight lower than its MTOW** (option a). Neither the MTOW nor W_{OE} changes.

7 Refined sizing and optimization

Once the revised sketch makes sense, the remaining elements are characterized: aerodynamics, weights, propulsion, stability and control, structures and cost, ... Cost and technological availability are critical **before** moving on to preliminary design. The process is repeated for **several configurations**, and only when candidates have been designed is the one with the best properties chosen. Many designs fulfill the same mission (slide 22 compares an Avro with a B-52).

8 What usually appears on the exam

WHAT USUALLY APPEARS ON THE EXAM
There are **few test questions per midterm**, almost always conceptual; the numerical ones are short:

- What to start with (specifications) and what each phase does (conceptual: weights and viability; preliminary: freeze; detail: parts).
- Weight composition: what W_{OE} includes (crew yes; fuel, passengers and suitcases no) and the payload (without crew or seats).
- Breguet: proportionality (V, L/D, $1/c_{sp}$, logarithm of the weight ratio); the problem of changing c_{sp} and L/D; round-trip route with MTOW.
- Configuration: fixed gear in slow aircraft; high/low wing; mission-dependent wing planform; typical taper ratio; aileron for roll.

The essentials: $W_{TO} = W_E + W_{crew} + W_F + W_{PL}$; $W_{OE} = W_E + W_{crew}$; $R \approx \frac{V}{c_{sp}} \frac{L}{D} \ln \frac{W_i}{W_f}$; $AR = b^2/S$; $\lambda = c_t/c_r$; $\bar{c} = S/b$.

UNIT 4

Aircraft structures

1 What a structure does and the vocabulary you need

The structure of an aircraft is its **skeleton and skin**: it holds everything else (engines, cargo, passengers, landing gear) and carries forces from one place to another without breaking or deforming excessively. Its design is always a compromise: it must be **very strong and very stiff yet as light as possible**, because every kilogram of structure is one kilogram less of payload or fuel (which is why the operating empty weight, OEW, was so important in Unit 3).

MINIMUM GLOSSARY FOR THIS UNIT

- **Stress** (σ or τ): force per unit area inside the material, $\sigma = F/A$ (Pa or N/mm^2). It measures *how hard the material is working*.
- **Strain** (ε): relative elongation, $\varepsilon = \Delta L/L$ (dimensionless). It measures *how much the material has stretched*.
- **Strength**: maximum stress the material withstands before deforming permanently or breaking. **Stiffness**: how hard it is to deform. **They are not the same**: a steel cable is very strong but not at all stiff in bending; a thick plastic ruler is stiff but not very strong.
- **Beam**: long, slender member. **Cantilever**: beam fixed at one end and free at the other, like a diving board. Structurally, a wing is a cantilever fixed to the fuselage.
- **Longitudinal** = along the long direction of the member; **transverse** = perpendicular to it. **Spar** = main longitudinal beam of the wing.

2 Stresses, stiffness and buckling

2.1 Types of stress

Any structural member undergoes one of these stresses or a combination of them (Figure 4.1).

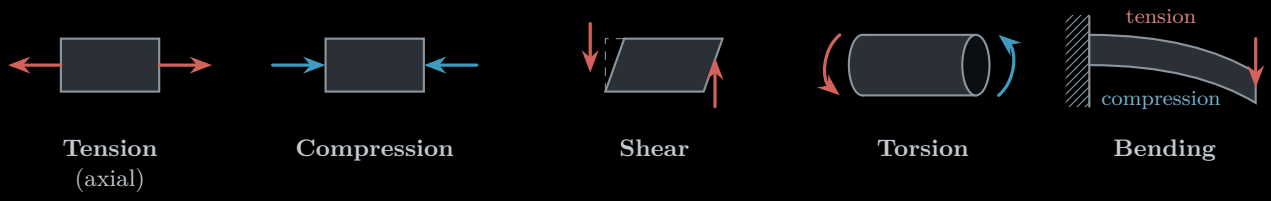


Figure 4.1. Basic stresses: tension, compression, shear, torsion and bending (tension on one side and compression on the other). Source: Unit 4, slide 4.

- **Tension** (also called axial stress): stretches. **Compression**: squashes, and in slender members it causes buckling.

- **Shear**: two parallel, opposite forces try to slide one section relative to the adjacent one, like the blades of a pair of scissors.
- **Torsion**: twisting, like wringing out a towel.
- **Bending**: flexing. In a bent beam, one face stretches (tension) and the opposite face shortens (compression); in between there is a neutral fibre with no stress. That is why the *useful* material in bending is the material far from the centre (see the I shape below).

2.2 Hooke's law and the concept of stiffness

A linear spring obeys $F = kx$ (applied force F, spring stiffness k, extension x). A beam under axial load behaves in the same way in the elastic region:

$$\sigma = E\,\varepsilon \qquad \Longrightarrow \qquad \frac{F}{A} = E\,\frac{\Delta L}{L} \qquad \Longrightarrow \qquad k_{axial} = \frac{EA}{L}$$

with E the Young (elastic) modulus (how much the material "resists stretching", in Pa) and A the cross-sectional area. For shear the same holds with the **shear modulus** G and the shear strain γ_{xy}:

$$\tau_{xy} = G\,\gamma_{xy} \qquad\qquad k_{shear} = \frac{GA}{L}$$

For bending, in a cantilever with a force F at the free end:

$$y_{max} = \frac{FL^3}{3EI_x} \qquad \Longrightarrow \qquad k_{bending} = \frac{3EI_x}{L^3}$$

In the cantilever the maximum moment is at the fixed end, $M_{z,max} = -FL$, and I_x is the **moment of inertia** (second moment of area) of the section about its axis, which measures how the area is distributed relative to the neutral fibre.

Stress	Stiffness	What increases it
Axial	EA/L	$E\uparrow$, $A\uparrow$, $L\downarrow$
Shear	GA/L	$G\uparrow$, $A\uparrow$, $L\downarrow$
Bending (cantilever)	$3EI/L^3$	$E\uparrow$, $I\uparrow$, $L\downarrow\downarrow$ (cubic)

KEY IDEA
The higher the stiffness, the less the structure deforms under a given load. High stiffness is sought to limit deformations, but *elastic* deformations are a "good" mechanism (the wing flexes and recovers its shape, absorbing gusts). What must be avoided are **permanent deformations** and breakage. Stiffness depends on E (or G), on the geometry (A, I) and on the length L, which enters cubed in bending.

The stress–strain curve

As Figure 4.2 shows, as long as the stress stays below P (the **elastic limit**) the material returns to its shape when the load is removed, like a spring. If P is exceeded, on unloading the curve descends *parallel to the elastic region* and a permanent deformation S remains. The *yield stress* is the stress beyond which this happens appreciably, and the maximum stress reached is the ultimate strength.

Common mistake: confusing the *elastic limit* (permanent damage begins) with the *ultimate strength* (fracture). In design, members are sized to stay always below the elastic limit.

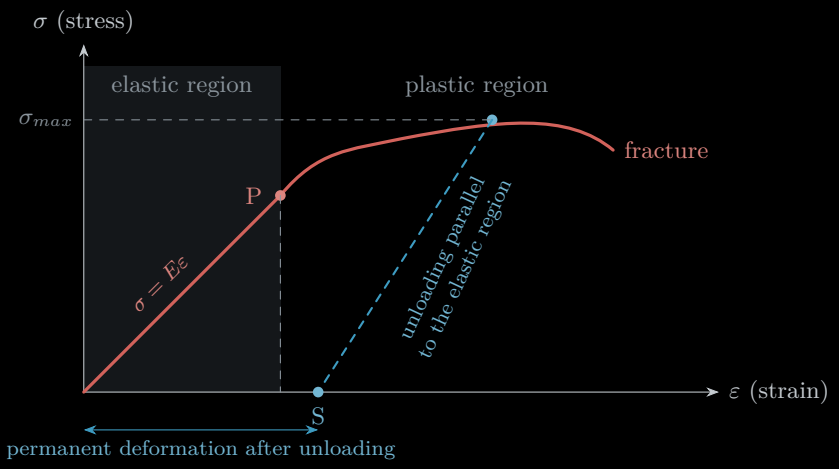


Figure 4.2. Material behaviour: linear and reversible up to the elastic limit (P); beyond P, unloading leaves a permanent deformation (S). Source: Unit 4, slide 6.

Moment of inertia: why beams are I-shaped

Supplement (not developed in the slides). For a rectangle of base b and height h (the dimension in the bending direction), $I = bh^3/12$. The factor h^3 appears: **doubling the height multiplies the bending stiffness by 8** while only doubling the material, whereas doubling the width only doubles it. That is why the I-beam (Figure 4.3; flanges far from the axis, thin web) or the tube is so efficient: it puts the material where it works hardest and wastes no weight at the centre.

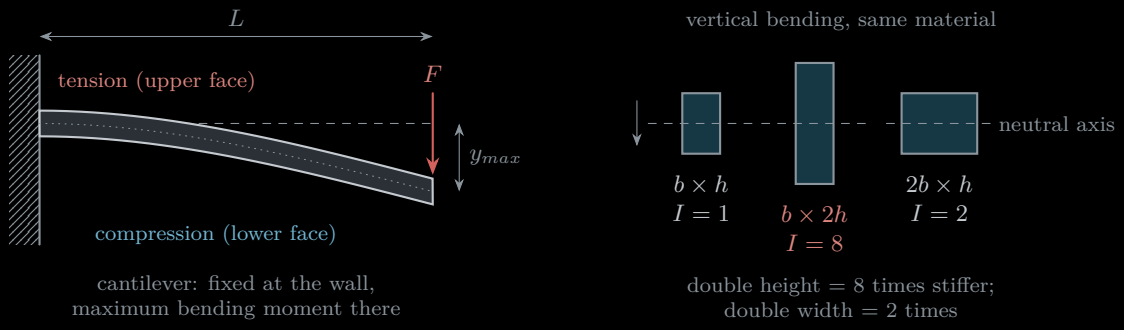


Figure 4.3. Left: a cantilever beam loaded at the free end (tension on one face, compression on the other, maximum moment at the wall). Right: the same material and the same bending direction; $I = bh^3/12$ grows with the cube of the height (own supplement).

2024/25 EXAM, TEST 1, QUESTION 7: INCREASING BENDING STIFFNESS
For a member subjected to a bending moment, stiffness is proportional to EI/L^3. Effect of each option:

Option	Factor on stiffness	Comment
Doubling the length L	$1/2^3 = 0.125$	becomes 8 times worse
Cross-sectional area ×3	depends on how	A does not appear in $3EI/L^3$; it only matters if I grows
Material with a 20 % higher E	1.2	small improvement
Moment of inertia ×2	2	**the best**

Answer: increase I by a factor of 2 (option d). The area option is misleading: more area does not imply a larger moment of inertia.

2.3 Buckling

A slender column under **compression** can suddenly give way by bowing sideways (*buckling*), without the material having reached its strength. This is what happens with a plastic ruler squeezed between the hands: it bends long before it is crushed. The same happens with the skin and stringers under compression or shear. It is an *instability* (a sudden change of shape), not a failure of the material. Figure 4.4 shows how the support conditions change the effective length.

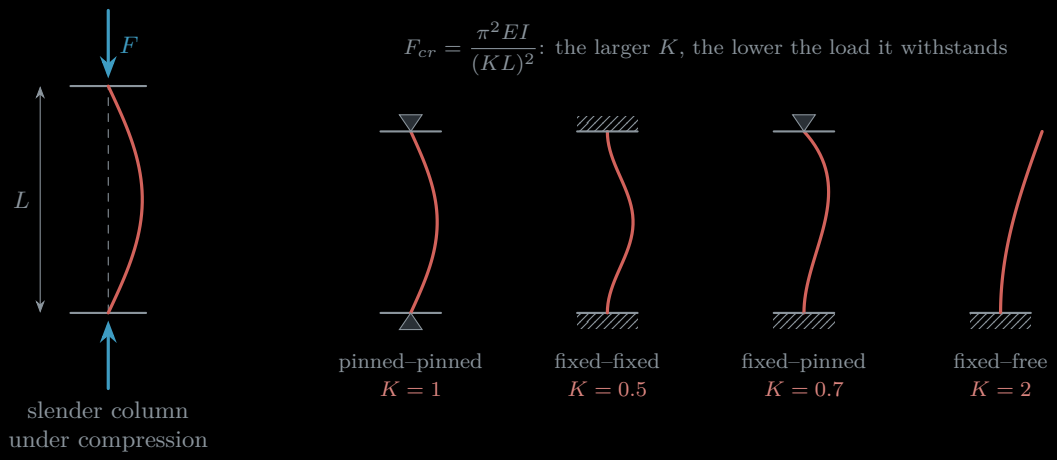


Figure 4.4. Buckling of a column and effect of the support conditions on the effective length KL (the values of K are the classical ones from Euler theory; the slide only says that K depends on the supports). Source: Unit 4, slide 9.

$$F_{cr} = \frac{\pi^2 \, E \, I}{(K \, L)^2}$$

E Young modulus, I moment of inertia, L length, K effective length factor (depends on the supports). Consequences:

- The critical buckling load **may be lower than the load that would reach the elastic limit**: members must be sized taking buckling into account.
- The parameters are the same ones that give stiffness (E, I, L), and similar relations hold for plates (the skin). That is why stringers and frames are added: they **shorten the free length** of the panel and prevent it from buckling.
- **Length is the most powerful parameter**, because it enters squared: half the length between supports = four times the buckling load.

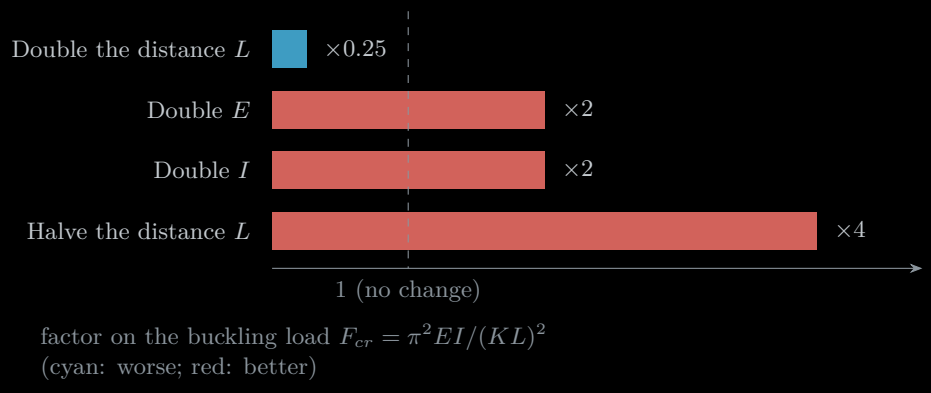


Figure 4.5. Effect of each modification on the buckling load (2023/24 exam, question 10).

2023/24 EXAM, TEST 1, QUESTION 10: HOW TO INCREASE THE BUCKLING LOAD OF A STRINGER?
With total cost and mass held constant, the options are compared (Figure 4.5; all relative to the initial situation, $F_0 = \pi^2 EI/(KL)^2$):

- Doubling the distance between supports: $L \to 2L$, $F \to F_0/4$.
- **Halving the distance between supports: $L \to L/2$, $F \to 4F_0$.**
- Doubling the Young modulus: $F \to 2F_0$. Doubling the moment of inertia: $F \to 2F_0$.

Answer: halve the distance between supports (option b).

3 Fuselage structure

The fuselage houses the cargo, passengers and crew and joins the wings and tail. It bears bending (due to its weight and to the forces from the wings and tail), torsion, shear and, if pressurized, the internal pressure. It has gone through several historical architectures.

3.1 Truss structures

A **truss structure** is a set of bars in which each bar undergoes only axial stress (tension or compression), **never bending**. It usually has at least 4 **main longerons** running along the whole fuselage, joined by bars that form the lattice. **The skin** (usually fabric, sometimes wood or metal) **does not contribute**: it only provides the aerodynamic shape. It is light but not very aerodynamic and does not make use of the skin. The two classic layouts, Pratt and Warren, are compared in Figure 4.6.

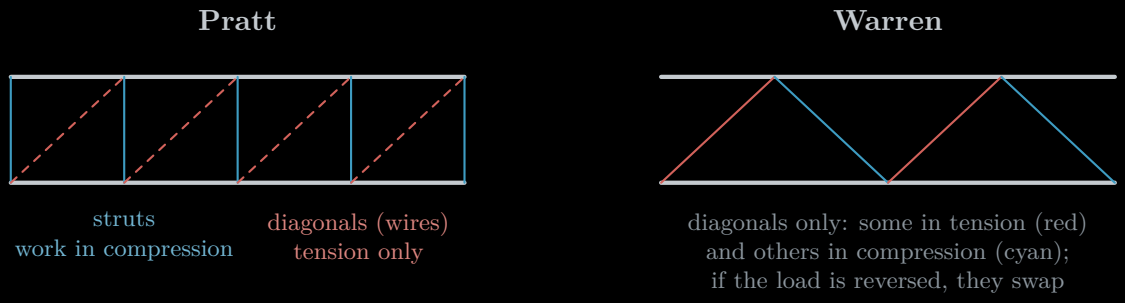
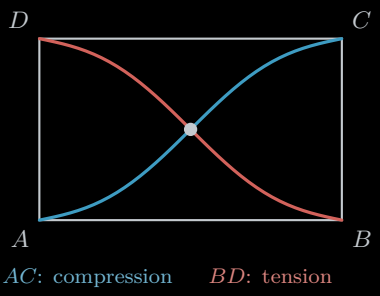


Figure 4.6. Fuselage truss structures. Pratt: vertical struts (in compression) and diagonals that are wires (working only in tension). Warren: diagonals only, working in tension or compression depending on the load. Source: Unit 4, slides 11 to 13.

- **Pratt**: the longerons are joined by vertical **struts**, which work mainly in compression, and **diagonals** that work only in tension and are usually wires. Examples: **Blériot XI**, **Wright Flyer** (first controlled aircraft of the 20th century). Biplanes used this *wire-braced* structure in their wings too.
- **Warren**: the longerons are joined only by **diagonals** that work in both tension *and* compression: under a given load some members are in tension and others in compression, and if the load is reversed they swap roles. Example: Piper PA-18.

3.2 Geodesic structure

It is made of **geodesic** arcs (minimum-length lines over a curved surface). In each section one member works in compression and another in tension, crossing each other (Figure 4.7).



1. AC is compressed and its curvature increases.
2. Tension appears in BD, which tries to straighten AC.
3. They are joined at the centre: the forces oppose and balance each other (equilibrium, also in torsion).

Figure 4.7. Principle of the geodesic structure: the compression of AC and the tension of BD balance each other because they are joined at the centre. Source: Unit 4, slides 14 and 15.

- **Torsional loads are cancelled**.
- It achieves **maximum strength with minimum weight**, and **it still performs well even if part of the structure is lost** (as stated in a test from previous years: the geodesic structure is the one that best withstands the loss of part of it).
- **Problem**: it is very slow to build; it was restricted to World War II (Vickers Wellington, Vickers Warwick).

3.3 Monocoque and reinforced monocoque

Monocoque ("a single shell"): **the skin bears the stresses** and is the main structural element. It is a thin layer, ideally without longitudinal or transverse load-bearing members. Advantages: stability and aerodynamic efficiency. It was originally built in plywood (Deperdussin) and is now made of light alloys.

Reinforced monocoque: the skin is stiffened transversally by vertical rings, the **frames**. Since the skin withstands tension but *not compression* (it buckles), angular profiles are added to the frames. Today it is only used where there are no openings (windows) or where they are few and small, so that the stress distribution is uniform: it is light but **difficult to build and repair** and limits the design. The evolution from the monocoque to the semi-monocoque is summarised in Figure 4.8.

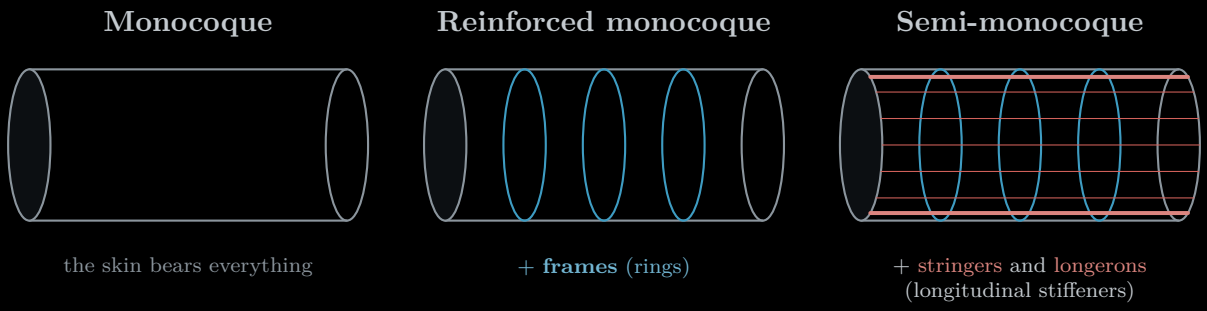


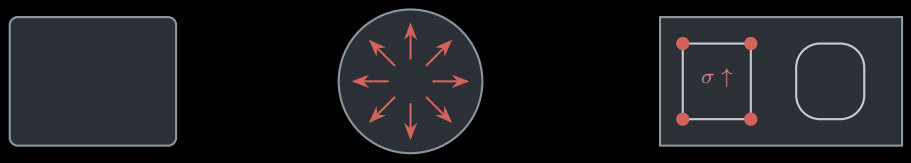
Figure 4.8. Evolution of the skin: monocoque, reinforced monocoque (frames) and semi-monocoque (frames plus longitudinal stiffeners). Source: Unit 4, slides 16 to 18.

3.4 Semi-monocoque

This is the current architecture: a **thinner skin** (with a structural role) with **frames** and also **stringers** and, sometimes, **longerons** as longitudinal stiffeners. It greatly increases buckling resistance at low weight.

- **Non-pressurized**: it can have a **square** shape (makes better use of the space; example: CASA C-212).

- **Pressurized**: the **internal overpressure** is added as an extra load; the fuselage inflates like a balloon and therefore has a **rounded** shape, as Figure 4.9 illustrates (a cylindrical section distributes the pressure uniformly: in a square box, the pressure would try to round it off and concentrate stresses at the corners). Concentrated shear stresses appear on the frames: **floating frames** are used so that the shell can expand uniformly. Any discontinuity causes problems (stress concentrators, sealing issues, corrosion) and **the windows are still the weakest point**.



Non-pressurized
square shape acceptable

Pressurized
round section
(inflates like a balloon)

Windows
sharp corner = stress concentrator;
rounded: stress spread out

Figure 4.9. Why the pressurized fuselage is round and the windows have rounded corners. Source: Unit 4, slides 19 and 20.

> **CASE STUDY: THE 1953–54 COMETS**
> In 1953 and 1954 three de Havilland Comets broke up in flight soon after taking off, killing everyone on board. Construction techniques were improved, the windows were changed from **square to rounded**, and other improvements made later versions safe. Sharp corners concentrate stress (fatigue cracks) and rounded ones do not. (Slide 21; the slide only mentions "square windows"; fatigue as the mechanism is added information.)

Parts of a semi-monocoque fuselage

Figure 4.10 shows the main elements in a side section; the table below gives the function of each one.

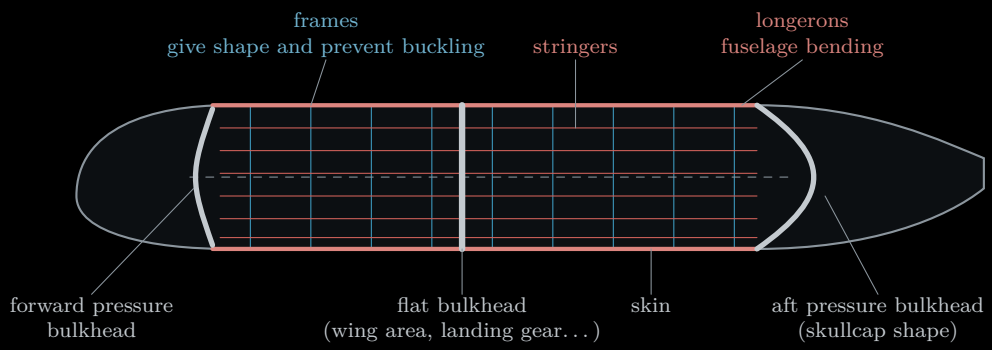


Figure 4.10. Parts of a pressurized semi-monocoque fuselage, in side section (schematic; in reality there are many more frames). Source: Unit 4, slides 22 to 25.

Element	Function
Skin	Bears most of the tension, compression and torsion, and **maintains the cabin pressure**.
Frames (or *former rings*)	Transverse members that give the skin its **circular shape**. They support the skin and stringers, **preventing them from buckling**.
Stringers	Small longitudinal beams. They support the skin and prevent it from buckling; they bear tension and compression, including the **fuselage bending**.
Longerons	Larger longitudinal beams that bear bending and distribute axial loads. **They do not always exist.**
Pressure bulkheads	Close the cylinder at the front and rear to maintain pressurization. The rear one has a **skullcap** shape.
Flat bulkheads	Structural planes in stress-concentration zones (engine pylons, landing gear, wings, tail surfaces).

All of them are joined with **rivets or bolts**. Other elements shown in the illustrations of slide 23: seat track, floor beam, window frames and columns.

SUMMARY OF FUSELAGE TYPES

Type	What bears the loads	Notes
Pratt	Bars (struts in compression, wire diagonals in tension); skin does not contribute	Wright Flyer, Blériot XI
Warren	Diagonal bars, in tension or compression depending on the load	Piper PA-18
Geodesic	Crossed arcs (compression and tension)	Cancels torsion; slow to build (WWII)
Monocoque	The skin	Aerodynamic; thin skin without stiffeners
Reinforced monocoque	Skin + frames	No windows or very few
Semi-monocoque	Skin + frames + stringers (+ longerons)	Current; pressurized or not

4 Wing structure

The wing must **generate lift and bear loads**: it behaves as a **beam** (a cantilever fixed to the fuselage) that must resist the stresses. The loads come from:

- **Aerodynamic loads** (lift and drag).
- **Engine thrust** (if the engines are mounted on the wing).
- **Landing gear reaction**.
- **Deflection of the high-lift devices** (flaps, slats).

In addition there are the self-weight and the fuel inside. In flight, lift bends the wing upwards: the upper surface is compressed and the lower one is in tension (the upper flange of the spar is in compression and the lower one in tension). Figure 4.11 shows the main parts of the wing in plan view.

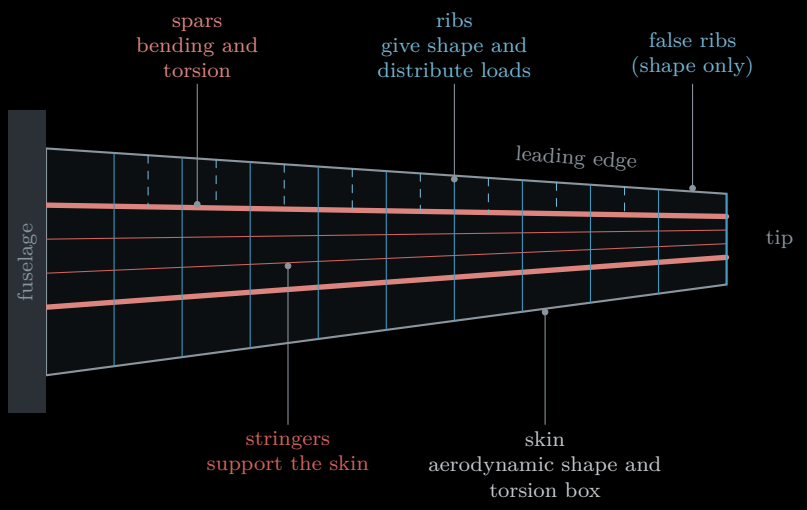


Figure 4.11. Parts of a semi-monocoque wing seen in plan view (half-wing): spars, ribs, stringers, false ribs and skin. Source: Unit 4, slides 28 to 31.

4.1 Parts of a semi-monocoque wing

Element	Function
Skin	Gives the **aerodynamic shape** and maintains it along the wingspan; it **can contribute to the structural strength**.
Spars	Beams running along the wing: **the main support of the structure**. They bear **bending and torsion**.
Ribs	Transverse members. They give the **airfoil shape** and transmit the loads from the skin to the spars.
Stringers	Longitudinal members that transmit the load from the skin to the ribs and **prevent the skin from buckling**.
Fittings	Metal joining parts: wing–fuselage, control surfaces–wing... They bear stresses, vibrations and deflections.

The closed region between the spars and the skin forms the **torsion box**, which resists torsion and also serves as a fuel tank (the ribs create baffles in the tanks). Figure 4.12 shows it in cross-section.

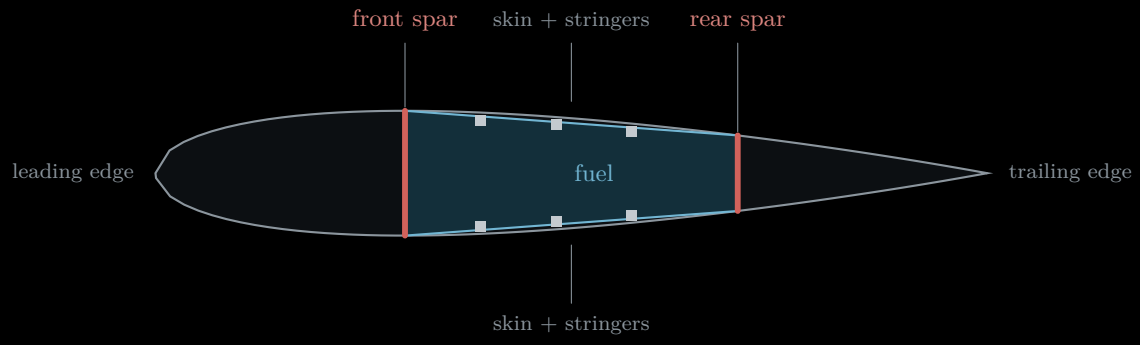


Figure 4.12. Cross-section of a wing: the torsion box (between the two spars, closed by the upper and lower skin with stringers).

Spars

A spar has two main parts: the **web**, which bears the **shear stresses**, and the **flanges**, upper (compression in flight) and lower (tension), which bear the tension and compression due to bending.

It is like an I-beam: the flanges do the bending work and the web keeps them apart and absorbs the shear. Figure 4.13 shows typical spar sections and the two spar types according to the web.

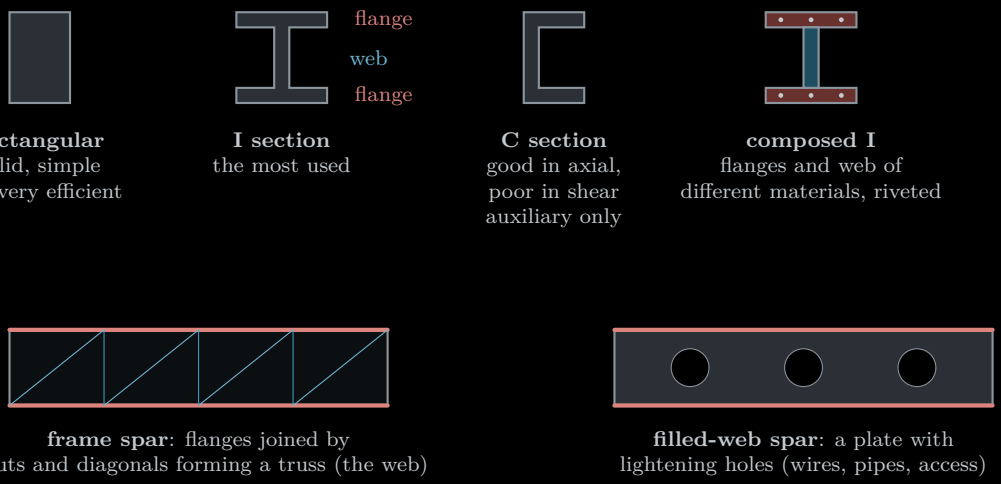


Figure 4.13. Typical spar sections and spar types according to the web. Source: Unit 4, slides 32 to 34.

- **Frame spars**: the flanges are joined by vertical and/or diagonal members (riveted, bolted or welded) that form a truss which is the web.
- **Filled-web spars**: the flanges are joined by a plate. If the web is very tall, it carries vertical reinforcements. **Holes** are made in it to reduce weight, to pass wires and pipes through, and to make access easier for maintenance.

The forces vary along the span, so the spar section can be **variable** (smaller towards the tip) to save weight. Its shape depends on the airfoil, its thickness, the required strength and the material:

- **Rectangular**: solid, economical and simple, not very efficient.
- **I section**: upper and lower flanges joined by the web.
- **C section**: good for axial stresses, but **unstable under shear loads**; it is only used as an **auxiliary spar**.
- **Composed I section**: the flanges are made of one material and the web of another, joined with rivets.

Ribs

- They keep the **airfoil shape** (thickness, spacing between spars).
- They **transmit the aerodynamic forces** to the spars.
- They **distribute concentrated loads** (landing gear, tanks...).
- They **close the wing cells** (torsion box) and create baffles in the fuel tanks.

Type (by function)	Purpose
Compression ribs	Link the spars and distribute the stresses to them; placed where there are **local stresses** (engine mounts, landing gear).
Master ribs	Keep the distance between spars and give stiffness to the skin and stringers, **preventing their buckling**.
Common ribs	Less robust: they keep the airfoil shape and transmit the internal forces to the spars.
False ribs	Only used to **keep the shape of the skin**; located between the spar and the leading or trailing edge.

Like spars, ribs can be of the **truss** or **filled-web** type, and have a **web** (stiffness against vertical and diagonal deformations; if metallic, it is manufactured by pressing) and **nerves**. In swept wings they can be placed **parallel to the fuselage axis** or **perpendicular to the spars** (Figure 4.14).

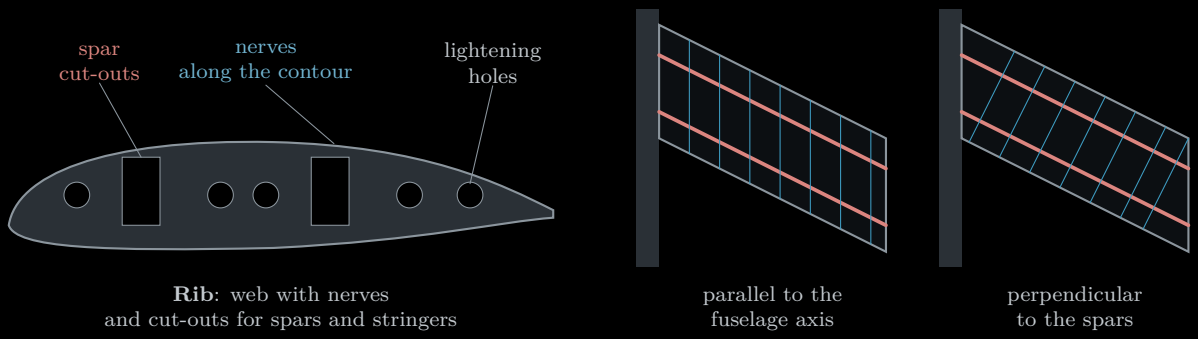


Figure 4.14. Anatomy of a rib and arrangement in swept wings. Source: Unit 4, slides 35 to 37.

5 Materials

5.1 Requirements for aircraft structural materials

- **Specific strength** (strength/weight) **and stiffness** *at the same time* (the combination, not just one).
- **Predictable** behaviour.
- **Good ageing** (against the environment: corrosion…).
- **Fireproof**.
- Easy to obtain, readily available and at a **reasonable cost**.
- **Fatigue resistance** (thousands of load cycles) and **thermal resistance**.

Specific strength $= \sigma_{\text{strength}}/\rho$: how much stress the material withstands per unit of density. The same applies to specific stiffness E/ρ. This is what matters because the goal is to withstand what is needed with the least weight.

5.2 Materials used

- **Origins of aviation**: **wood** (problems with humidity), **fabric** and **steel wire** (very dense).
- **Light aluminium alloys**: high specific strength with low density; they are the most widely used.
- **Titanium alloys**: where **thermal loads** are expected.
- **Stainless steel**: extremely strong, but with a high density; only a few components (the **landing gear**).
- **Composite materials**: fabric sheets (carbon fibre) bound by a **resin matrix** (generally epoxy), sometimes embedded in a honeycomb core (sandwich panel). **Extremely light**, but with problems of **impact resistance and lightning resistance**.

Figure 4.15 shows how a sandwich panel is built.

Treatments and coatings required: **aluminium** against **corrosion**; **composites** against **humidity**.

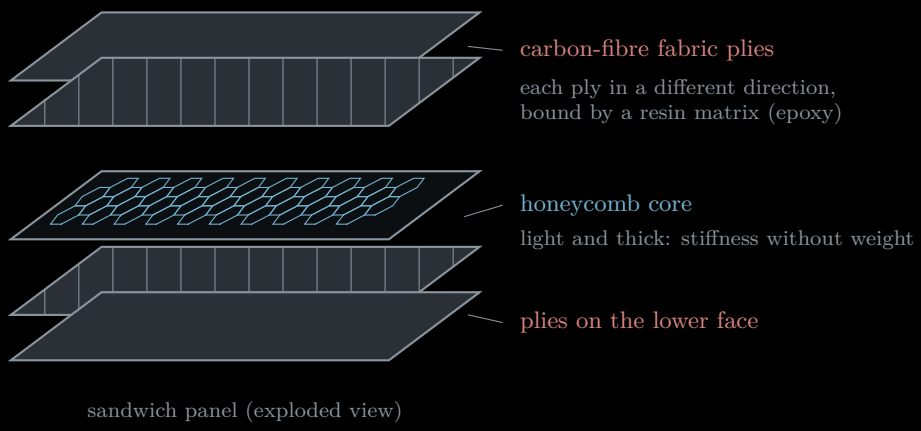


Figure 4.15. Sandwich panel of composite material: fabric plies bound by resin on both faces of a honeycomb core.

Material	Advantage	Drawback / use
Wood, fabric, wire	Available; light (wood)	Humidity (wood); wire very dense. Early aviation.
Aluminium alloy	High specific strength, low density	Corrosion: needs treatment. Most aircraft.
Titanium alloy	Withstands thermal loads	Thermal loads (expensive: added information, from exams of other years).
Stainless steel	Very strong	Dense: landing gear only.
Composites (CFRP)	Very light	Impact, lightning, humidity.

2023/24 EXAM, TEST 1, QUESTION 9: WING SKIN THAT ALWAYS WORKS IN TENSION

If the region always works in tension it cannot buckle, so buckling does not matter. What must not be exceeded is the elastic limit (so that no permanent deformation remains). With the goal of "high strength at low weight", the **specific yield strength** σ_y/ρ is maximized. (This is the answer given in the students' compilation; the official key is not in the exam PDF.)

2024/25 EXAM, TEST 1, QUESTION 8: WHICH ONE IS NOT CORRECT

"Aluminium alloys are used in most aircraft because they are the materials with the highest strength" is the false statement: they are used for their high **specific** strength with low density, not because they have the highest absolute strength (steel is stronger, but much heavier). Steel in the landing gear is correct (high strength per unit volume), titanium for thermal loads is also correct, and composites do need a coating against humidity.

WHAT USUALLY APPEARS ON THE EXAM (UNIT 4)

All past exams are **multiple-choice tests** (4 options, a wrong answer deducts 1/3). From this unit there are **few questions per midterm**, almost always conceptual:

- **Fuselage**: which type *cancels torsion* (geodesic) and which *withstands the loss of part of the structure best* (geodesic); which type was used by the first controlled aircraft (Pratt, Wright Flyer); what bears the pressurization in a semi-monocoque (the skin, and also the pressure bulkheads; *not* the flat bulkheads nor the stringers as such); what the frames bear (they give shape and prevent buckling, *not* the bending moment); **the spacing between windows is chosen according to the spacing between frames**.
- **Wing**: the spar **web** bears **shear**, the **flanges** bear tension and compression due to

bending; the spars bear the wing bending (a 2022/23 test, however, gives as correct that *the skin* bears the bending moment from lift: in a semi-monocoque the skin and stringers work together with the spars); the **ribs** give shape and distribute loads; stringers prevent skin buckling; false ribs only maintain the shape.

- **Stiffness and buckling, with numbers**: $k = 3EI/L^3$ in bending (doubling I doubles it; doubling L divides it by 8); $F_{cr} \propto EI/(KL)^2$ (half the distance between supports $\rightarrow \times 4$, the most effective option).
- **Materials**: aluminium for *specific* strength; titanium for thermal loads; steel only in the landing gear; composites with a coating against humidity (and problems with impact and lightning); surface treatments against corrosion and humidity.

Essentials: $\sigma = E\varepsilon$; $k_{axial} = EA/L$; $k_{bending} = 3EI/L^3$; $F_{cr} = \pi^2 EI/(KL)^2$; skin + frames + stringers = semi-monocoque; web = shear, flanges = bending; specific strength = strength/density.

UNIT 5

Fundamentals of atmospheric flight

1 Introduction: where the force that keeps an aircraft flying comes from

The air surrounding a vehicle exerts forces on it. There are two big families, which give rise to the most basic classification of aircraft:

- **Aerostats**: the force is **independent of the velocity** of the air relative to the body. It is Archimedes' principle (buoyancy from being lighter than the displaced air): balloons and airships.
- **Aerodynes**: the force **depends on the relative velocity** between air and body (aerodynamic force): aeroplanes, helicopters, drones. These are the ones studied in this unit. An aerodyne does not have to be lighter than air; it needs to move relative to it.

Common mistake: saying that aerostats need flow around an airfoil to stay aloft (false) or that an A320 is an aerostat (false: it is an aerodyne). Aerostats also cannot fly fast because of their large aerodynamic drag (exam question).

MINIMUM GLOSSARY FOR THIS UNIT

- **Airfoil**: the shape of the section of a wing cut by a plane parallel to the direction of motion. A wing of infinite span is a 2D airfoil.
- **Angle of attack** α: angle between the airfoil chord and the incident airstream.
- **Coefficient** (lift, drag…): the force divided by $\frac{1}{2}\rho V^2 S$, that is, the force *without units* and independent of how much air there is, how fast it moves or how large the body is. It depends mainly on the shape and on the angle of attack.
- **Mach number** $M = U_\infty / a_\infty$: flight speed divided by the speed of sound ($a = \sqrt{\gamma RT} \approx 340\,\text{m/s}$ at sea level; Unit 2).
- **Reynolds number** Re: ratio of inertial to viscous forces; it indicates the relative importance of the air viscosity at that scale.
- **Boundary layer**: very thin layer of air attached to the surface where viscosity slows the air down until it stops at the wall.
- **Stall**: the flow separates from the wing and a lot of lift is lost.

1.1 Aerodynamic forces and axes

Four forces act on an aircraft: **lift** L (perpendicular to the airstream), **drag** D (parallel to the airstream and in the same direction), **thrust** T (from the engine) and **weight** $W = mg$ (Figure 5.1). The first two are written with coefficients:

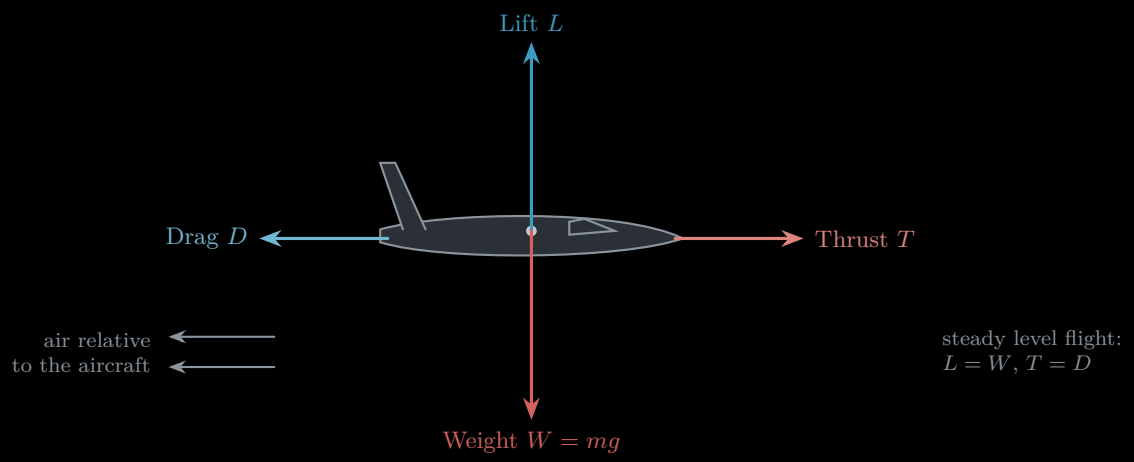


Figure 5.1. The four forces in steady level flight. Source: Unit 5, slide 5.

$$L = \tfrac{1}{2}\rho V^2 S\, C_L \qquad D = \tfrac{1}{2}\rho V^2 S\, C_D \qquad W = mg$$

ρ air density, V velocity, S reference area (the wing area), C_L and C_D coefficients (dimensionless). The quantity $\frac{1}{2}\rho V^2$ is the **dynamic pressure**, which measures how much kinetic energy per unit volume the air carries. In steady level flight $L = W$ and $T = D$.

Three axes passing through the centre of gravity are defined (Figure 5.2): the **longitudinal** or roll axis (from tail to nose), the **lateral** or pitch axis (from wingtip to wingtip, towards the right) and the **vertical** or yaw axis (perpendicular to both, pointing to the ground). The three **aerodynamic moments** are roll, pitch and yaw.

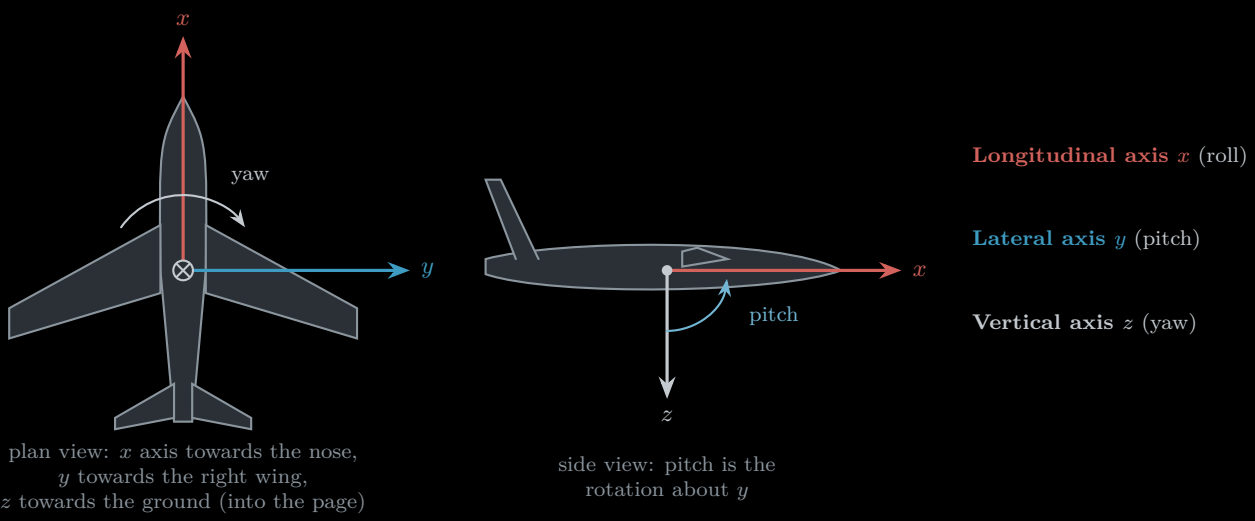


Figure 5.2. Aircraft axes and rotations. Source: Unit 5, slide 6.

1.2 Blunt and streamlined bodies

A **blunt body** (a plate perpendicular to the wind) has **a lot of drag**. A **streamlined body** (like a wing) has **little drag** and can generate **a lot of lift**: this is the one of interest for the flight of aerodynes (Figure 5.3).

1.3 Flight regimes

Behaviour changes greatly with the Mach number $M_\infty = U_\infty / a_\infty$ (Figure 5.4):

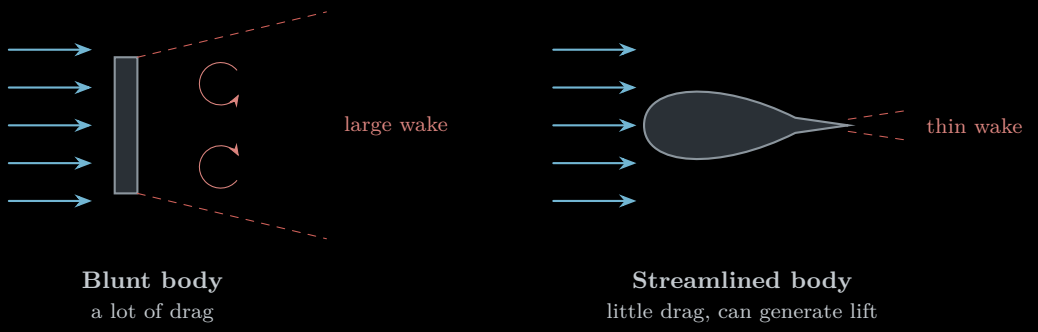


Figure 5.3. Blunt versus streamlined body: the blunt one leaves a large wake behind it.

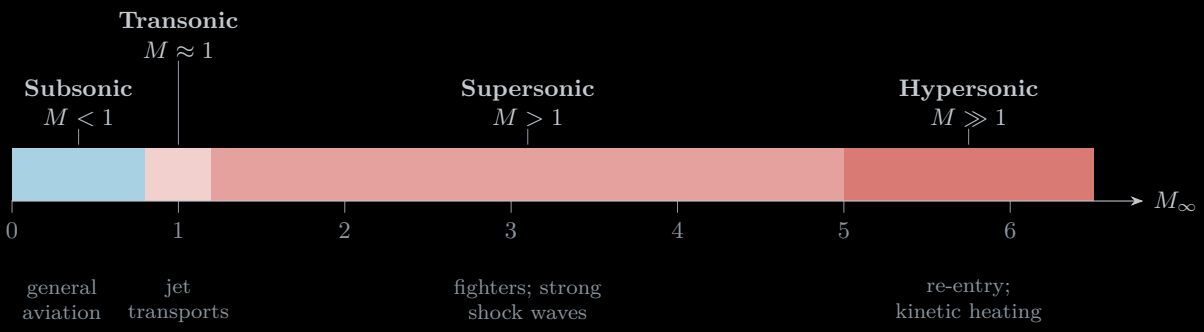


Figure 5.4. Regimes according to the Mach number (the numerical limits are indicative; the slides only give the conditions $M < 1$, $M \approx 1$, $M > 1$ and $M \gg 1$). Source: Unit 5, slide 8.

- **Subsonic** ($M < 1$): the usual case in general aviation.
- **Transonic** ($M \approx 1$): almost the case of commercial jet transports.
- **Supersonic** ($M > 1$): typical of fighters; strong shock waves may appear.
- **Hypersonic** ($M \gg 1$): typical of re-entry vehicles; there are always strong shock waves and kinetic heating (Section 6) is a problem.

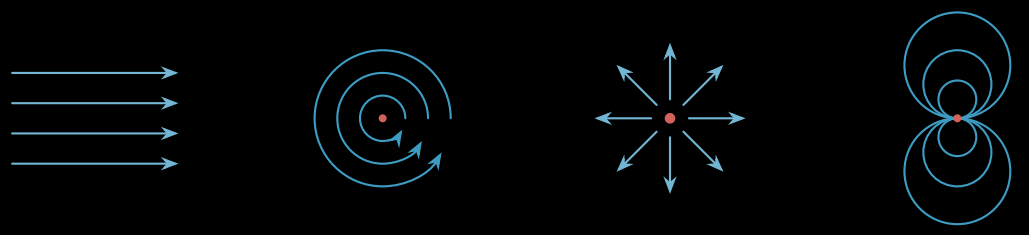
1.4 Potential flow, boundary layer and Reynolds number

Far from streamlined bodies the velocity field **derives from a potential**, just like a gravitational or electric field. The flow is assumed incompressible ($\nabla \cdot \mathbf{v} = 0$) and irrotational ($\nabla \times \mathbf{v} = 0$), so that there exists a φ with $\mathbf{v} = \nabla\varphi$ and the **Laplace equation** holds:

$$\nabla^2\varphi = \frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2} + \frac{\partial^2\varphi}{\partial z^2} = 0 \qquad U_x = \frac{\partial\varphi}{\partial x},\ U_y = \frac{\partial\varphi}{\partial y},\ U_z = \frac{\partial\varphi}{\partial z}$$

Since the equation is linear, solutions can be **added**: combining simple flows yields complex flows (e.g. uniform flow + doublet = flow around a cylinder). The elementary solutions (slide 10) are drawn in Figure 5.5 and summarised in the table below:

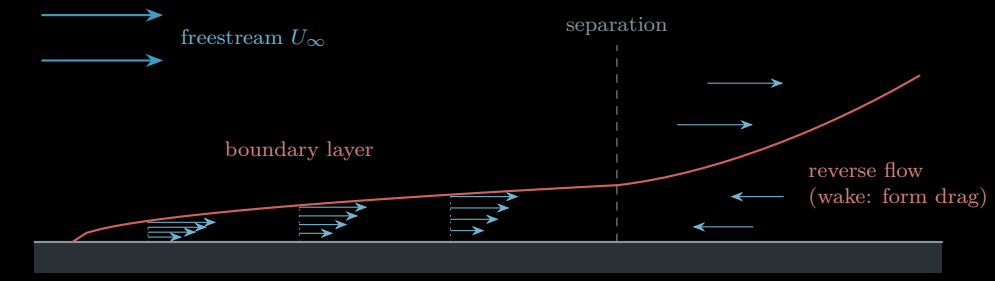
Flow	Potential	Velocity
Uniform	$\varphi = U_0 x$	$U_x = U_0$
Line vortex	$\varphi = \dfrac{\Gamma}{2\pi}\theta$	$U_\theta = \dfrac{\Gamma}{2\pi r}$
Source/sink	$\varphi = \dfrac{Q}{2\pi}\ln r$	$U_r = \dfrac{Q}{2\pi r}$
Doublet (dipole)	$\varphi = -\dfrac{\mu}{r}\cos\theta$	–



uniform $\varphi = U_0\, x$ | **line vortex** $\varphi = \dfrac{\Gamma}{2\pi}\theta$ | **source / sink** $\varphi = \dfrac{Q}{2\pi}\ln r$ | **doublet (dipole)** $\varphi = -\dfrac{\mu}{r}\cos\theta$

Figure 5.5. Elementary solutions of potential flow. The slide writes the source with a minus sign, but then $U_r = Q/(2\pi r)$ would come out negative; the usual sign is used here. Source: Unit 5, slides 9 to 11.

Potential flow **has no viscosity**: it only gives forces perpendicular to the surface, from pressure. But close to the body, viscous effects are important: this is the **boundary layer**, where the air goes from being at rest at the wall to the freestream velocity over a very short distance; it generates **tangential (shear) stresses**. Figure 5.6 shows how the velocity profile grows downstream and what happens at separation.



velocity goes from 0 at the wall to U_∞ over a small thickness: shear stress (friction drag).
With an adverse pressure gradient the layer separates.

Figure 5.6. Boundary layer: velocity profile growing downstream and separation with reverse flow. Source: Unit 5, slides 12 to 14.

$$Re = \frac{\rho_\infty U_\infty\, c}{\mu_\infty}$$

ρ_∞ density, U_∞ velocity, μ_∞ dynamic viscosity and c characteristic length (the wing chord). If Re is **small**, viscous effects matter and the boundary layer is **thick**; if Re is **large**, the layer is **thin**. The aerodynamic behaviour depends on Re.

- The boundary layer can be **laminar** (ordered layers) or **turbulent** (chaotic mixing). A laminar layer *transitions* to turbulent with roughness and with high Re.
- Boundary layers can **separate** from the body; this happens more easily if they are laminar. The higher Re is, the later the flow separates.

WHAT IS THE VALUE OF Re ON A TYPICAL WING
Chord $c = 1.5\,\mathrm{m}$, $V = 50\,\mathrm{m/s}$, sea level ($\rho = 1.225\,\mathrm{kg/m^3}$, $\mu = 1.789 \cdot 10^{-5}\,\mathrm{Pa\,s}$):

$$Re = \frac{1.225 \cdot 50 \cdot 1.5}{1.789 \cdot 10^{-5}} \approx 5.1 \cdot 10^6$$

This is a "high" value: viscous effects are limited to a very thin layer and the flow outside it is almost potential. In a wind-tunnel lab session, with chords of a few centimetres and low speeds, Re is several orders of magnitude smaller, so the boundary layer matters more.

2 Two-dimensional aerodynamics (airfoils)

2.1 Airfoil nomenclature

The wing section is called an **airfoil**; it is essentially a wing of infinite span, and much of the wing behaviour is extrapolated from that of the airfoil. Airfoils usually have low thickness and low camber. Their nomenclature is shown in Figure 5.7.

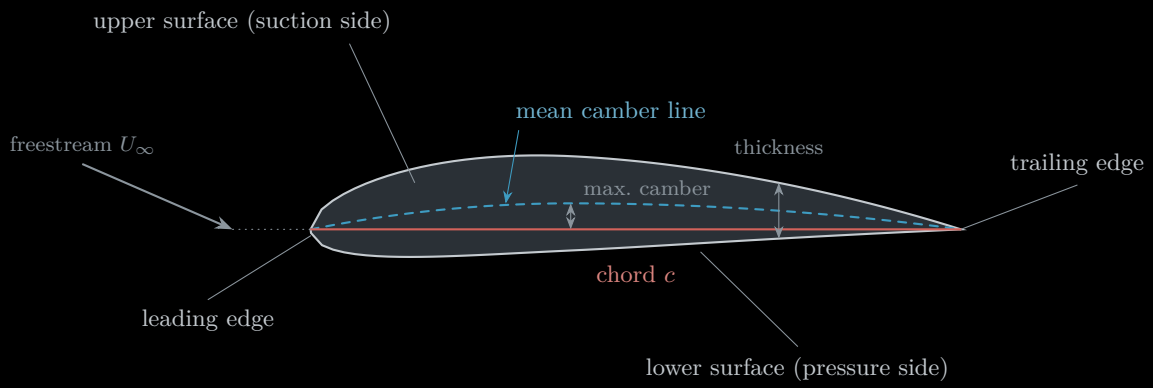


Figure 5.7. Nomenclature of a cambered airfoil (here with exaggerated camber and thickness). Source: Unit 5, slide 17.

- **Leading edge**: front part, also defined by its radius. **Trailing edge**: rear part, normally sharp.
- **Chord**: line joining both edges; its length is also called the chord c.
- **Upper surface** (suction side): top face. **Lower surface** (pressure side): bottom face.
- **Mean camber line**: line equidistant from the upper and lower surfaces. **Maximum camber**: measured relative to the chord (0 = symmetric airfoil); also its position.
- **Thickness**: distance between the upper and lower surfaces (perpendicular to the camber line in the American convention, perpendicular to the chord in the British one). **Maximum thickness** and its position, relative to the chord.

2.2 Airfoil aerodynamic coefficients

The forces and moments on an airfoil change greatly with the conditions (density, velocity…), but once non-dimensionalized they depend almost **only on the airfoil shape and its orientation relative to the airstream**. Lowercase letters are used for the airfoil (per unit span: l, d in N/m, m in N·m/m):

$$c_l = \frac{l}{\frac{1}{2}\rho_\infty U_\infty^2\, c} \qquad c_d = \frac{d}{\frac{1}{2}\rho_\infty U_\infty^2\, c} \qquad c_m = \frac{m}{\frac{1}{2}\rho_\infty U_\infty^2\, c^2}$$

This makes it possible to have airfoil **databases** (airfoils have been tested in wind tunnels for a century).

2.3 NACA airfoils

The NACA (National Advisory Committee for Aeronautics, founded in 1915) tested a great variety of airfoils in wind tunnels and created families of great historical interest (some are still in use). The most important ones here are the **4-digit and 5-digit** series; Figure 5.8 shows some examples.



NACA 0006 symmetric, thin | NACA 0015 symmetric, thick | NACA 2412 camber 2 % | NACA 4412 camber 4 %

Figure 5.8. Some 4-digit NACA airfoils: the first digit is the maximum camber as a % of the chord, the second the position of that camber in tenths of the chord, and the last two the maximum thickness as a %. Source: Unit 5, slides 20 and 21 (own drawings).

Series	Meaning of the digits
4-digit (e.g. 2412)	1st: **maximum camber** as a % of the chord. 2nd: **position** of the maximum camber in tenths of the chord. 3rd and 4th: **maximum thickness** as a % of the chord.
5-digit (e.g. 24012, 24112)	1st $\times 0.15$ = **design lift coefficient**. 2nd $\times 5$ = position of the maximum camber as a % of the chord. 3rd: 0 = normal camber; 1 = reflex camber (with a minimum near the trailing edge). 4th and 5th: **maximum thickness** as a % of the chord.

Examples: NACA 2412 → camber 2 % at 40 % of the chord, thickness 12 %. NACA 0012 → symmetric (camber 0), thickness 12 %. NACA 24012 → design c_l of $2 \cdot 0.15 = 0.3$, maximum camber at 20 % of the chord ($4 \cdot 5$), normal camber, thickness 12 %.

> **2018/19 AND 2021/22 EXAMS: "A NACA 21518 AIRFOIL"**
> The options were: (a) $c_{l,max} = 0.3$; (b) maximum thickness of 15 %; (c) maximum camber at 0.15 of the chord; (d) none is correct. With the 5-digit rule: 1st digit 2 → *design* $c_l = 0.3$ (not the maximum: it is the c_l for which the camber is optimized); 2nd digit 1 → maximum camber at $1 \cdot 5 = 5$ % of the chord (not 15 %); last two digits, 18 → thickness of 18 % (not 15 %). Answer: (d). (Note: the middle digit, 5, is neither 0 nor 1 as the slides' rule requires; it is a deliberately tricky question.)

2.4 Lift generation

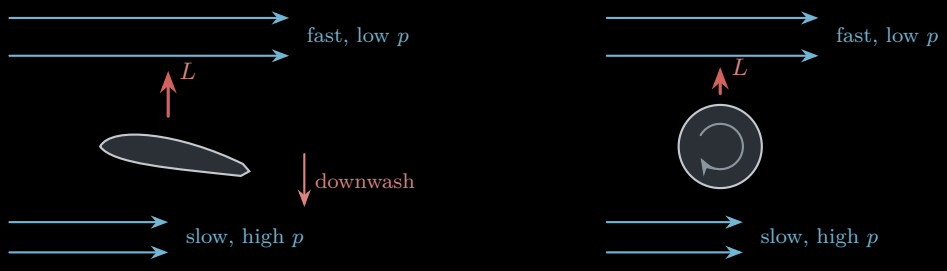
An airstream reaching the airfoil exerts an upward force on it; the airfoil exerts a force **downwards on the air** (action and reaction: the wing deflects the air downwards, *downwash*). A **circulation** Γ appears in the fluid around the airfoil (this is what the potential-flow panel method computes). It is observed (and can be proven from first principles) that the air **accelerates over the upper surface, where its pressure drops**, and decelerates under the lower surface, where it rises.

Beware of causality: the velocity changes because the pressure changes, but the pressure distribution appears because there is relative motion between the air and the airfoil. For this to happen, the airfoil must have **camber** or **a certain angle of attack** α relative to the airstream. *An everyday example is the Magnus effect*: a spinning cylinder or ball drags the air along and creates circulation; that is why balls with spin curve (Figure 5.9).

Subsonic: lift

In purely potential flow it can be shown that the only force is lift (the suction force at the leading edge cancels the drag): this is **d'Alembert's paradox**, because real airfoils do have drag (due to viscosity). **Most of the lift is generated by the upper surface, near the leading edge**: the suction peak, as the pressure distribution in Figure 5.10 shows.

The c_l–α relation is **linear** (found experimentally and provable):



Airfoil
circulation Γ appears and
the air is deflected downwards

Magnus effect
a spinning cylinder drags the air
along: circulation, and so lift

Figure 5.9. Circulation around an airfoil and around a spinning cylinder: the air goes faster (lower pressure) on one side and slower (higher pressure) on the other.

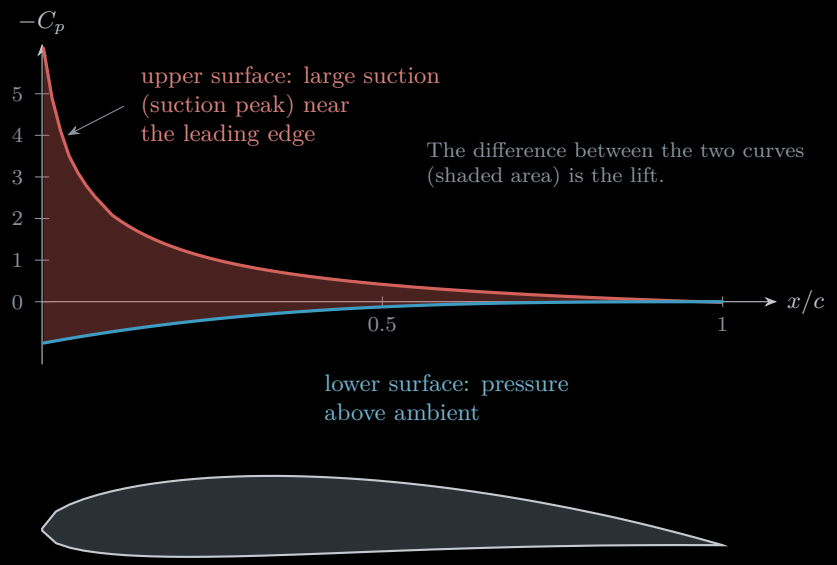


Figure 5.10. Schematic pressure distribution ($-C_p$): large suction on the upper surface near the leading edge; the lower surface contributes less. It is an illustrative curve, not a measured one. Source: Unit 5, slide 26.

$$c_l = c_{l,0} + c_{l,\alpha}\,\alpha \qquad c_{l,\alpha} \approx 2\pi \text{ (with } \alpha \text{ in radians)} \approx 0.11 \text{ per degree}$$

$c_{l,0}$ is the coefficient at zero angle of attack. It is **0 for symmetric airfoils and > 0 for airfoils with positive camber**. For a cambered airfoil, there is a negative angle of attack (the *zero-lift angle*) where $c_l = 0$ (Figure 5.11).

Stall. Downstream of the suction peak the pressure rises again: an **adverse pressure gradient** appears on the upper surface (the air moves towards higher pressure and slows down) and the boundary layer may **separate**; where there is separation, the upper surface stops "working". If the separation is massive, there is a **stall**; the **critical angle of attack** is the stall angle and gives $c_{l,max}$.

- It tends to be **higher for thick airfoils**; airfoils with **large camber** usually have a higher $c_{l,max}$.
- The **higher** Re **is, the later the flow separates** (higher $c_{l,max}$).

Among the NACA airfoils of one exam (0006, 0010, 2410, 2414) the one with the highest $c_{l,max}$ is the **2414**: it has camber *and* it is the thickest of the cambered ones.

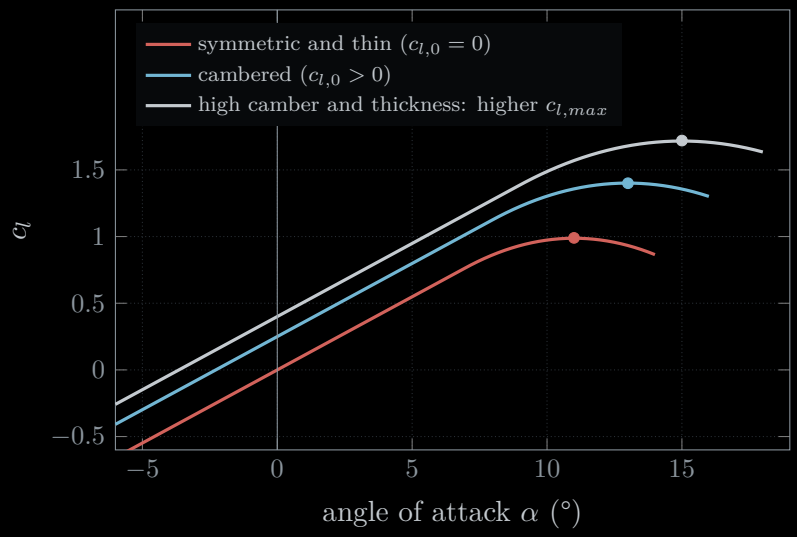


Figure 5.11. Lift curve of three airfoil types: same slope for all, different $c_{l,0}$ and different $c_{l,max}$; the dots mark the stall. Qualitative sketch inspired by slide 28.

Subsonic: drag

- **Skin friction drag**: due to viscosity (shear stresses in the boundary layer).
- **Form drag** (pressure drag): in the separated-flow region a wake forms; the larger it is, the more drag. At low angles of attack it is smaller than the friction drag.
- Together they make up the **parasitic drag**: $c_{d,par}$ = friction + form. It decreases as Re rises and is larger with a turbulent boundary layer than with a laminar one. For constant Re it is **very constant up to large angles of attack** (Figure 5.12).

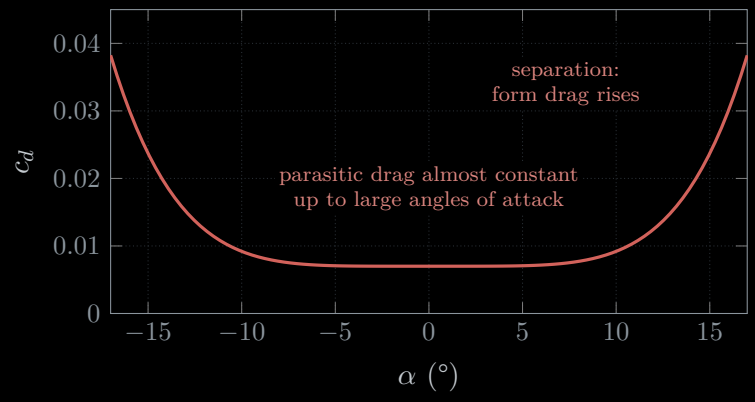


Figure 5.12. Airfoil drag coefficient versus α: almost flat and then it shoots up when the flow separates. Qualitative sketch (slide 31).

Subsonic: pitching moment

The aerodynamic forces are applied at a point, the **centre of pressure** (CP). There is another point, the **aerodynamic centre** (AC), about which the moment coefficient **does not change when α varies**. Once the AC and the moment about it are known, it is trivial to find the moment about any other point. The AC is **approximately at 1/4 of the chord** and **coincides with the CP in symmetric airfoils**.

- **Symmetric**: $c_{m,ac} = 0$ (far from stall).
- **Positive camber**: $c_{m,ac} < 0$, a moment that makes the nose **pitch down** (nose-down).

With $c_{m,ac} < 0$, the CP position is $x_{cp}/c \approx \frac{1}{4} - c_{m,ac}/c_l$ (supplement; it comes from taking moments). Answering the questions on slide 33 (Figure 5.13 plots $c_{m,ac}$ and x_{cp}/c against c_l):

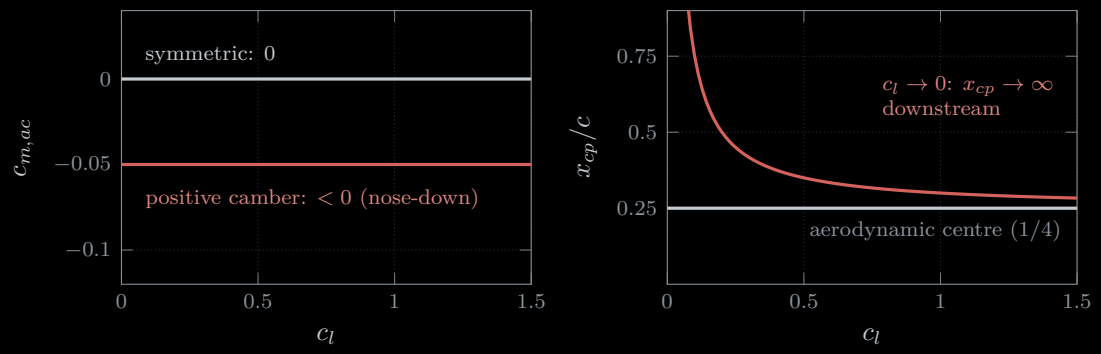


Figure 5.13. Moment about the aerodynamic centre and centre-of-pressure position (the value $c_{m,ac} = -0.05$ is illustrative). Source: Unit 5, slides 32 and 33.

- *At the zero-lift α of a cambered airfoil:* $c_l = 0$ but there is a moment, so the force is zero and the moment is not: the CP is **at infinity downstream**.
- *When α increases from the zero-lift value:* the CP **comes from infinity downstream towards the aerodynamic centre** (and approaches it as c_l grows). This is the question that appears in the tests (option "from infinity downstream towards the aerodynamic centre").
- *A 2D airfoil in level flight without drag:* moment equilibrium requires (own derivation, the slide only poses the question) the CP to coincide with the point of application of the weight.

Common mistake: getting the direction wrong. In airfoils with positive camber the CP does *not* move away from the AC towards the front as α increases; it approaches it *from behind*.

2.5 Airfoils in the transonic regime

The flow accelerates over the upper surface, so sonic conditions are reached at some point even when the flight is at $M < 1$. Two important Mach numbers:

- **Critical Mach number** M_{crit}: the flight Mach number at which $M = 1$ is reached *locally* at some point of the airfoil.
- **Drag divergence Mach number** M_{DD}: above the critical Mach number, shock waves appear on the upper surface; when they interact with the boundary layer they may separate it and the drag **grows a lot** from M_{DD}. This is the origin of the so-called "sound barrier". Once the transonic regime is passed, the drag falls again (Figure 5.14 shows the drag rise).

Always $M_{crit} < M_{DD}$. Aircraft usually fly between the two (there are shock waves but weak ones, still without massive separation); it is not true that they cannot exceed M_{crit}. The presence of shocks and separation also strongly affects the pitching moment: **controlling transonic flight is difficult**. Both problems are mitigated and delayed to higher Mach numbers with **supercritical airfoils** (Figure 5.15). **Their main goal is to increase** M_{DD} (not M_{crit}). In addition, a *thinner* airfoil delays compressibility effects (that is why a NACA 2412 is preferable to a 2416 in transonic flight).

2.6 Airfoils in supersonic and hypersonic flow

- **Supersonic, lift**: the pressure force is roughly normal to the *mean line* (in subsonic flow it was almost normal to the airstream). In a symmetric airfoil, the upper and lower surfaces contribute equally. $c_{l,max}$ decreases.

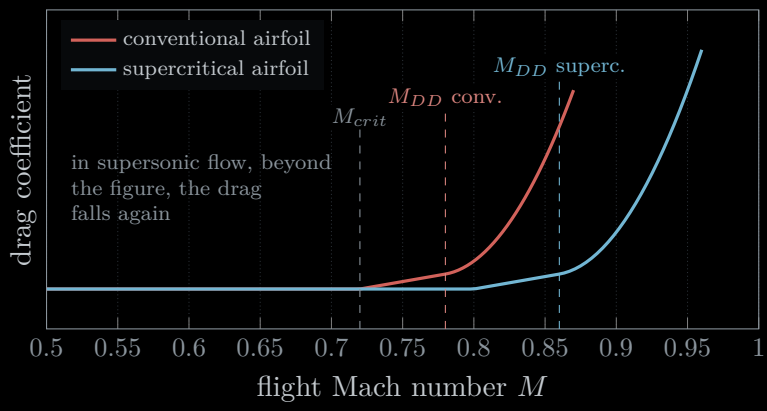


Figure 5.14. Drag rise in the transonic regime and effect of the supercritical airfoil. Qualitative sketch (slides 35 and 36).

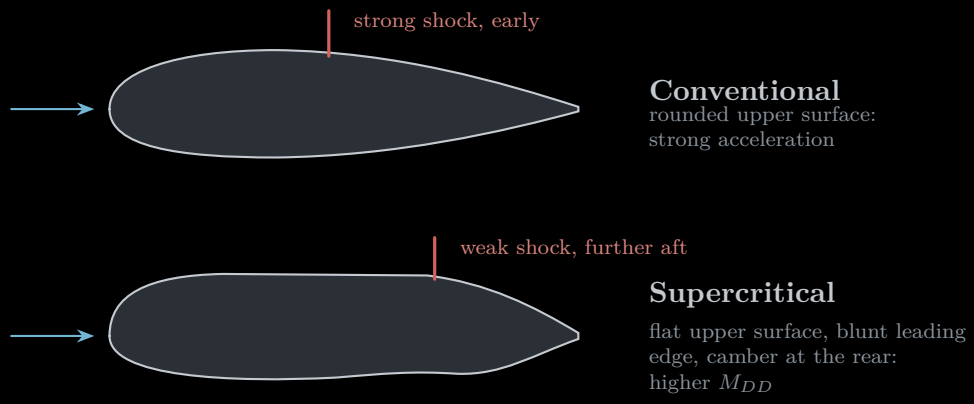
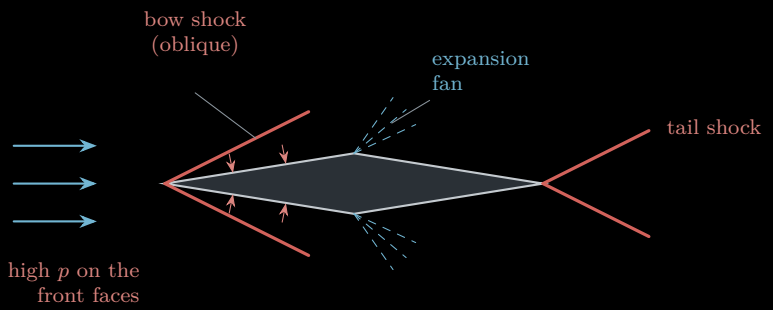


Figure 5.15. Conventional versus supercritical airfoil (schematic, own drawing): the flatter upper surface makes the shock weaker and moves it aft.

- **Supersonic, drag**: there is no d'Alembert paradox here: drag appears even in potential flow, the **wave drag** (form drag and drag due to lift; it is a potential drag). Even at zero lift there is an increase due to the shock waves. The more lift, the more drag. **A thinner airfoil reduces the form wave drag.** The origin of the wave drag is sketched in Figure 5.16.
- **Hypersonic**: the upper surface stops generating lift ("surface in shade"); c_l stops being linear with α and is very low; the drag is very high.



supersonic flow over a diamond airfoil: the pressure on the front faces is higher than on the rear faces, so there is *wave drag* even without viscosity (no d'Alembert paradox)

Figure 5.16. Supersonic flow over a diamond airfoil (schematic, own drawing).

2.7 Aerodynamic efficiency

$$AE = \frac{c_l}{c_d} \quad \text{(airfoil)} \qquad\qquad E = \frac{L}{D} = \frac{C_L}{C_D} \quad \text{(aircraft)}$$

It is the lift-to-drag ratio. If it is high, **less thrust** is needed to fly (because $T = D = W/E$ in level flight). Figure 5.17 shows how it varies with α for an airfoil.

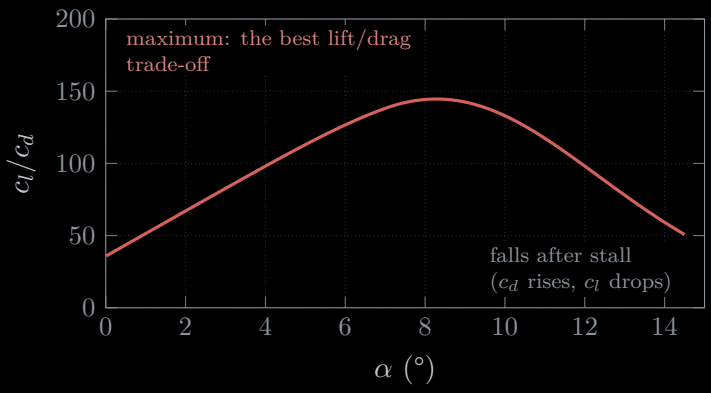


Figure 5.17. Shape of c_l/c_d versus α for a subsonic airfoil: maximum at a moderate angle, falling at stall. Qualitative sketch (slide 40).

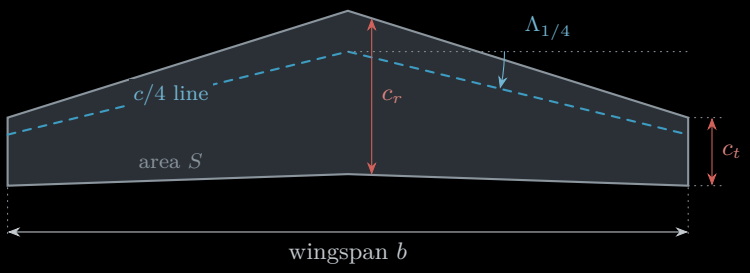
Regime	Airfoil (2D)	Aircraft/wing (3D)
Subsonic	very high, 50–100 or more	10–20 (gliders: up to 70)
Supersonic	≈ 10	≈ 10
Hypersonic	0.5–4	0.5–3

Exam value: the cruise aerodynamic efficiency of a subsonic jet airliner is **between 10 and 20** (option "20" in the test as a typical value). It is always lower in 3D than in 2D because induced drag appears.

3 Three-dimensional aerodynamics (the complete wing)

3.1 Wing geometric parameters

Only more or less conventional wings are studied. A wing is described by its position (low, mid or high) and by its **planform**, whose parameters are defined in Figure 5.18 and in the table below.



$$AR = \frac{b^2}{S} \quad \text{(aspect ratio)}$$

$$\lambda = \frac{c_t}{c_r} \quad \text{(taper ratio)}$$

$$\bar{c} = \frac{S}{b} \quad \text{(mean geometric chord)}$$

Figure 5.18. Planform parameters. Source: Unit 5, slide 43.

Parameter	Definition
Wing area S	Planform area of the wing (reference area for C_L and C_D).
Wingspan b	Distance from tip to tip.
Root chord c_r, tip chord c_t	Chords at the junction with the fuselage and at the tip.
Taper ratio λ	$\lambda = c_t/c_r$.
Aspect ratio AR	$AR = b^2/S$. Long, narrow wing = high aspect ratio.
Sweep angle $\Lambda_{1/4}$	Angle of the quarter-chord line relative to the perpendicular to the aircraft axis.
Dihedral	Inclination of the half-wings relative to the horizontal, seen from the front.
Mean geometric chord $\bar{c}$	$\bar{c} = S/b$: planform area divided by the span.
Mean aerodynamic chord (MAC)	Equivalent chord of the whole wing in aerodynamic terms (the slides refer to a separate document for computing it). **It is not the same as the geometric mean.**

Wing twist: variation of the incidence angle of the local chord along the span (**geometric twist**) or of the airfoil shape (**aerodynamic twist**) to modify the lift distribution. The usual goal is to **increase the margin to the critical angle of attack at the wingtip** so that the stall does not start there. Some wings are not trapezoidal; for example the **elliptic wing** (like the Spitfire's).

QUICK CALCULATION OF PLANFORM PARAMETERS

Trapezoidal wing of root chord $c_r = 2\,\text{m}$, tip chord $c_t = 1\,\text{m}$ and span $b = 10\,\text{m}$:

$$S = b\,\frac{c_r + c_t}{2} = 10 \cdot 1.5 = 15\,\text{m}^2 \qquad \bar{c} = \frac{S}{b} = 1.5\,\text{m}$$

$$AR = \frac{b^2}{S} = \frac{100}{15} = 6.67 \qquad \lambda = \frac{c_t}{c_r} = \frac{1}{2} = 0.5$$

(Supplement: in a trapezoid the mean aerodynamic chord is $\text{MAC} = \frac{2}{3} c_r \frac{1+\lambda+\lambda^2}{1+\lambda} = 1.56\,\text{m}$, slightly larger than $\bar{c}$.)

3.2 Subsonic: wingtip vorticity and 3D lift

The pressure difference between the lower surface (high) and the upper surface (low) favours **air recirculating around the wingtip**, from the lower to the upper surface. **Vorticity** is generated (wingtip vortices) which induces a change in the velocity field (*downwash*) in their vicinity, as Figure 5.19 shows.

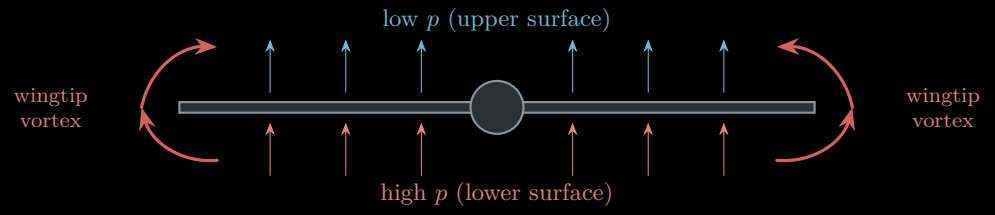


Figure 5.19. Origin of the wingtip vortices. Source: Unit 5, slide 46.

For the complete wing:

$$C_L = \frac{L}{\frac{1}{2}\rho_\infty U_\infty^2 S} \qquad C_L = C_{L,0} + C_{L,\alpha}\,\alpha \qquad C_{L,\alpha} = \frac{c_{l,\alpha}}{1 + \frac{c_{l,\alpha}}{\pi\,AR}} \text{ (long wings)}$$

Everything is the same as in 2D (linear relation with α), but:

- The **slope is lower** than that of the airfoil, and the lower AR is, the lower the slope (Figure 5.20).
- The **stall usually occurs at lower angles of attack** than for the airfoil.

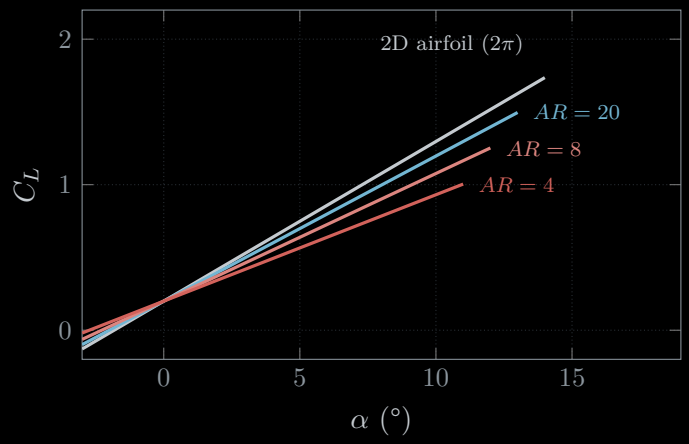


Figure 5.20. Slope of the wing C_L for several aspect ratios, with $c_{l,\alpha} = 2\pi$ (sketch; same $C_{L,0}$ for comparison). With $AR = 8$: $C_{L,\alpha} = 2\pi/(1 + 2/8) = 5.03$ rad^{-1} $= 0.088$ °$^{-1}$.

3.3 Subsonic: 3D drag

$$C_D = \frac{D}{\frac{1}{2}\rho_\infty U_\infty^2 S} \qquad C_D = C_{D,0} + C_{D,i}$$

There is **parasitic drag** $C_{D,0}$ (similar to the 2D one: friction + form) and **induced drag** $C_{D,i}$ is added, due to the vortices thrown into the wake. There is also **interference drag** between elements (wing–fuselage, etc.), which is normally added to the parasitic drag.

Parasitic drag estimation: the flat-plate analogy

The $C_{D,0}$ of the complete aircraft is estimated with the **flat-plate method**:

1. The friction coefficient C_f of a flat plate with the **same wetted area** S_{wet} (area in contact with the air) is estimated.
2. It is corrected for **3D effects** with form factors FF (from 1 to 1.35).
3. It is corrected for the **interference** between elements with interference factors FI.
4. It is "translated" into a drag coefficient by changing the reference area ($S_{wet} \to S$).

1) Flat-plate friction coefficient.

$$\text{Laminar (Blasius): } C_f = \frac{1.328}{\sqrt{Re}}$$

$$\text{Turbulent: } C_f = \frac{0.445}{(\log_{10} Re)^{2.58}\,(1 + 0.144M^2)^{0.65}}$$

(The slide writes 0.445 in the numerator; in the classical correlation from the design literature 0.455 usually appears. The slide's value is used; the difference is 2 %.) C_f **decreases as** Re **increases**

(Figure 5.21). In a turbulent layer (only there) there is a point beyond which it stops decreasing: if $Re > Re_{cutoff}$ the coefficient stays constant, and this minimum **depends on the surface finish**:

$$\text{Subsonic: } Re_{cutoff} = 38.25 \left(\frac{c}{k}\right)^{1.053}$$

$$\text{Transonic: } Re_{cutoff} = 44.62 \left(\frac{c}{k}\right)^{1.053} M^{1.16}$$

with k the equivalent surface roughness. Slide 51 gives (in feet) $k \approx 3.3 \cdot 10^{-5}$ for camouflage paint on aluminium, $2.1 \cdot 10^{-5}$ smooth paint, $1.3 \cdot 10^{-5}$ production sheet metal, $0.5 \cdot 10^{-5}$ polished sheet metal and $0.17 \cdot 10^{-5}$ smooth moulded composite; and low-Mach C_f ranges of 0.008–0.010 (propeller, fixed landing gear), 0.0045–0.007 (propeller, retractable landing gear), 0.0035–0.0045 (jet, nacelle-mounted engines) and 0.0030–0.0035 (jet, internal engines) (read from the scanned table on the slide; check them if precision is needed).

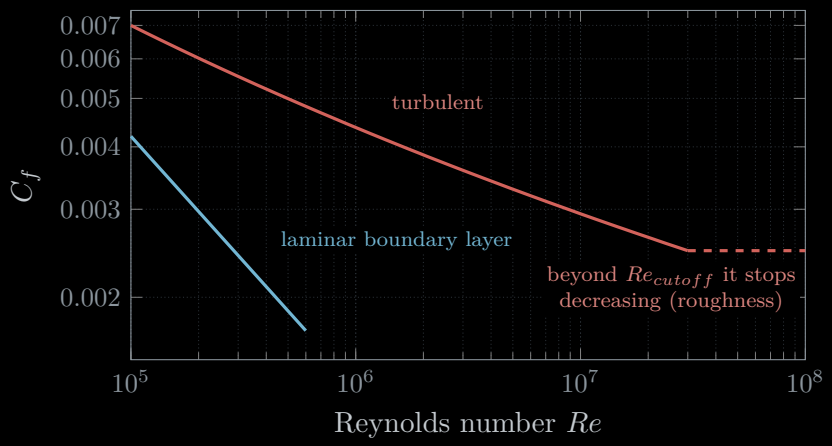


Figure 5.21. C_f of a flat plate versus Re: the laminar value is lower than the turbulent one at equal Re; in the turbulent case a plateau due to roughness is reached (the value of Re_{cutoff} here is illustrative). Source: Unit 5, slides 51 and 52.

2) Form factors.

$$FF_{fus} = \left(1 + \frac{60}{f^3} + \frac{f}{400}\right), \quad f = \frac{L}{D_{eq,max}} = \frac{L}{\sqrt{\frac{4}{\pi} A_{max}}}$$

$$FF_{wing,stab.} = \left[1 + \frac{0.6}{(x/c)_m}\frac{t}{c} + 100\left(\frac{t}{c}\right)^4\right]\left[1.34\, M^{0.18} (\cos \Lambda_m)^{0.28}\right]$$

$$FF_{engine,ext.} = 1 + \frac{0.35}{f}$$

(fuselage and cabin; wing, horizontal and vertical stabilizer, where m indicates the position of maximum thickness; engines and external elements.) **There is no need to memorize them**: what matters is to understand that they increase the flat-plate friction the thicker the element is.

3 and 4) Sum over all elements.

$$C_{D,0} = \frac{\sum_i S_{wet,i}\, C_{f,i}\, FF_i\, FI_i}{S}$$

Other additional terms may be added (protuberances, roughness, etc.).

Slide 56 also has a bar chart with the **typical cruise drag breakdown of a transport jet**: flat-plate friction (of the order of 45 %, values estimated from the chart) and induced drag (around 30 %)

dominate; the rest (profile, roughness, excrescences, interference, 3D effects and compressibility) are small contributions of a few per cent each.

ILLUSTRATIVE EXAMPLE (MADE-UP DATA FOR PRACTICE): $C_{D,0}$ OF WING + FUSELAGE

Light aircraft at $V = 60\,\text{m/s}$ at sea level ($M = 0.176$) with $S = 16\,\text{m}^2$. Wing and fuselage only, with $FI = 1$.

- **Wing**: $c = 1.6\,\text{m}$, $S_{wet} \approx 2.04S = 32.6\,\text{m}^2$, $t/c = 0.12$, $(x/c)_m = 0.3$, $\Lambda_m = 0$. $Re = \dfrac{1.225 \cdot 60 \cdot 1.6}{1.789 \cdot 10^{-5}} = 6.57 \cdot 10^6$; $C_f = \dfrac{0.445}{(6.82)^{2.58}(1 + 0.144 \cdot 0.176^2)^{0.65}} = 3.14 \cdot 10^{-3}$. $FF = [1 + 0.6/0.3 \cdot 0.12 + 100 \cdot 0.12^4] \cdot [1.34 \cdot 0.176^{0.18}] = 1.261 \cdot 0.980 = 1.236$.
- **Fuselage**: $L = 8\,\text{m}$, $A_{max} = 1.1\,\text{m}^2 \Rightarrow D_{eq} = \sqrt{4A_{max}/\pi} = 1.18\,\text{m}$, $f = 6.76$, $S_{wet} = 20\,\text{m}^2$. $Re = 3.29 \cdot 10^7$, $C_f = 2.44 \cdot 10^{-3}$, $FF = 1 + 60/6.76^3 + 6.76/400 = 1.211$.

$$C_{D,0} = \frac{32.6 \cdot 3.14 \cdot 10^{-3} \cdot 1.236 + 20 \cdot 2.44 \cdot 10^{-3} \cdot 1.211}{16} = 0.0079 + 0.0037 \approx 0.0116$$

Tail surfaces, landing gear, engine, interference… would still be missing: the real value would be considerably higher (up to ∼0.02–0.03 for a light aircraft). Friction decreases slightly with Re and the wing dominates because of its large wetted area.

Induced drag

It has been observed (and can be proven) that:

$$C_{D,i} = \frac{C_L^2}{\pi\, AR\, e} = \frac{C_L^2}{\pi\, AR}(1 + \delta)$$

where e is the **Oswald factor**: $e = 1$ for the elliptic wing (the optimal, reference case) and $e < 1$ otherwise. Consequences:

- **How can it be reduced?** By increasing the aspect ratio AR, by approaching an elliptic distribution (larger e), and with wingtip devices (they increase e).
- **It varies with C_L^2**: in cruise at higher speed less C_L is needed (because $L = W$ fixes $C_L \propto 1/V^2$), so $C_{D,i} \propto 1/V^4$, and the force $D_i \propto 1/V^2$ (the red curve in Figure 5.22). That is why it decreases as speed increases.
- The **parasitic** $C_{D,0}$ is more or less **constant with speed** (as a coefficient); the force $D_0 \propto V^2$.
- **The sum** (total drag) has a minimum: the **minimum-drag speed** V_{md}. At lower speeds the induced drag dominates; at higher speeds, the parasitic drag.

Common mistake (exam): wingtip vorticity does *not* improve efficiency; and with $e = 1$ the induced drag does *not* disappear (it is just the minimum possible for that AR and C_L).

The parabolic drag polar

The relation between C_L and C_D is called the **drag polar**. With the above, it is a parabola:

$$C_D = C_{D,0} + k\, C_L^2, \qquad k = \frac{1}{\pi\, AR\, e}$$

$$\text{Maximum efficiency when } C_{D,0} = kC_L^2: \qquad C_L^* = \sqrt{\frac{C_{D,0}}{k}}, \qquad \left(\frac{L}{D}\right)_{max} = \frac{1}{2\sqrt{k\, C_{D,0}}}$$

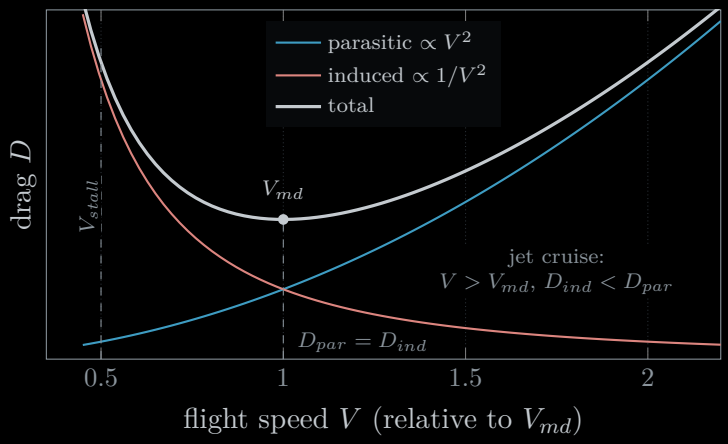


Figure 5.22. Parasitic ($\propto V^2$) and induced ($\propto 1/V^2$) drag versus speed: the sum has a minimum at V_{md} (minimum-drag speed), where both are equal. Source: Unit 5, slide 56.

(The last two, which give the tangency point of a line through the origin, are a supplement not developed in the slides, but they justify the exam question on the minimum-drag speed.) Figure 5.23 shows the polar of the example below.

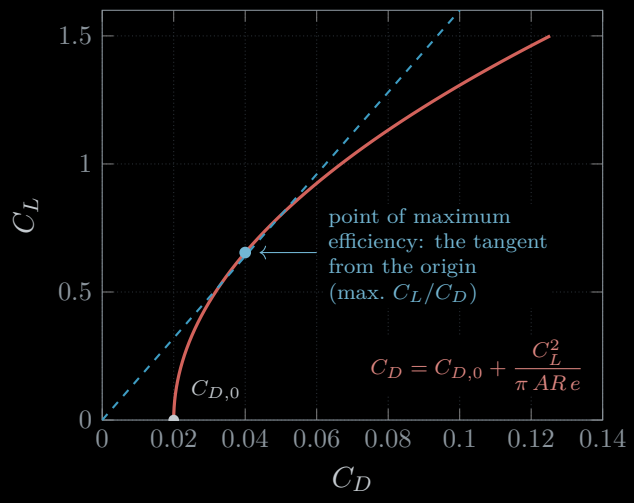


Figure 5.23. Parabolic C_L–C_D polar of the example ($C_{D,0} = 0.02$, $AR = 8$, $e = 0.85$); the tangent from the origin marks the maximum efficiency. Source: Unit 5, slide 57 (own sketch).

MAXIMUM EFFICIENCY AND MINIMUM-DRAG SPEED

Wing with $C_{D,0} = 0.02$, $AR = 8$, $e = 0.85$, $S = 16\,\mathrm{m}^2$, $m = 1000\,\mathrm{kg}$ ($W = 9810\,\mathrm{N}$):

$$k = \frac{1}{\pi \cdot 8 \cdot 0.85} = 0.0468 \qquad C_L^* = \sqrt{\frac{0.02}{0.0468}} = 0.654 \qquad \left(\frac{L}{D}\right)_{max} = \frac{1}{2\sqrt{0.0468 \cdot 0.02}} = 16.3$$

The corresponding speed comes from $L = W$:

$$V_{md} = \sqrt{\frac{2W}{\rho S C_L^*}} = \sqrt{\frac{2 \cdot 9810}{1.225 \cdot 16 \cdot 0.654}} = 39.1\,\mathrm{m/s} \approx 141\,\mathrm{km/h} \text{ (sea level)}$$

At 5 000 m ($\rho = 0.736$): $V_{md} = 50.5\,\mathrm{m/s} \approx 182\,\mathrm{km/h}$. The **maximum efficiency does not change with altitude**, but the speed at which it is achieved **does increase** (approximately $\propto 1/\sqrt{\rho}$), and also with $\sqrt{W/S}$. That is why in cruise at higher altitude one flies faster for the same C_L (a statement that appears in a test).

Stall on the wing

The 3D stall depends on the planform (Figure 5.24).

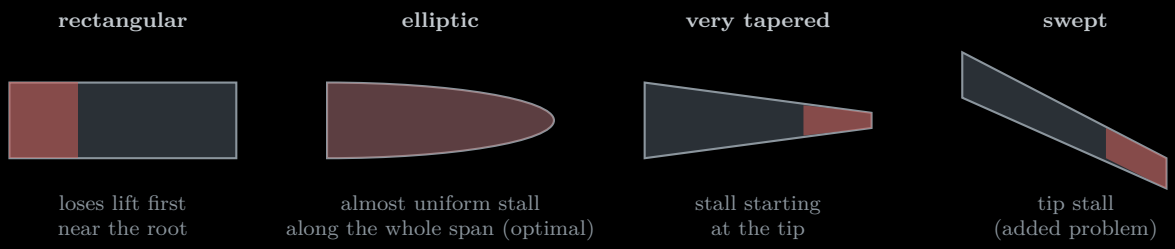


Figure 5.24. Where the stall starts depending on the planform (shaded region). Qualitative sketch based on slides 58 and 67; for the rectangular wing, the root is what the tests give.

- **Rectangular wing**: the stall starts near the **root**.
- **Elliptic wing**: uniform lift distribution; separation appears almost simultaneously along the whole span.
- **Very low taper ratio (delta type)** and swept wings: the stall starts at the **tip**, which is worse (roll control is lost).

Twist is used to avoid tip stall: with **geometric twist** the zero-lift angle of each airfoil is modified (the tip at a lower incidence, *washout*, right panel of Figure 5.26); with **aerodynamic twist** the airfoil shape (e.g. its camber) is changed along the span. A test asks which is which: *aerodynamic = changes the airfoil; geometric = changes the angle.*

3.4 Transonic: sweep delays M_{DD}

With a **swept wing** the airstream is decomposed (Figure 5.25) into a component **normal** to the leading edge, $V\cos\Lambda$, and a **tangential** one, $V\sin\Lambda$. Only the normal one accelerates the flow over the airfoil and causes compressibility effects: the wing "feels" a lower Mach number $M\cos\Lambda$. Therefore the velocity normal to a transonic shock wave is lower than the incident one and the M_{DD} **is delayed**.

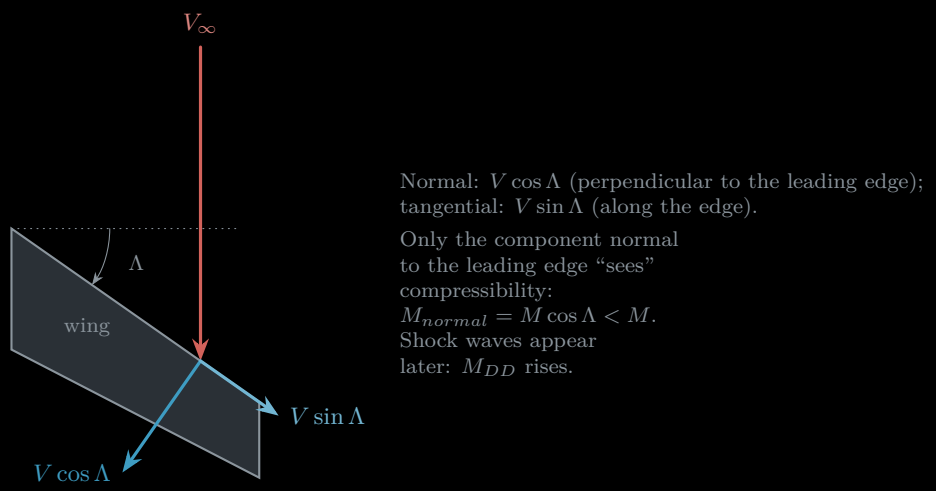


Figure 5.25. Decomposition of the velocity on a swept wing. Source: Unit 5, slides 59 and 60.

The rule $M_n = M\cos\Lambda$ is the standard model of "simple sweep theory" (slide 60 only gives sweep angles and curves, not formulas). If $M = 0.8$ and $\Lambda = 30°$: $M_n = 0.8\cos 30° = 0.69$. Slide 60 shows that as the sweep increases (0°, 10.5°, 40°, 49.25°) the transonic drag peak decreases and shifts to

a higher Mach number. Drawbacks: more structural weight, risk of tip stall, it may destabilize the aircraft and it complicates the landing gear (slide 68).

3.5 Supersonic and 3D efficiency

- **The wingtip recirculation disappears: there is no induced drag**.
- As in 2D, there is **wave drag** (form drag and drag due to lift).
- If the **sweep** is high enough: leading-edge suction appears, and the drag decreases (the photo on the slide is the Concorde).
- 3D efficiency: **lower than that of the airfoil**: subsonic 10–20 (gliders up to 70), supersonic ≈ 10, hypersonic 0.5–3.

3.6 How the wing design parameters influence it

There is no universally optimal wing: **the optimum depends on the mission**.

Parameter	High / large value	Low / small value
Area S	More space for fuel; worse behaviour in **gusts**	Low weight and structural loads; **higher stall speed** (and longer runway)
Aspect ratio AR	**Reduces induced drag**; increases parasitic drag	More fuel in the wing; easier to store the landing gear; better against **flutter** and more manoeuvrability; lower wave drag
Thickness t/c	More stiffness, more fuel volume, **higher** $c_{l,max}$ at low speed	Lower compressibility drag
Taper ratio λ	High ($\approx$ rectangular): easier to manufacture and maintain. Moderate ($\approx$ elliptic): less drag	Low ($\approx$ delta): stall from the tip, **lower structural weight**, easier to store landing gear and fuel, harder to manufacture
Sweep Λ	Raises M_{DD} and lowers drag; generates leading-edge suction in supersonic flow	Structurally simpler; (with high sweep: more weight, tip stall, may destabilize, difficult landing gear)

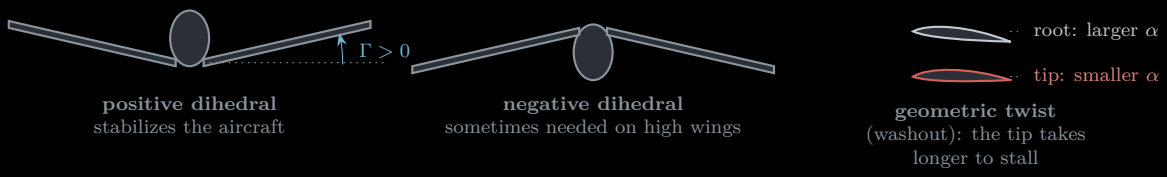


Figure 5.26. Dihedral and geometric twist. Source: Unit 5, slides 63 and 69.

Dihedral (Figure 5.26, left and centre): if **positive**, it tends to **stabilize** the aircraft (roll corrects itself); on high wings **negative dihedral** may be needed. The **typical taper ratio** of a subsonic passenger transport is **moderate** (in one test, the correct option was "around 0.6").

4 Movable surfaces, high-lift devices and other elements

4.1 High-lift devices

The minimum flight speed is linked to the stall:

$$U_{stall} = \sqrt{\frac{W}{\frac{1}{2}\rho_\infty S\, C_{L,max}}} = \sqrt{\frac{2W}{\rho_\infty S\, C_{L,max}}}$$

In practice one flies somewhat above it, as a margin for manoeuvring. To lower U_{stall} one must **raise** $C_{L,max}$; but a wing with an extreme C_L is not optimized for cruise. Solution: **temporarily** modify the wing with **high-lift devices**: **passive** (they just move surfaces) or **active** (they use the engine jet, suck the boundary layer…, with better results and higher cost). Figure 5.27 shows two of the most widely used passive devices.

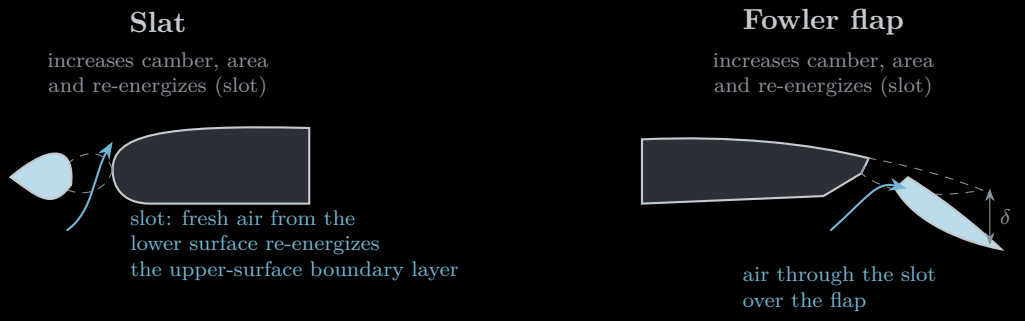


Figure 5.27. Two of the most widely used high-lift devices, schematic. Source: Unit 5, slides 72 to 75.

What each one does (table from slides 72 and 74). The columns are: "increases camber", "increases chord (area)" and "**Slot**" = re-energizes the boundary layer through the slot effect:

Position	Device	Camber	Area	Slot
Leading edge	Slot			X
	Leading-edge flap	X		
	Slat	X	X	X
	Krueger flap	X	X	
Trailing edge	Simple flap	X		
	Simple flap (slotted)	X		X
	Fowler flap (slotted; double or triple slotted)	X	X	X
	Lower-surface flap	X		

The devices not drawn above (slot, leading-edge flap, Krueger, simple, slotted and lower-surface flaps) are shown in Figure 5.28.

Their effect on the lift curve is sketched in Figure 5.29.

- They are used in **take-off and landing**; in landing they are deployed further. They are *not* used in cruise.
- They cost: **weight** and **high manufacturing and maintenance costs**; they also **generate drag** (hence larger deflections are used for landing, where the drag is welcome).
- Improvement of $C_{L,max}$: from **50 %** (simple flaps or slats) to **100 %** (double-slotted Fowler).

Frequent test mistakes: the Krueger flap *increases camber and* $C_{L,max}$; slots carry air from the lower to the upper surface (not the other way round) to delay separation; simple flaps increase camber but *not* the critical angle; the Fowler is a **trailing-edge** device and increases camber *and* area; the lower-surface flap only increases camber, not area.

4.2 Other movable surfaces

Figure 5.30 shows where the control and lifting surfaces are located; the table below gives the function of each.

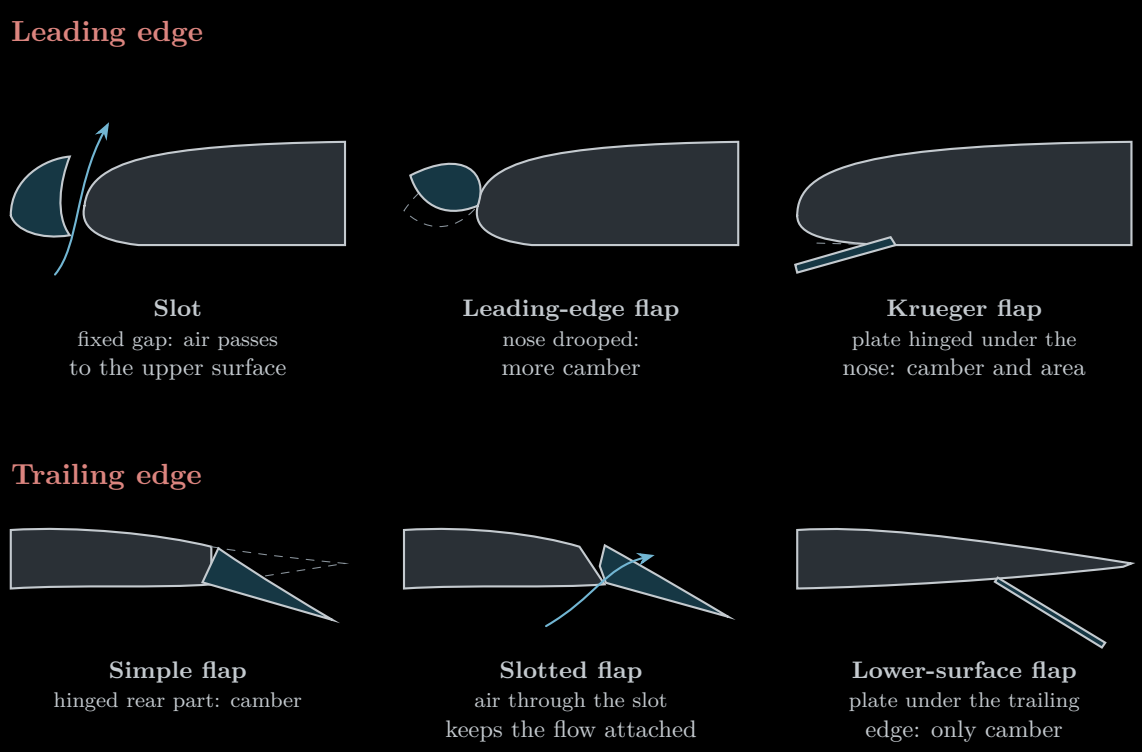


Figure 5.28. The remaining leading-edge and trailing-edge devices of the table (schematic, own drawing; dashed: stowed position).

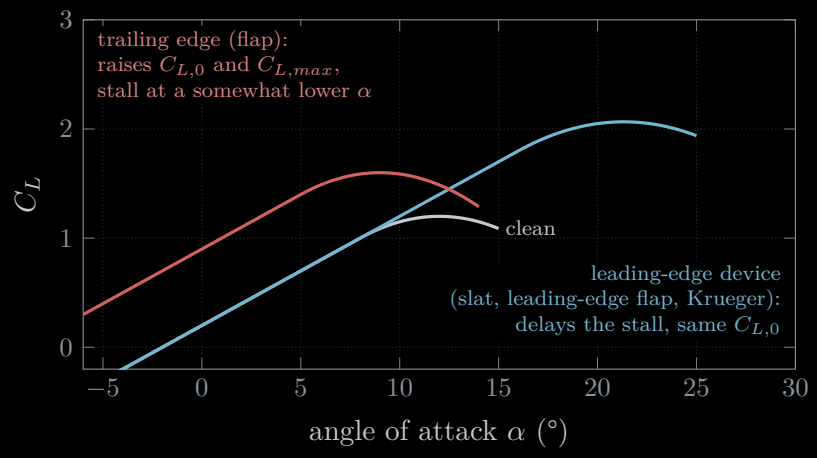


Figure 5.29. Effect on the C_L–α curve (qualitative sketch): leading-edge devices *delay the stall* (extend the α range); trailing-edge devices *raise the curve* and shift the stall to a lower α. Source: Unit 5, slides 73 and 75.

Surface	Function
Ailerons	Control the **roll** moment; close to the tip (at low speed) or close to the root (at high speed). They deflect in opposite directions.
Elevator	Controls the **pitch** moment.
Rudder	Controls the **yaw** moment.
Trim tabs	Set the neutral position of the controls (the stick).
Spoilers	Destroy lift (*dump lift*) and brake; if they act as ailerons they are called *spoilerons*.

Other elements: **wingtip devices** (winglets, Hoerner wingtips, wingtip fences…), which also **increase the Oswald factor** (they reduce induced drag), and **boundary-layer control**: strakes or LEX, dogtooth, wing fences. Wingtip devices are drawn in Figure 5.31 and boundary-layer control elements in Figure 5.32.

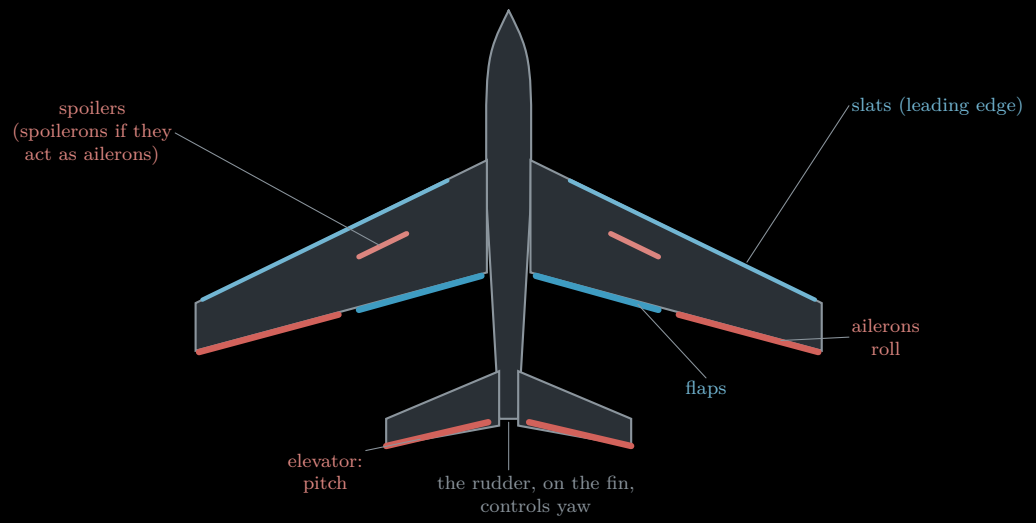
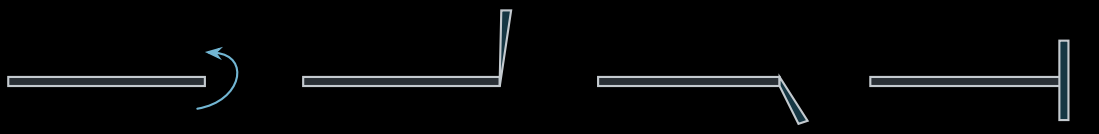


Figure 5.30. Control and lifting surfaces. Source: Unit 5, slides 77 to 79.



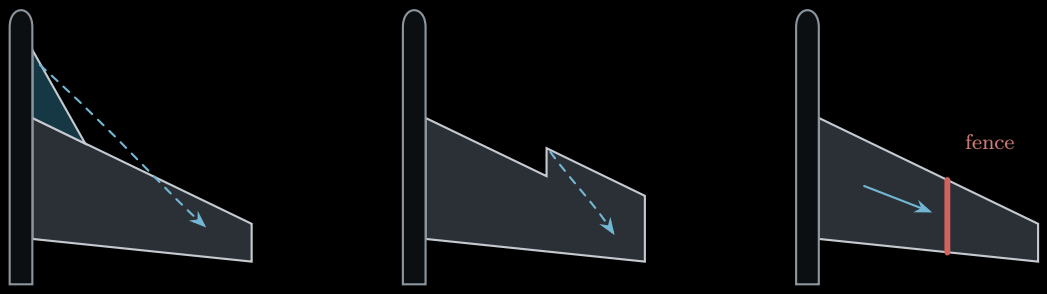
Plain tip
air curls from the lower to the upper surface

Winglet
small vertical surface at the tip

Hoerner tip
tip bent downwards

Tip fence
plate above and below the tip

Figure 5.31. Wingtip devices, front view of the tip (schematic, own drawing).



Strake / LEX
fixed vortex over the wing: flow stays attached

Dogtooth
notch in the leading edge sheds a vortex

Wing fence
blocks the spanwise flow on the swept wing

Figure 5.32. Boundary-layer control elements, plan view of half an aircraft (schematic, own drawing).

5 Solved exam-type problems

Past exams are 4-option tests; the "calculations" are quick and are solved with proportions. Here are the recurring ones.

1. STALL SPEED WHEN $C_{L,max}$ IS DOUBLED (2023/24 EXAM, QUESTION 16)
An aircraft has $V_{stall} = 280\,\text{km/h}$ in the clean configuration. What if high-lift devices that **double** $C_{L,max}$ are deployed?

$$V_{stall} = \sqrt{\frac{2W}{\rho S C_{L,max}}} \propto \frac{1}{\sqrt{C_{L,max}}} \quad \Rightarrow \quad V'_{stall} = \frac{280}{\sqrt{2}} = 198\,\text{km/h} \approx 200\,\text{km/h}$$

Frequent mistake: the option "140" comes from dividing by 2 (wrong: the speed is not proportional to $C_{L,max}$ but to its square root).

2. CLIMBING AT CONSTANT SPEED: WHAT DOES α DO? (2023/24 EXAM, QUESTION 4)
Solved in Unit 2 (Section 1.1). Summary: in nearly level flight $C_L = 2W/(\rho V^2 S)$; when climbing, ρ decreases with V, W and S constant, so C_L increases and, since $C_L = C_{L,0} + C_{L,\alpha}\alpha$ grows with α, α **increases during the 2 km** (option b).

3. BETTER FUEL CONSUMPTION BUT WORSE AERODYNAMIC EFFICIENCY (2023/24 EXAM, QUESTION 8)
Solved in Section 6.2. Summary: with R and V fixed, the Breguet equation requires $\ln(W_i/W_f) \propto c/(L/D)$, which increases by a factor $0.97/0.95 = 1.021$ (the consumption c is defined in Unit 6). Slightly more fuel is needed and, with the same payload and empty mass, the **take-off mass increases** (option c); the payload cannot increase because the design mission is fixed.

4. REQUIRED THRUST FROM THE EFFICIENCY
In level cruise $T = D = W/E$. For a 70 t aircraft with $E = 17$: $T = \dfrac{70\,000 \cdot 9.81}{17} \approx 40\,\text{kN}$ in total. If the efficiency rises to 20, $T \approx 34.3\,\text{kN}$: 15 % less thrust, and therefore less fuel. Hence the interest in increasing L/D.

5. CRUISE ABOVE THE MINIMUM-DRAG SPEED (2023/24 EXAM, QUESTION 14)
If the jet flies at $V > V_{md}$, to the right of the minimum of the drag curve (Figure 5.22): the **parasitic** drag ($\propto V^2$) **exceeds** the induced drag ($\propto 1/V^2$). Answer: **the induced drag is lower than the parasitic drag**. At V_{md} they are equal; below it the induced drag dominates.

6. WHERE IS THE CENTRE OF PRESSURE? (QUESTION REPEATED IN 2023/24 AND 2024/25)
Airfoil with positive camber and α increasing from the zero-lift value. $x_{cp}/c = \frac{1}{4} - c_{m,ac}/c_l$ with $c_{m,ac} < 0$: when $c_l \to 0^+$, $x_{cp} \to +\infty$ (downstream); as c_l grows it approaches $c/4$ from behind. Answer: **it moves from infinity downstream towards the aerodynamic centre**.

6 Summary and what usually appears on the exam

WHAT USUALLY APPEARS ON THE EXAM (UNIT 5)
This is **the unit that contributes the most questions**: in past midterms, **a large share of the 20 questions** are about aerodynamics and wings (first midterm) and the data recur almost every year. Everything is multiple-choice; the "problems" are proportionalities.

- **Airfoils**: decoding a NACA code (2412, 0012, 21518…); which has the highest $c_{l,max}$ (cambered and thicker: 2414); 0012 is used for aerobatics and stabilizers (symmetric), 2412 has $c_{l,0} > 0$; **the region that lifts most is the upper surface near the leading edge**; **CP movement** (from infinity downstream to the AC); $c_{l,max}$ of the lab's NACA 0015 ≈ 1 (near 14°); c_l of a symmetric airfoil at 4° ≈ 0.4, not 0.
- **Boundary layer and Re**: viscous effects are more noticeable at *low* Re; layers are thin at large Re; separation depends on the adverse gradient and on viscosity and can be influenced by compressibility; potential flow does not represent drag (d'Alembert).

- **Transonic**: $M_{crit} < M_{DD}$; M_{DD} is reached when the shock separates the boundary layer; supercritical airfoils raise M_{DD}; sweep and low thickness delay the effects.
- **Drag**: induced drag is *only* 3D; $C_{D,i} = C_L^2/(\pi AR\,e)$; at V_{md} parasitic = induced; $V > V_{md} \Rightarrow$ induced smaller; the minimum-drag speed grows with altitude and with $\sqrt{W/S}$.
- **Efficiency**: transport aircraft 10–20 (option "20" as the typical value, not 0.2 nor 200); gliders up to 70; hypersonic < 4.
- **High-lift devices**: table of effects; Krueger; Fowler = trailing edge + camber + area; $V_{stall} \propto 1/\sqrt{C_{L,max}}$ (280 km/h $\rightarrow$ 198); used in take-off and landing.
- **Wing parameters**: aerodynamic twist = varying airfoil; geometric = angle; high aspect ratio $\rightarrow$ less induced but more parasitic drag; low taper ratio = less structural weight and easier to store landing gear and fuel, but tip stall; typical taper ratio on a transport ~ 0.6; positive dihedral stabilizes.
- **Surfaces**: the **aileron controls roll**, the elevator pitch, the rudder yaw.

Essentials: $L = \frac{1}{2}\rho V^2 S C_L$; $c_{l,\alpha} \approx 2\pi$; $C_{L,\alpha} = c_{l,\alpha}/(1+c_{l,\alpha}/\pi AR)$; $C_D = C_{D,0}+C_L^2/(\pi AR\,e)$; $V_{stall} = \sqrt{2W/(\rho S C_{L,max})}$; $AR = b^2/S$; $Re = \rho U c/\mu$; $M = U/a$.

DOUBTS AND OPEN POINTS (UNIT 5)

- Slide 51 writes 0.445 in the turbulent C_f (the literature has 0.455) and the point-source potential with a minus sign; these have been corrected/flagged in the text.
- In one of the students' compilations (2018/19) the official key for "which aircraft has the highest cruise aerodynamic efficiency" marks the B787 over the Orlican VT-16 glider; the slides say gliders reach 70, so **that key is not trusted**.
- The values of k and of the C_f ranges on slide 51 and the percentages in the chart on slide 56 were read from scanned images: the order of magnitude is right, exact figures are not guaranteed.

UNIT 6

Propulsive systems

1 Introduction: what an aircraft engine does and how it is measured

To fly in level flight, the aerodynamic drag must be overcome (Unit 5): that is the job of **thrust**. Propulsive systems fall into two groups (Figure 6.1):

- **Air-breathing** (non-autonomous): **they use the atmosphere** (air is the working mass). These are the engines of aircraft and most drones.
- **Non-air-breathing** (autonomous): they carry **their own reaction mass** (and oxidizer), e.g. missiles and rockets.

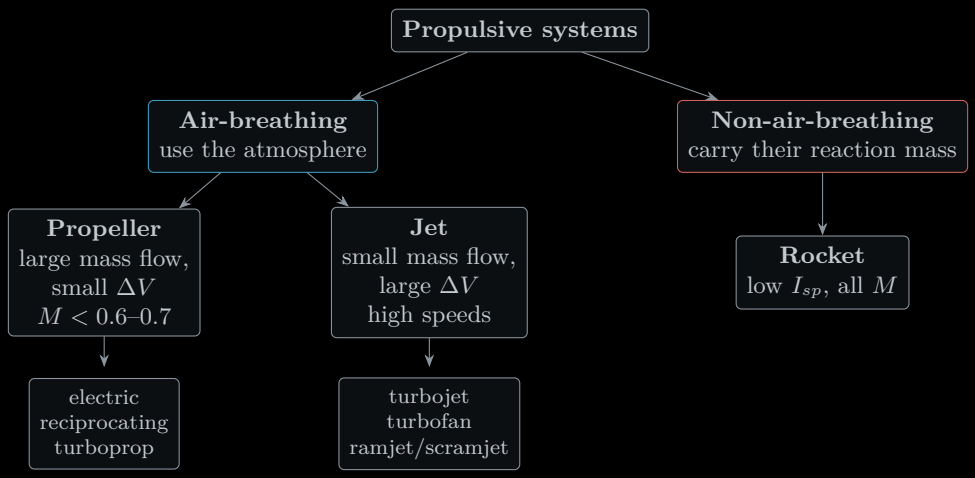


Figure 6.1. Classification of the propulsive systems studied in this unit. Source: T6, slides 3, 7 and 21.

1.1 Operating principle

The principle of air-breathing systems has two steps: (1) the system **increases the kinetic energy and momentum** of a flow of air taken from the atmosphere; (2) as a **reaction**, the atmosphere "pushes" the system (action and reaction, like a boat pushing water backwards). With the mass flow rate $\dot{m}$ (kg/s) through the engine:

$$T = \dot{m}\,V_{out} - \dot{m}\,V_{in} + p\,A_{out} - p\,A_{in} \;\approx\; \dot{m}\,(V_{out} - V_{in}) = \dot{m}\,\Delta V$$

The last two terms are the pressure thrust, which is neglected to a first approximation (it can be important in jets). Hence **the exit velocity must be greater than the inlet velocity**, $V_{out} > V_{in}$. There are two ways to obtain the same thrust (Figure 6.2):

- **a large mass flow** $\dot{m}$ with a **small** velocity increase ΔV (propeller, fan);
- **a small mass flow** with a **large** ΔV (jet).

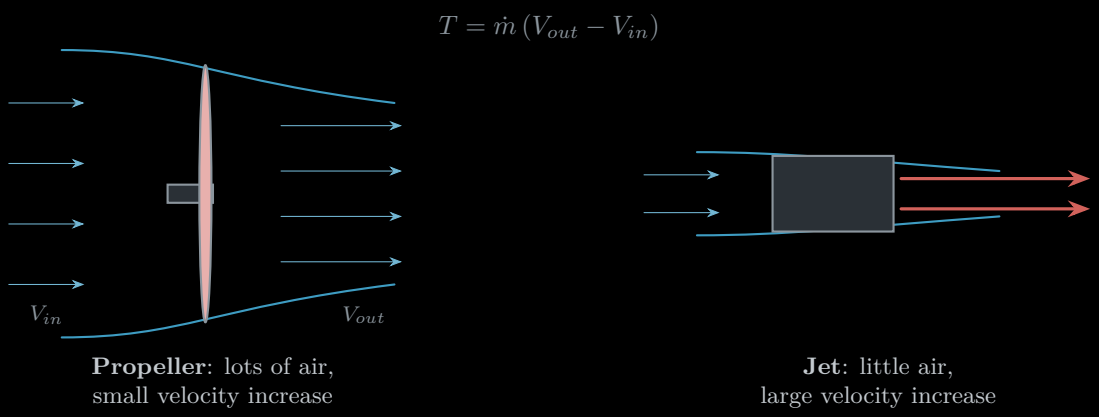


Figure 6.2. The same thrust is obtained by accelerating a lot of air a little (propeller) or a little air a lot (jet). Source: T6, slides 4 and 5.

1.2 Propulsive efficiency

Neglecting the enthalpy change, the **propulsive efficiency** η_p compares the useful power ($T \cdot V_\infty$: what actually pushes the aircraft) with the power given to the air (the increase in kinetic energy of the flow):

$$\eta_p = \frac{T \cdot V_\infty}{\frac{1}{2}\left[(\dot{m}V^2)_{out} - (\dot{m}V^2)_{in}\right]} \simeq \frac{2}{1 + \dfrac{V_{out}}{V_{in}}}$$

If $V_{out} \to V_{in}$, $\eta_p \to 1$; if V_{out} is much greater, $\eta_p \to 0$. Intuition: the kinetic energy left in the wake is energy *lost*: for the same thrust, "**a little velocity to a lot of air**" leaves less energy in the wake than "**a lot of velocity to a little air**". Figure 6.3 plots the formula: the efficiency falls steadily as V_{out}/V_{in} grows.

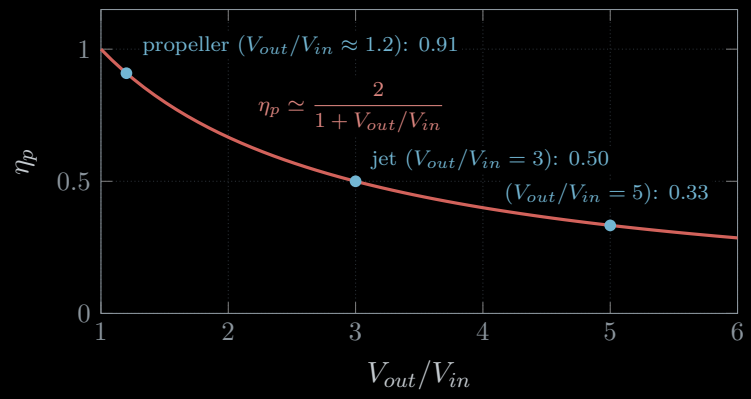


Figure 6.3. Propulsive efficiency versus V_{out}/V_{in} (from the formula on slide 6; the dots are examples). Source: T6, slides 5 and 6.

KEY IDEA

Higher propulsive efficiency = smaller velocity change with a larger mass flow: less kinetic energy is given to the atmosphere for the same thrust. Hence:

- **Propeller**: large mass flow, small ΔV, almost constant pressure between inlet and outlet. Very efficient at low speed (below Mach 0.8 and normally below 0.6), **impossible at high speed**.
- **Jet**: small mass flow, large ΔV, the pressure change can be large. Suitable for high-speed flight.

1.3 Figures of merit: specific fuel consumption and specific impulse

Two measures are used to compare engines ($\dot{m}_{fuel}$ is the fuel or propellant flow rate):

$$\text{BSFC} = \frac{\dot{m}_{fuel}}{P} \text{ (propeller)} \qquad \text{TSFC} = \frac{\dot{m}_{fuel}}{T} \text{ (jet)} \qquad I_{sp} = \frac{T}{\dot{m}_{fuel}\, g}$$

- **BSFC** (*Brake Specific Fuel Consumption*): fuel burned per unit of **shaft power** P (propeller); typical units g/kWh or kg/(W·s).
- **TSFC** (*Thrust Specific Fuel Consumption*): fuel per unit of **thrust** (jet); typical units kg/(N·s) or lb/(lbf·h).
- **Specific impulse** I_{sp}: thrust divided by the fuel weight flow rate. **Depending on whether g is included, it has different units**: with g it comes out in **seconds**; without g it comes out in m/s.

High specific consumptions are bad (a lot of fuel for the same power or thrust); by contrast **a high I_{sp} is good**: $I_{sp} = 1/(\text{TSFC} \cdot g)$. *Common mistake*: thinking that "more is better" for both. Link with Unit 3: the Breguet specific consumption, referred to weight, is $c_{sp} = \text{TSFC} \cdot g = 1/I_{sp}$ (in 1/s); with $\text{TSFC} = 1.70 \cdot 10^{-5}$ kg/(N s) it gives $1.67 \cdot 10^{-4}$ s^{-1}, the value in the Breguet example.

CONVERTING TSFC TO I_{sp} AND BSFC TO EFFICIENCY (ILLUSTRATIVE EXAMPLE)
A turbofan with $\text{TSFC} = 0.6\,\text{lb/(lbf} \cdot \text{h)}$. Since $1\ \text{lb/(lbf} \cdot \text{h)} = 2.83 \cdot 10^{-5}$ kg/(N · s):

$$\text{TSFC} = 1.70 \cdot 10^{-5}\ \text{kg/(N} \cdot \text{s)} \ \Rightarrow\ I_{sp} = \frac{1}{1.70 \cdot 10^{-5} \cdot 9.81} \approx 6\,000\ \text{s}$$

(of the order of the Boeing 747 turbofan marked on the chart on slide 23).
A reciprocating engine with BSFC = 270 g/kWh = $7.5 \cdot 10^{-8}$ kg/J and a fuel of 42 MJ/kg (the slide 23 value for hydrocarbons) has an efficiency

$$\eta = \frac{1}{\text{BSFC} \cdot \text{LHV}} = \frac{1}{7.5 \cdot 10^{-8} \cdot 42 \cdot 10^{6}} \approx 0.32$$

which agrees with the 0.30–0.35 in the table of Unit 1 of the UAS Propulsion notes. (Both numbers are examples; no slide gives a specific BSFC.)

2 Propeller propulsion

The **propeller** works as a **small rotating wing**: it pushes the air mass backwards, increases its velocity and obtains thrust as a consequence. The blades can be turned by an **electric** motor, a **reciprocating** engine or a **turbine**. The blade tips operate in the transonic regime, and this **sets the maximum flight speed with a propeller (up to M 0.7)**; at higher speeds complex 3D geometries are needed (the A400M).

- **Electric motor** (battery or fuel cell → electric motor → propeller): a high-efficiency conversion, but the problem is **storing the electrical energy**. The specific energy of batteries is very low: **1 kg of fuel contains the same energy as 70–80 kg of LiPo batteries**, which would give an unacceptable weight in large aircraft. It is used in **small unmanned aircraft**.
- **Reciprocating engine**: similar to automotive engines. Most are **spark-ignition** (gasoline); some are **compression-ignition** (diesel). The fuel energy is released by burning it in a **variable-geometry combustion chamber** (the cylinder with its piston) and is transmitted

to the propeller through a **connecting-rod mechanism** (Figure 6.4). Typical of ultralights and general aviation.

- **Turboprop**: **continuous-flow** combustion with a kerosene-derived fuel (typically **Jet A-1**). A turbine extracts part of the energy to drive the propeller; the same turbine also drives the **compressor**, which draws in the air and pressurizes it for efficient combustion (not all of the turbine energy reaches the propeller). It may be less efficient than a reciprocating engine, but it is **lighter**. Typical of regional aircraft (Figure 6.5).

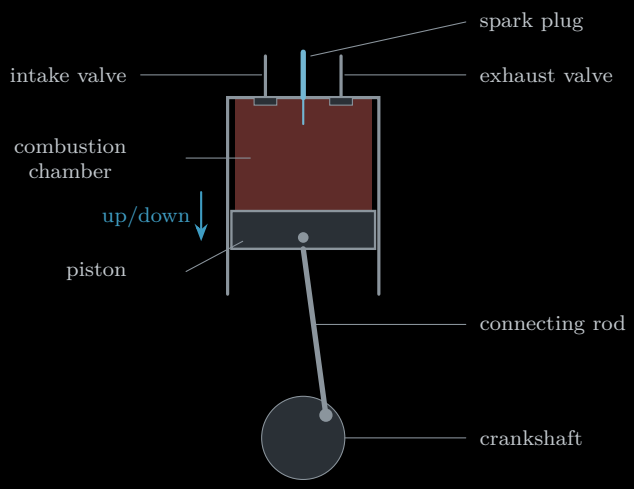


Figure 6.4. Reciprocating engine cylinder (one cylinder, four-stroke, schematic). The burning gas pushes the piston down; the connecting rod and the crank turn that linear motion into the rotation that drives the propeller.

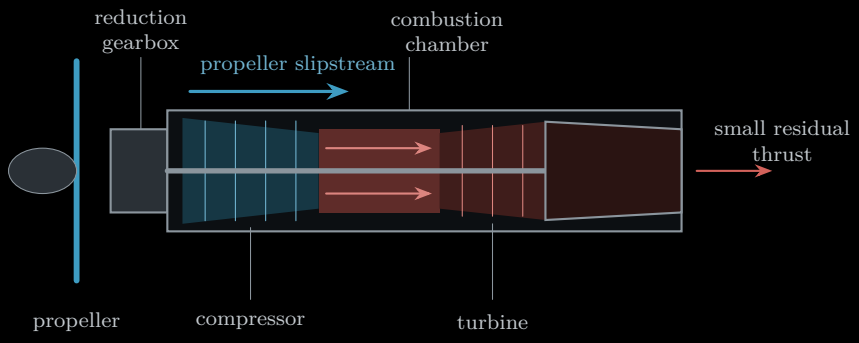


Figure 6.5. Turboprop: the turbine drives both the compressor and, through a reduction gearbox, the propeller, which produces most of the thrust. Schematic.

Common mistake: in a turboprop it is *not* the pressure difference between engine inlet and outlet that gives the thrust; the thrust comes from the propeller, which accelerates a lot of air (the exhaust jet contributes little).

2.1 Propeller coefficients (laboratory supplement)

This is not in these slides (it comes from the lab sessions and is asked in tests), but it helps in understanding the questions. For a propeller of diameter D rotating at n revolutions per second with advance (forward) speed V:

$$J = \frac{V}{nD} \qquad C_T = \frac{T}{\rho n^2 D^4} \qquad C_P = \frac{P}{\rho n^3 D^5} \qquad \eta = \frac{J\,C_T}{C_P}$$

J is the **advance ratio** (how far the aircraft advances per revolution, relative to the diameter). With a **fixed-pitch** propeller turning at constant rpm, **as flight speed increases J grows and**

C_T **always *decreases*** (each blade sees the air at a smaller angle of attack): **the thrust drops**. The efficiency, in contrast, first *rises* and then falls to zero when the thrust vanishes; it is not monotonic, so "the efficiency always increases/decreases" is false (Figure 6.6).

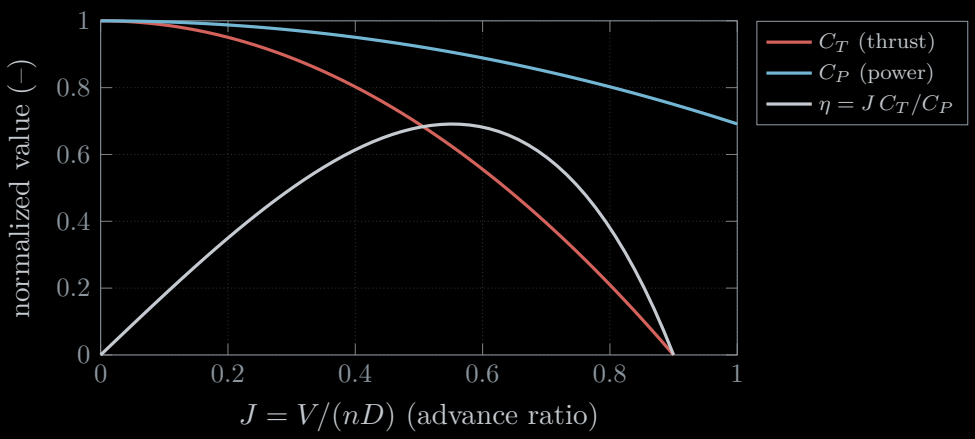


Figure 6.6. Typical curves of a fixed-pitch propeller (schematic, no real values; C_T and C_P are normalized to their value at $J = 0$).

WHICH ENGINE FOR WHICH AIRCRAFT (TEST QUESTIONS)

- Small UAS (5 kg payload, 30 km): **electric**.
- 500 km ultralight: **reciprocating engine + propeller**.
- Regional aircraft (Valencia–Mallorca): **turboprop**.
- High, fast commercial aircraft (M 0.8): **turbofan**.
- Mach 3 at 10 km altitude: the one with the highest I_{sp} is the **ramjet**.

3 Jet propulsion

When the aircraft flies so fast that the propeller suffers transonic effects, its efficiency drops rapidly. The flow can then be accelerated **internally** by means of a **thermodynamic cycle**: air is compressed, energy is added by burning fuel, and the gas is expanded. Figure 6.7 shows the elements of the simplest such engine, the turbojet, numbered as in the list below.

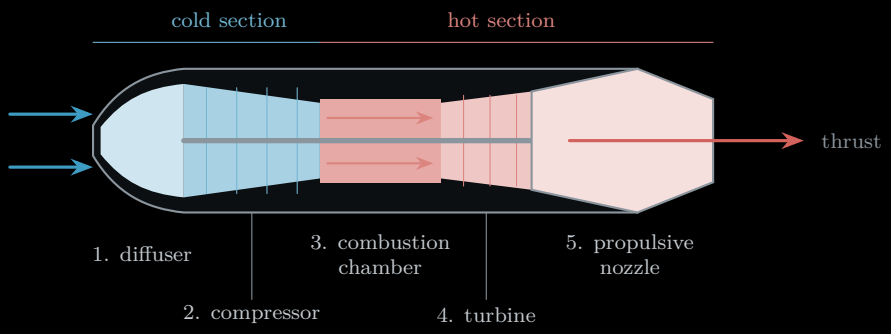


Figure 6.7. Elements of a turbojet. The air enters through the diffuser, is compressed, is heated in the combustion chamber, drives the turbine (which drives the compressor), and the remaining energy is converted into velocity in the nozzle. The blue (cold) part is upstream of the combustion chamber and the red (hot) part is downstream. Source: T6, slide 15.

1. **Diffuser**: decelerates the incoming air and raises the pressure.

2. **Compressor**: raises the pressure much further.
3. **Combustion chamber**: the fuel is burned and the temperature rises.
4. **Turbine**: extracts just enough energy to drive the compressor.
5. **Propulsive nozzle**: accelerates the flow (extracting the energy released in combustion) to high velocities.

A remark from slide 16: the propulsive nozzle accelerates the flow, but *the thrust is produced by "pushing" on the compressor blades* (that is, the force is transmitted to the engine and from there to the aircraft). And in a turbofan, **the element that increases the kinetic energy and momentum of the air is the compressor rotor** (not the stator, nor the turbine) (test question).

3.1 Nozzle

The propulsive nozzle can have two basic shapes, depending on the flight regime (Figure 6.8).

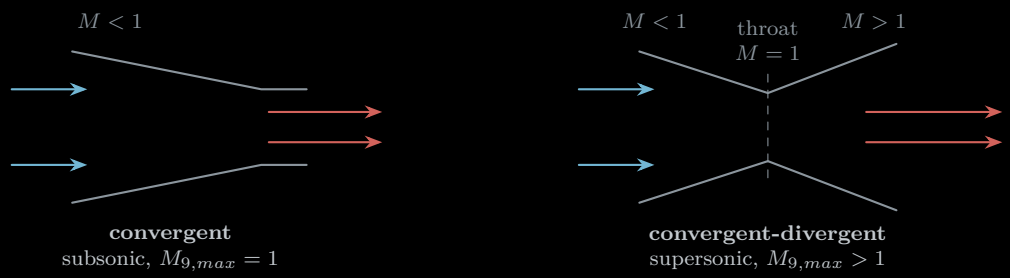


Figure 6.8. Convergent and convergent-divergent nozzles. Source: T6, slide 16.

- **Convergent**: **subsonic** case. The flow cannot exceed $M = 1$ at the exit.
- **Convergent-divergent**: **supersonic** case; the flow reaches $M = 1$ at the throat and accelerates to $M > 1$ in the divergent part.

(The subscript 9 denotes the nozzle exit: V_9 and M_9 are the exit velocity and Mach number. The detailed calculation of V_9 and of the adapted nozzle belongs to other subjects.)

3.2 Turbofan, turbojet and afterburner

The **turbofan** is something in between a turboprop and a turbojet: the air first goes through the first row of compressor blades, the **fan** (somewhat like a propeller), and **one part** goes through the engine **core** while **the rest bypasses** the core (Figure 6.9). High efficiencies are achieved with a high **bypass ratio** (more flow around the core, less through it): this is again the "a lot of air, small ΔV" principle.

When flying above the speed of sound, **the fan becomes useless** (low efficiency in supersonic flow, little thrust for its size). Only the core is then kept: the **turbojet**, or in some cases a low-bypass turbofan. The turbojet is **less efficient**, but it was developed before the turboprop and the turbofan.

To obtain extra thrust for a limited time (military applications only) an **afterburner** (*reheat*) is used: it burns more fuel after the turbine to raise the thrust, but it **considerably reduces the efficiency while in use**.

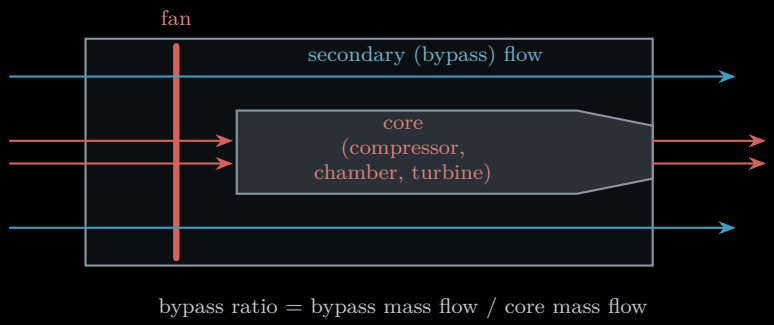


Figure 6.9. Turbofan: two streams, the core stream (red) and the bypass stream (blue). Source: T6, slide 17.

3.3 Ramjet and scramjet

At supersonic speed the intake air can be compressed **without moving parts**, using the shock waves in the diffuser (Figure 6.10). Such an engine consists only of a **diffuser, a combustion chamber and a nozzle**. It is very simple and its efficiency is low, but it produces thrust at higher speeds than the turbojet.

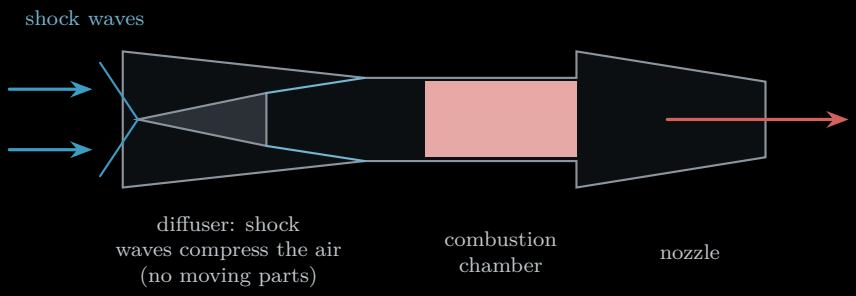


Figure 6.10. Ramjet: the diffuser compresses with shock waves; no compressor or turbine is needed. Source: T6, slides 19 and 20.

- **Ramjet**: combustion takes place at **subsonic** speed.
- **Scramjet** (supersonic-combustion ramjet): combustion takes place in **supersonic** flow.

A ramjet **has no turbine** (it does not need a starter motor to spin up its turbines; this error appears in tests) but **it cannot operate at rest**: it must be launched at supersonic speed by another system; this is why a missile such as the **Meteor** needs a **rocket** (*booster*) to reach the speed at which the ramjet ignites.

3.4 Rocket

A **rocket** is a **non-air-breathing** system: it uses the atmosphere neither to react with the fuel nor as reaction mass. It has a **low specific impulse** because it carries **all of the reaction mass and the oxidizer**, not just the fuel. It is suitable **for the whole Mach range**, from almost 0 to high values.

4 Summary: which system for which speed

Figure 6.11 places each engine type on a specific impulse versus Mach chart (the higher the I_{sp}, the less fuel for the same thrust); the table below gives the same information in words.

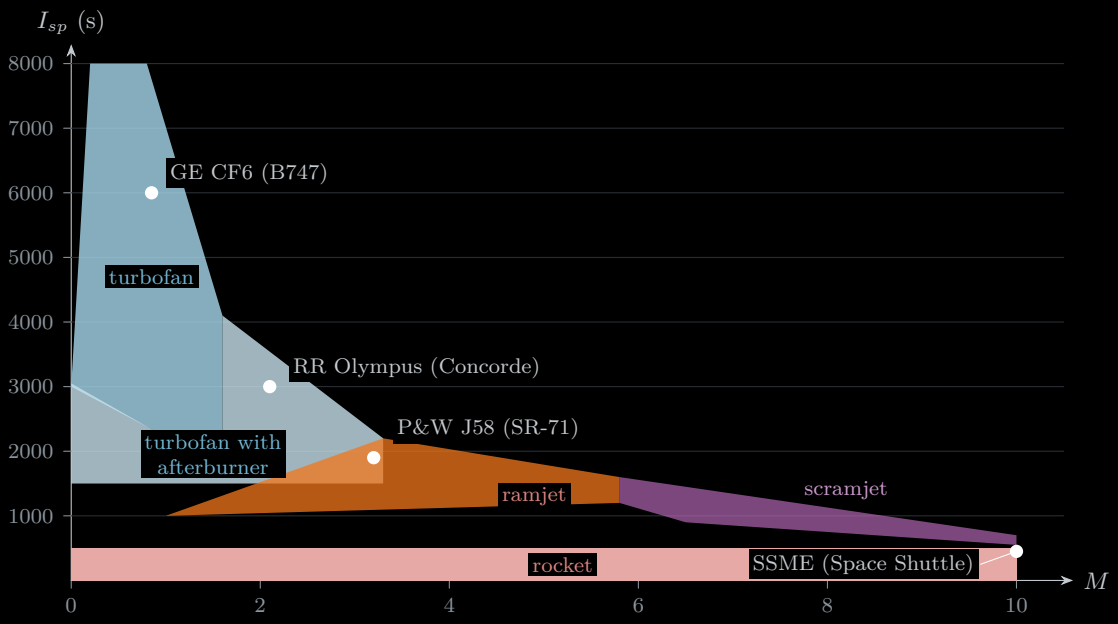


Figure 6.11. Specific impulse versus Mach for each engine type, redrawn and simplified from slide 23 (the regions are approximate; the dots are real engines: GE CF6 of the B747, RR Olympus of the Concorde, P&W J58 of the SR-71 and SSME of the Space Shuttle). The slide also shows the theoretical maxima of hydrocarbons (42 MJ/kg) and hydrogen (143 MJ/kg) in air, which have not been drawn.

System	Operating range	Characteristics
Propeller (electric, reciprocating or turboprop)	$M < 0.6$–0.7 (up to ~ 0.8)	Large mass flow, small ΔV: highest η_p. Limit: transonic blade tips.
Turbofan	Transport at $M \approx 0.8$; on the slide 23 chart it extends up to $M \approx 1.6$	Higher I_{sp} than the turbojet in that range; **high bypass**.
Turbojet (and with afterburner)	Up to $M \approx 3$ (slide 23 chart)	No fan; less efficient; the afterburner gives extra thrust.
Ramjet	Supersonic, from $M \approx 1$ to ≈ 6 (the chart's I_{sp} is the highest at $M \approx 3$–6)	No moving parts; does not work at rest.
Scramjet	$M \gtrsim 6$	Supersonic combustion.
Rocket	All M	Non-air-breathing; very low I_{sp}.

5 Solved exam-style problems

1. PROPULSIVE EFFICIENCY OF A PROPELLER AND OF A JET
Same thrust and same flight speed $V_{in} = 200\,\text{m/s}$.

- **Propeller** accelerating the air to $V_{out} = 240\,\text{m/s}$: $\eta_p = \dfrac{2}{1 + 240/200} = \dfrac{2}{2.2} = 0.91$.
- **Jet** accelerating it to $V_{out} = 600\,\text{m/s}$: $\eta_p = \dfrac{2}{1+3} = 0.50$.

To give the same thrust with $\dot{m}\,\Delta V = T$, the propeller needs **much more mass flow**: $\dot{m}_{propeller}/\dot{m}_{jet} = 400/40 = 10$ (with ΔV of 40 and 400 m/s). The propeller moves 10 times more air, accelerating it 10 times less, and wastes much less kinetic energy.

2. THRUST OF A TURBOJET

An engine takes in $\dot{m} = 20\,\text{kg/s}$ at $V_{in} = 250\,\text{m/s}$ and expels it at $V_{out} = 650\,\text{m/s}$ (pressure terms and the added fuel are neglected):

$$T = \dot{m}\,(V_{out} - V_{in}) = 20 \cdot 400 = 8\,000\ \text{N}$$

Useful power $T \cdot V_\infty = 8\,000 \cdot 250 = 2.0$ MW; power given to the air $= \frac{1}{2}\dot{m}(V_{out}^2 - V_{in}^2) = \frac{1}{2}\cdot 20\cdot(650^2 - 250^2) = 3.6$ MW; $\eta_p = 2.0/3.6 = 0.56$, which agrees with $2/(1+650/250) = 0.556$.

3. BATTERIES VERSUS FUEL (SLIDE 11)

The specific energy of fuel is $\approx 42\,\text{MJ/kg} = 11.7\,\text{kWh/kg}$. If 1 kg of fuel contains the energy of 70–80 kg of LiPo batteries, each kilogram of battery stores $\approx 11.7/75 \approx 0.16\,\text{kWh} = 160\,\text{Wh/kg}$. But this compares *stored energy*; in practice the electric motor converts better (0.72 versus 0.30–0.35 in the UAS Propulsion notes), so the real mass ratio for the same useful energy is somewhat smaller (about 30–35 times instead of 75, indicative).

4. 2023/24 AND 2024/25 TESTS ON PROPULSION

- *"The efficiency of turbofans is higher than that of turbojets at M 0.8."* **True**: more mass flow, smaller ΔV, higher η_p.
- *"Ramjets require a starter motor to accelerate their turbines."* **False**: they have no turbines.
- *"Rockets outperform turbojets in specific consumption at supersonic speeds."* **False**: the rocket has a low I_{sp} (much higher consumption).
- *"Mach 3 at 10 km: highest I_{sp}?"* **Ramjet**, not turboprop, rocket or scramjet.
- *"The thrust of commercial turboprops is due to the inlet-outlet pressure difference."* **False**: it is given by the propeller.

WHAT USUALLY APPEARS ON THE EXAM (T6)

In the first midterms of 2023/24 and 2024/25 there are **0 to 1 questions** on propulsion. When they appear, they are **conceptual multiple-choice, no calculations**.

- **Choice of system according to vehicle and speed**: piston engine for ultralights, turboprop for regional aircraft, electric for small UAS, turbofan at M 0.8, ramjet at M 3.
- $\eta_p \simeq 2/(1 + V_{out}/V_{in})$: more mass flow and less ΔV = more efficient (turbofan > turbojet; propeller > jet).
- **Turbojet elements** and nozzle (convergent subsonic, convergent-divergent supersonic); **the compressor rotor** is what gives kinetic energy to the air.
- **Ramjet/scramjet**: no moving parts, subsonic/supersonic combustion; they do not work at rest (the Meteor carries a booster rocket).
- **Rocket**: non-air-breathing, low I_{sp}, whole Mach range.
- **Fixed-pitch propeller at fixed rpm**: as speed rises, *the thrust (and C_T) always drops*; the efficiency does not always.
- **Figures of merit**: BSFC (propeller, per power), TSFC (jet, per thrust), $I_{sp} = T/(\dot{m}_{fuel} g)$ in seconds; high specific consumption = poor efficiency; high I_{sp} = good.
- **Fuel versus battery**: 1 kg of fuel $\approx$ 70–80 kg of LiPo.

The essentials: $T \approx \dot{m}\,\Delta V$; $\eta_p \simeq 2/(1 + V_{out}/V_{in})$; BSFC $= \dot{m}_f/P$; TSFC $= \dot{m}_f/T$; $I_{sp} = T/(\dot{m}_f g)$; propeller up to M 0.6–0.7; jet at high speeds.

OPEN QUESTIONS AND LOOSE ENDS (T6)

- Slide 23 (the I_{sp}-Mach chart) is an image: the regions and the position of the points have been estimated from the image and the own drawing is approximate.
- In the slides, the propeller Mach limit appears as "less than 0.8, normally less than 0.6" (slide 7) and as "Mach 0.7" (slide 10); both are given.
- The BSFC and TSFC examples are illustrative: the slides give no numerical values.